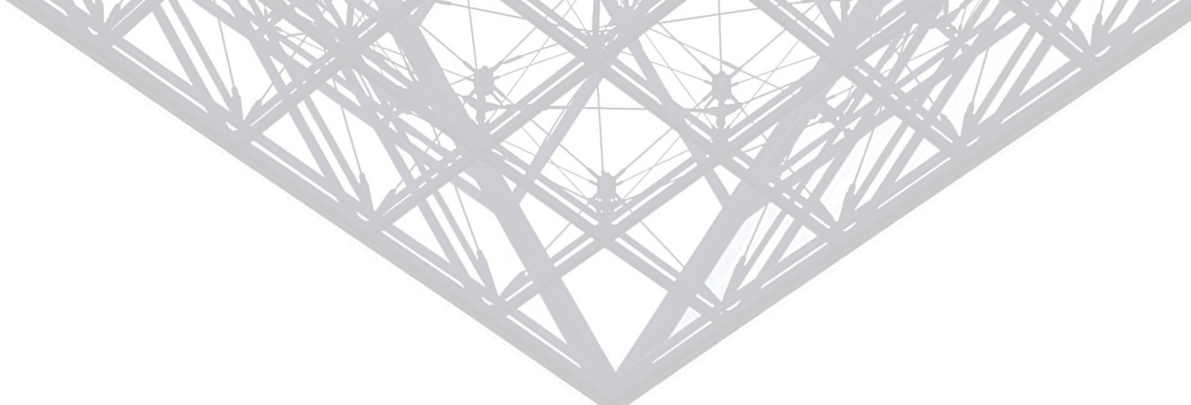

Ex-Ante Structural Diagnosis of Climate Project Performance

VERA-TCV™ Framework Applied to 15 GCF Projects

In climate systems, structure determines outcome. Everything else is noise.

ENHANCE Institute

May 2026



ENHANCE

Enhancing the Systems that sustain our world

Ex-Ante Structural Diagnosis of Climate Project Performance

This report utilises artificial intelligence–assisted information extraction and deterministic computational evaluation to analyse the structural integrity of publicly available climate project documentation. Artificial intelligence systems were employed solely to assist in the structured extraction of textual information from project documents.

All structural assessments were conducted using a deterministic Python-based implementation of the VERA-TCV™ framework — comprising VERA™, ToC:f(x)™, C-FAIR™, and VORTA™ — each of which evaluates a distinct structural layer of climate project design based on predefined structural criteria.

The evaluation process combines:

- Three-agent parallel extraction with Structured Convergence Integration (SCI)
- Protocol-constrained structural inference with Divergence Resolution Protocol (DRP)
- Deterministic Python-based scoring via independent model kernels

This hybrid analytical architecture ensures that project scores are generated through rule-based computational procedures rather than probabilistic AI judgement.

About ENHANCE

ENHANCE Institute is the research division of ENHANCE Co., Ltd. It designs sustainable policies, projects, and systems based on deterministic architectural principles.

It develops integrated architectures that ensure climate interventions are structurally viable, financially aligned, and capable of delivering real-world outcomes.

To support this, ENHANCE operates proprietary evaluation frameworks — including VERA™, ToC:f(x)™, C-FAIR™, and VORTA™ — which verify the structural integrity, risk allocation, and value formation of designed systems.

ENHANCE also develops quantitative methodologies and analytical tools to improve the transparency, reliability, and design quality of climate and sustainability projects.

The Fifteen Projects

Executive Summary	1
How to Read This Report	3
Opening	6

PART I – THE DIAGNOSTIC FRAMEWORK

Section 1	Why Structure Determines Performance	
	1.1 The Problem is Not Intent	7
	1.2 The Behavioral Premise	7
	1.3 Structure as the Unit of Analysis	7
Section 2	VERA-TCV™ Architecture	
	2.1 Four Independent Models	8
	2.2 VERA™ – Intent Fidelity at the Input Layer	9
	2.3 ToC:f(x)™ – Causal Structure at the Activity-Output Boundary	9
	2.4 C-FAIR™ – Delivery Environment at the Output-Outcome Boundary	9
	2.5 VORTA™ – Value Realization at the Outcome-Impact Boundary	10
	2.6 The Independence Principle	10
Section 3	Ex-Ante Methodology	
	3.1 The Diagnosis Precedes the Evidence	11
	3.2 Annual Performance Report as Comparator, Not Input	11
	3.3 Multi-Agent Extraction Protocol	11
	3.4 Epistemological Boundary	12
	3.5 Indicator Hierarchy and Exception Disclosure	12
Section 4	Scope and Limitations	
	4.1 Selection Criteria	12
	4.2 What This Study Is Not	13
	4.3 On the Relationship to the SSRN Paper	14

PART II — PORTFOLIO COVERAGE

Section 5	Fifteen Projects: Composition and Coverage	
	5.1 Portfolio Architecture	15
	5.2 Continental and Thematic Distribution	15
	5.3 Implementing Entity Diversity	16
	5.4 Verification Horizon	16
Section 6	Portfolio Scorecard	
	6.1 VERA™ Distribution — Design Authenticity	17
	6.2 ToC:f(x)™ Distribution — Causal Structure	18
	6.3 C-FAIR™ Distribution — Delivery Environment	19
	6.4 VORTA™ Distribution — Value Realization	19
	6.5 The Consolidated Pattern	20
Section 7	Document Set Standard	
	7.1 Three-Document Architecture	22
	7.2 Funding Proposals — The Diagnostic Source	22
	7.3 Annual Performance Reports — The Independent Comparator	22
	7.4 Evaluation Reports — External Verification	23
	7.5 Document Availability by Project	24

PART III — SIX CASE ANALYSES

Case 1 — FP034 · Uganda · Building Resilient Communities, Wetlands Ecosystems and Associated Catchments	25
Case 2 — FP072 · Zambia · Strengthening Climate Resilience of Agricultural Livelihoods	26
Case 3 — FP009 · El Salvador · Energy Savings Insurance for Private EE Investments by SMEs	27
Case 4 — FP039 · Egypt · GCF-EBRD Egypt Renewable Energy Financing Framework	29
Case 5 — FP069 · Bangladesh · Enhancing Adaptive Capacities of Coastal Communities	30
Case 6 — FP026 · Madagascar · Sustainable Landscapes in Eastern Madagascar	32

PART IV — CROSS-CASE FINDINGS

Section 8	Four Structural Patterns	34
Section 9	Three Portfolio-Level Findings	
	Finding 1 — C-FAIR Grade C is the structural justification for concessional climate finance	38
	Finding 2 — VORTA™ VCS F means structurally different things across the portfolio	38
	Finding 3 — No project achieved ToC:f(x)™ TIER I	39
Section 10	Predictive Validity Assessment	40

PART V — IMPLICATIONS

Section 11	What Ex-Ante Diagnosis Changes	43
Section 12	Verification Architecture Requirements	
	Requirement 1 — Measurement design as approval condition	45
	Requirement 2 — Chronic vs acute signal classification	45
	Requirement 3 — ToC Q5 pre-approval gate	46
Conclusion		47

APPENDICES

Appendix A	VERA™ v2.2: Structural Overview	48
Appendix B	ToC:f(x)™ SIM-v4.1: Structural Overview	48
Appendix C	C-FAIR™ v3.0: Structural Overview	49
Appendix D	VORTA™ v2.5: Structural Overview	49
Appendix E	Project Universe: 15 Selected Projects	50
Appendix F	APR Extraction Protocol	52
Appendix G	Individual Structural Validation Reports: Access Protocol	54
	FP069 Bangladesh Case - Public Excerpt	55
Appendix H	Predictive Validity Matrix	59
Appendix I	KEY TERMS	63
Methodological Q&A		65

CLOSING ESSAY — Why Structure Must Come First

Ex-Ante Structural Diagnosis of Climate Project Performance

VERA-TCV™ Framework Applied to 15 GCF Projects

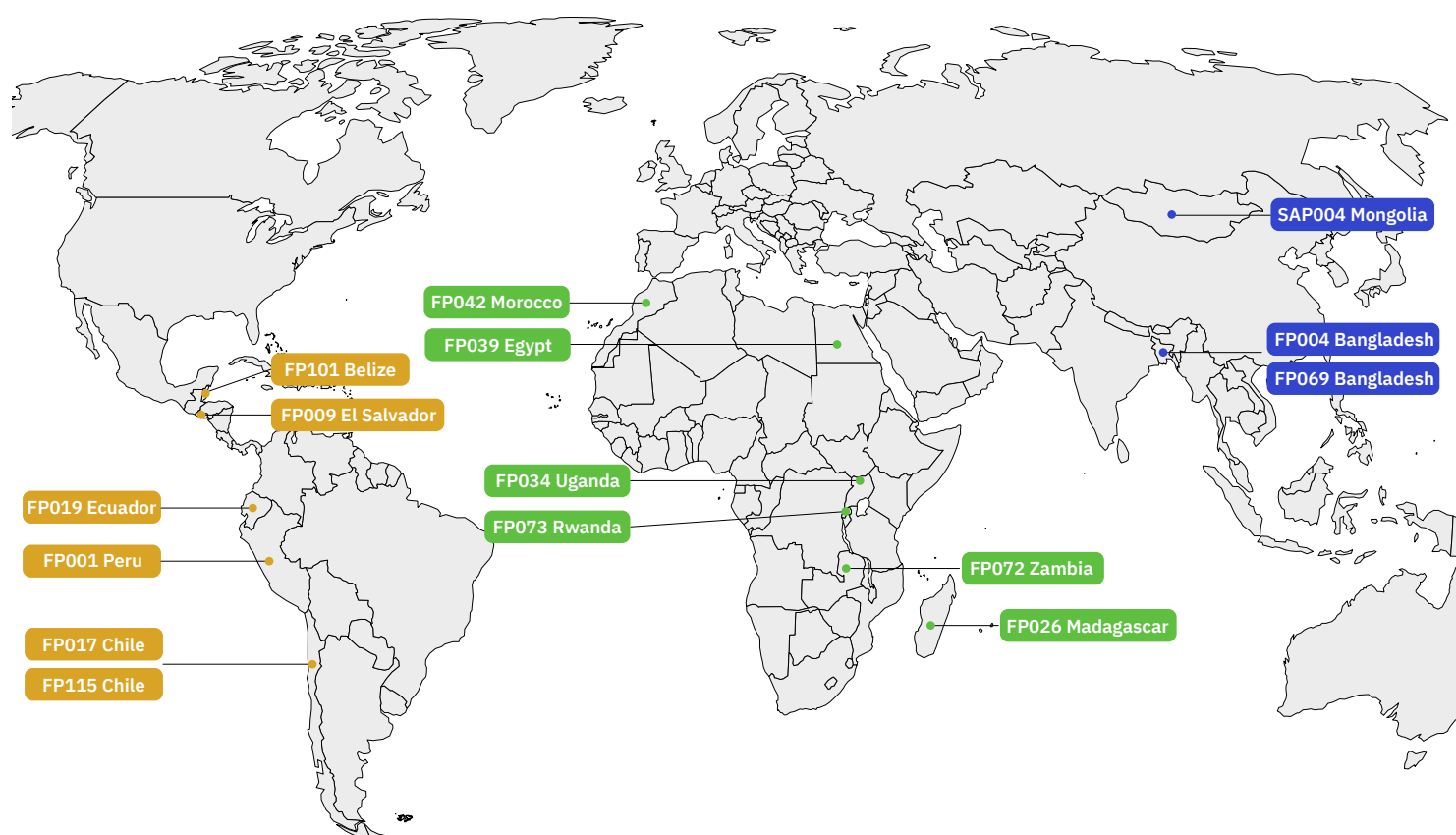
In climate systems, structure determines outcome. Everything else is noise.

THE FIFTEEN PROJECTS

Every project in this portfolio was approved by GCF's Board following a rigorous review process. Every project was submitted by accredited entities with documented track records. Every project carried a Theory of Change, a risk register, a financial model, and a logframe.

The fifteen projects presented here span four continents, five thematic areas, and an average of 5.7 years of implementation data. They were not selected because they failed. They were selected because their documentation was sufficient to test whether structural conditions present at the design stage were consequential for what happened next.

The VERA-TCV™ results shown on each page were generated from the funding proposal alone — before any implementation data was reviewed. What those results mean, and whether they held, is what this report is about.



Project	Region	Country	Classification	Size (USD)	GCF Financing	Status
FP001	LAC	Peru	Adaptation	9.1M	6.24M	Completed
FP004	Asia-Pacific	Bangladesh	Adaptation	81.0M	40.0M	Active
FP009	LAC	El Salvador	Mitigation	41.7M	21.7M	Completed
FP017	LAC	Chile	Cross-cutting	181.0M	39.0M	Completed
FP019	LAC	Ecuador	Cross-cutting	84.0M	41.1M	Completed
FP026	Africa	Madagascar	Cross-cutting	16.0M	15.2M	Active
FP034	Africa	Uganda	Adaptation	44.3M	24.1M	Active
FP039	Africa	Egypt	Mitigation	1.0B	154.7M	Completed
FP042	Africa	Morocco	Adaptation	88.8M	23.3M	Active
FP069	Asia-Pacific	Bangladesh	Adaptation	33.0M	24.9M	Active
FP072	Africa	Zambia	Adaptation	137.3M	32.0M	Active
FP073	Africa	Rwanda	Adaptation	33.2M	32.8M	Active
FP101	LAC	Belize	Adaptation	20.0M	8.0M	Active
FP115	LAC	Chile	Cross-cutting	1.1B	60M	Cancelled
SAP004	Asia-Pacific	Mongolia	Adaptation	21.5M	10.0M	Active



FP001: Building the Resilience of Wetlands in the Province of Datem del Marañón, Peru

REGION

Latin America and the Caribbean

COUNTRY

Peru

Financing	Public Sector	USD 9.1m
Size	Micro (≤ 10m)	

GCF financing 100% disbursed

Instrument	Amount
------------	--------

- Grant (68.5%) USD 6,240,000

TOTAL GCF FINANCING

USD 6,240,000

Co-financing

Co-financer	Instrument	Amount
-------------	------------	--------

- Co-financing Grant USD 1,070,000

- Co-financing Grant USD 1,800,000

TOTAL CO-FINANCING

USD 2,870,000

Approved
05 Nov 2015

To be completed
31 Dec 2024



Executive Summary

The project seeks to enhance the resilience capacity of the indigenous communities living in the rich carbon stock wetland ecosystem in the Province of Datem del Marañón (PDM) in the Region 1 of Loreto, Peru, improve their livelihoods and to reduce greenhouse gas (GHG) emissions from deforestation. The target population is primarily low-income indigenous peoples from seven ethnic groups. The project aims to create social capital and agreed plans to entrust the management of the natural resource base to the indigenous communities.

The proposed activities seek to: (a) facilitate the participatory preparation of land-use and operational management plans; (b) entrust natural resources management to indigenous communities and empower women in the decisionmaking process; and (c) strengthen and expand commercially viable and sustainable bio-businesses of non-timber forest products. All activities foster empowerment and community ownership, improve livelihoods, enhance learning opportunities, and consolidate indigenous peoples' basic rights. All the proposed activities are community-based adaptation and climate change mitigation good practices, which are complemented by strong monitoring and evaluation systems. The project includes a science and technology component to enhance the knowledge base in order to guide future activities aimed at preserving the carbon stock and enhancing community well-being.

The total carbon stock in the swamps of Datem del Marañón is estimated at around 3,780 million tonnes of carbon dioxide equivalent (Mt CO₂ eq). Based on data provided by MINAM, a deforestation reference scenario for the project area was defined as the average of the last 10 years of observation. The estimated avoided deforestation over a 10- year period amounts to 140.6 hectares (ha) of aguajales (palm swamp forest) and 4,720 ha of terra firma forests, equivalent to 2.63 Mt CO₂ eq. Climate change is expected to have profound impacts on forest and swamp ecosystems, with more frequent droughts and changes in primary productivity affecting peoples and ecosystems.

Project Information

Project Code	FP001
Classification	Adaptation
Accredited Entity	PROFONANPE (National DAE)
Board Approval	B.11 (Nov 2015)
Status	Completed
Beneficiaries	
APR Horizon	7 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier II (0.654)
ToC:f(x)™	TIER II (0.700)
C-FAIR™	Grade C (S2 Weak: Grant 30%+)
VORTA™	VCS F / GRANT_JUSTIFIED
Structural Pattern	A — Design–Environment Separation
Consistency Rate	79.2%



FP004: Climate-Resilient Infrastructure Mainstreaming in Bangladesh

REGION

Asia and the Pacific

COUNTRY

Bangladesh

Financing	Public Sector	USD 81.0m
Size	Medium (50m - 250m)	

GCF financing		32% disbursed
Instrument	Amount	
• Grant (49.4%)	USD 40,000,000	
TOTAL GCF FINANCING		USD 40,000,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Grant	USD 15,912,000
• Co-financing	Grant	USD 25,089,500
TOTAL CO-FINANCING		USD 41,001,500



Executive Summary

The Climate Resilient Infrastructure Mainstreaming (CRIM) project systematically integrates climate adaptation into infrastructure planning and maintenance of Bangladesh's Local Government Engineering Department (LGED), which manages over 10% of all annual public investments nationwide. A dedicated Centre of Excellence (CReLIC) is established within LGED as a permanent think tank to mainstream climate resilience across its entire infrastructure portfolio.

The project directly finances pilot climate-resilient infrastructure in three of Bangladesh's most vulnerable coastal districts — Bhola, Barguna, and Satkhira — including 45 new multipurpose cyclone shelters, rehabilitation of 20 existing shelters, 80 km of critical road connectivity, and urban infrastructure in Satkhira. Shelters are designed for year-round use as primary schools, supporting over 18,000 children and creating more than 1,700 full-time jobs.

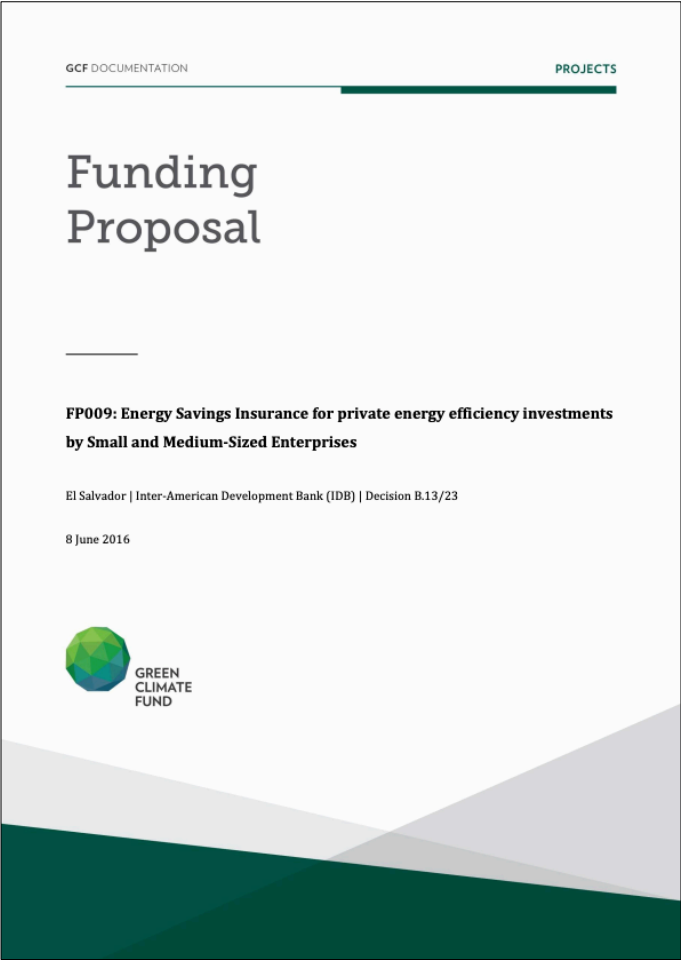
The project directly benefits 134,000 people and indirectly reaches 10.4 million (6.8% of Bangladesh's population) through systemic reform of LGED's climate resilience standards. Total project volume is USD 80M: USD 40M GCF grant, USD 15M KfW co-financing, and USD 25M Government of Bangladesh counterpart contribution.

Project Information

Project Code	FP004
Classification	Adaptation
Accredited Entity	KfW (Bilateral DFI)
Board Approval	B.11 (2015)
Status	Active
Beneficiaries	134,000 (direct) / 10.4M (indirect)
APR Horizon	5 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier I (0.800)
ToC:f(x)™	TIER II (0.750)
C-FAIR™	Grade C / Grant 30%+ (hazard-dominant)
VORTA™	VCS A / GRANT_JUSTIFIED
Structural Pattern	A — Design–Environment Separation
Consistency Rate	92.3%



FP009: Energy Savings Insurance for private energy efficiency investments by Small and Medium-Sized Enterprises

REGION

Latin America and the Caribbean

COUNTRY

El Salvador

Financing	Public Sector	USD 41.7m
Size	Small (10m - 50m)	

GCF financing		100% disbursed
Instrument	Amount	
• Loan (47.96%)	USD 20,000,000	
• Grant (4.08%)	USD 1,700,000	
TOTAL GCF FINANCING		USD 21,700,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 20,000,000
TOTAL CO-FINANCING		USD 20,000,000

Approved	To be completed
30 Jun 2016	22 Jul 2026



Executive Summary

FP009 promotes energy efficiency (EE) investment by SMEs in El Salvador through a two-tier financial mechanism. The IDB provides a sovereign-guaranteed GCF loan to BANDESAL, a national development bank, which blends GCF concessional resources with its own capital to offer financing to local financial institutions (LFIs) – which in turn lend to SMEs adopting eligible EE measures.

A complementary GCF grant funds non-financial instruments including performance contracts, MRV protocols for energy savings, and Energy Savings Insurance (ESI) products – creating the enabling environment for a bankable EE project pipeline. Total financing: GCF USD 21.7M (USD 20M loan + USD 1.7M grant), BANDESAL USD 20M, total USD 41.7M.

The project targets 562,037 tCO₂eq in emission reductions over 15 years. At 20% market penetration of 6,232 eligible firms, El Salvador could reduce energy-related emissions by 94,811 tCO₂eq annually – equivalent to 1.7% of the country's 2005 energy-related emissions. Full replication would yield 472,688 tCO₂eq annually, or 8.4% of the same baseline.

Project Information

Project Code	FP009
Classification	Mitigation
Accredited Entity	IDB Invest (MDB Private Window)
Board Approval	B.13 (2016)
Status	Completed
Beneficiaries	6,232 eligible SMEs (target)
APR Horizon	4 years*

*CY2021 missing; CY2023–2024 Section 2 confidential

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III (IF3=N/A; IF2=1.000)
ToC:f(x)™	TIER II (0.750)
C-FAIR™	Grade B / TIER IV (S2 Moderate)
VORTA™	VCS A / TIER I (IRRc +16.78%)
Structural Pattern	C — Measurement Architecture
Consistency Rate	80.0% (data-limited)



FP017: Climate Action and Solar Energy development Programme in the Tarapacá Region in Chile

REGION
Latin America and the Caribbean
COUNTRY
Chile

Financing Private Sector
Size Medium (50m - 250m) **USD 181.0m**

GCF financing		100% disbursed
Instrument	Amount	
• Loan (21.55%)	USD 39,000,000	
TOTAL GCF FINANCING		USD 39,000,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 40,000,000
• Co-financing	Equity	USD 82,000,000
• Co-financing	Loan	USD 20,000,000
TOTAL CO-FINANCING		USD 142,000,000

Approved 30 Jun 2016 To be completed 30 Oct 2022

Executive Summary

An enabling setting is made available by GCF and CAF to establish Chile's first national-level convergence between climate funding, private funding, and sector plans for low-emission sustainable development. The mechanism supports Atacama Solar, a project driven by Sonnedix, expanding photovoltaic solar energy in a region where the electricity generation matrix is highly intensive in thermoelectricity and GHG emissions.

Atacama Solar consists of a 143 MW solar photovoltaic plant in the Atacama Desert — an area with the highest solar radiation in South America — as Phase I of a planned 250 MW facility. Chile remains heavily dependent on fossil fuels and hydrological sources, with a commitment to achieve 20% non-conventional renewable power generation by 2025. The project delivers an estimated 193,949 tCO₂eq in annual reductions (3.7 Mt CO₂eq over 20 years) and provides the lowest cost per tCO₂eq among Chilean solar peers.

In an innovative financial design, the project is based on market-price cash flows without a fixed PPA, demonstrating that solar power can compete and displace fossil-fuel generation in Chile's capital markets. CAF complements the project finance with a strategic partnership to build climate resilience at community and regional level, establishing a national reference model for cross-cutting low-emission development in northern Chile.

Project Information

Project Code	FP017
Classification	Cross-cutting
Accredited Entity	IDB (MDB Public Window)
Board Approval	B.14 (2016)
Status	Completed
Beneficiaries	Regional electricity grid (SING)
APR Horizon	5 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III
ToC:f(x)™	TIER II (0.700)
C-FAIR™	Grade B (S2 Mod.; Grant <15%)
VORTA™	VCS A / TIER I (IRRc +48.71%)
Structural Pattern	C — Measurement Architecture
Consistency Rate	100%



FP019: Priming Financial and Land-Use Planning Instruments to Reduce Emissions from Deforestation

REGION

Latin America and the Caribbean

COUNTRY

Ecuador

Financing	Public Sector	USD 84.0m
Size	Medium (50m - 250m)	

GCF financing	100% disbursed
Instrument	Amount
• Grant (49.0%)	USD 41,172,739
TOTAL GCF FINANCING	
USD 41,172,739	

Co-financing	
Co-financer	Instrument Amount
• Co-financing	Grant USD 40,245,550
• Co-financing	Grant USD 1,769,458
• Co-financing	Grant USD 820,900
TOTAL CO-FINANCING	
USD 42,835,908	

Approved	To be completed
14 Oct 2016	22 Nov 2023



Executive Summary

Ecuador is finalizing its REDD+1 readiness process². A Forest Reference Emission Level (FREL) was submitted and technically assessed by the UNFCCC in 2015³ and a REDD+ Action Plan (REDD+ AP4) has been approved. The Action Plan presents the policies and measures prioritized to address the drivers of deforestation⁵. It has a national scope and includes the 5 REDD+ activities. It is designed to support the national objective of achieving zero net deforestation by 2020.

The GCF project will co-finance the REDD+ AP by providing targeted investment to control agricultural expansion into forest areas; optimize existing financial, economic mechanisms to implement agricultural and livestock production practices that reduce deforestation; align land-use zoning plans with national climate change-related targets; strengthen restoration, conservation and sustainable production in vulnerable watersheds; redesign public credit lines to orient them towards sustainable agricultural production practices; promote tax incentives for REDD-supportive activities; and strengthen purchasing policies for deforestation-free commodities, their certification and traceability.

The emission reductions that Ecuador will achieve by implementing its REDD+ AP during the GCF project's lifetime (2017-2021) will be assessed in 2018, 2020 and 2022, through the Biennial Update Reports to the UNFCCC⁶, with reference to the FREL

Project Information

Project Code	FP019
Classification	Cross-cutting
Accredited Entity	UNDP (UN Agency)
Board Approval	B.14 (2016)
Status	Completed
Beneficiaries	—
APR Horizon	6 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III (0.650)
ToC:f(x)™	TIER II (0.650)
C-FAIR™	Grade C (S2 Weak; Grant 30%+)
VORTA™	VCS F / REDESIGN_REQUIRED
Structural Pattern	D — Attribution Constraint
Consistency Rate	78.1% ‡

‡ Project-level causal exclusivity is structurally unavailable due to the national-level accounting architecture of the REDD+ Warsaw Framework.

*** Directional accuracy 100% — no contradictions recorded. Rate reflects attribution impossibility under REDD+ Warsaw Framework, not predictive failure.**



FP026: Sustainable Landscapes in Eastern Madagascar

REGION
Africa
COUNTRY
Madagascar

Financing	Private Sector	USD 16.0m
Size	Small (10m - 50m)	

GCF financing		100% disbursed
Instrument	Amount	
• Grant (95.0%)	USD 15,199,555	
TOTAL GCF FINANCING		USD 15,199,555

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Grant	USD 800,000
TOTAL CO-FINANCING		USD 800,000

Approved
14 Oct 2016

To be completed
08 May 2025



Executive Summary

Madagascar is one of the most vulnerable countries to climate change in the world^{1,2}. Home to vast numbers of extremely vulnerable smallholders³, the landscapes dominated by natural forest are also sources of greenhouse gas (GHG) emissions from deforestation⁴. The Project goal is to implement sustainable landscape measures to enhance resiliency of smallholders, reduce GHG emissions and channel private finance into climate-smart investments in agriculture and renewable energy that transform livelihoods. The Project aims to demonstrate a replicable model for addressing smallholder vulnerability that mobilizes both the public and private sector. Not-for-profit activities will prepare smallholders to ultimately access private sector investment, providing a pathway out of extreme vulnerability and dependency⁵. The Project goal and innovative model make it attractive for private investors and will leverage funds from the issuance of a first-of-its-kind Green/Climate Bond. All returns and profits from GCF's investments in the private sector activities will be re-invested to capitalise a Climate Change Trust Fund for Madagascar (subject to approval by GCF), to enable continued investment in landscape-level adaptation and mitigation activities. The public and private sector interventions will make major contributions towards Madagascar's adaptation and mitigation priorities as laid out in national policies and strategies. Among the high-level impacts are 10 MtCO₂eq of reduced emissions; 33 MW capacity of renewable energy generation; 562,000 direct beneficiaries, including 114,000 with improved resilience and 448,000 with access to low-emissions energy; and 1 million indirect beneficiaries.

Project Information

Project Code	FP026
Classification	Cross-cutting
Accredited Entity	Conservation International + EIB (NGO + MDB)
Board Approval	B.14 (2016)
Status	Active
Beneficiaries	—
APR Horizon	7 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier II (0.865)
ToC:f(x)™	TIER II (0.700)
C-FAIR™	Grade C / TIER V (S1 High; S2 Weak; S3 High)
VORTA™	VCS F / IRRc negative
Structural Pattern	B — Internal Mechanism Failure
Consistency Rate	98.9% §



FP034: Building Resilient Communities, Wetlands Ecosystems and Associated Catchments in Uganda

REGION

Africa

COUNTRY

Uganda

Financing	Public Sector	USD 44.3m
Size	Small (10m - 50m)	

GCF financing 100% disbursed

Instrument	Amount
• Grant (54.5%)	USD 24,140,160
TOTAL GCF FINANCING	
USD 24,140,160	

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 2,000,000
• Co-financing	Equity	USD 18,122,000
TOTAL CO-FINANCING		
		USD 20,122,000

Approved	To be completed
15 Dec 2016	31 Dec 2026



Executive Summary

Climate change, combined with human and environmental stressors, is accelerating the degradation of Uganda's wetlands and the ecosystem services they provide — directly threatening the livelihoods of approximately 4,000,000 people living in and around these areas, over 80% of whom depend on wetland resources for household food security. The project supports the Government of Uganda in integrating climate change considerations — increased variability, drought, floods, and extreme weather — into the management of critical wetland ecosystems.

The project targets two regions across 16 districts in South Western and Eastern Uganda, covering a total population of 3,946,366 and a land area of 13,000 km². At least 800,000 people will directly benefit through three intervention streams: restoration of wetland ecosystems using Ramsar Convention wise-use principles, support for resilient agricultural practices and alternative livelihoods, and strengthening of climate information and early warning systems. As a Low Income Economy and Least Developed Country, Uganda requests 100% GCF grant resources — the cost-benefit analysis confirms positive returns on investment, and the project is fully aligned with Uganda's National Adaptation Programme of Action and National Climate Change Policy.

Project Information

Project Code	FP034
Classification	Adaptation
Accredited Entity	UNDP (UN Agency)
Board Approval	B.15 (2016)
Status	Active
Beneficiaries	800,000 (direct) / 3,946,366 (total population)
APR Horizon	7 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier II (0.797)
ToC:f(x)™	TIER II (0.800)
C-FAIR™	Grade C / TIER V (S2 Weak; Grant 30%+)
VORTA™	VCS F / GRANT_JUSTIFIED (IRRm = -100%)
Structural Pattern	A — Design–Environment Separation
Consistency Rate	76.9% / 80.8% ¶

¶ Two rates are reported. The unadjusted rate (76.9%) reflects the automatic signal mapper. The adjusted rate (80.8%) applies chronic signal correction, recognising that VERA™ RA2 in LDC contexts operates as a multi-year cumulative delivery ceiling rather than a discrete annual event. Both are reported transparently. Zero directional contradictions recorded across all seven APR years under either convention.



FP039: GCF – EBRD Egypt Renewable Energy Financing Framework

REGION

Africa

COUNTRY

Egypt

Financing	Private Sector	USD 1.0b
Size	Large (250m +)	

GCF financing		100% disbursed
Instrument	Amount	
• Loan	USD 150,000,000	
• Grant (0.47%)	USD 4,700,000	
TOTAL GCF FINANCING		USD 154,700,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 350,000,000
• Co-financing	Loan	USD 125,000,000
• Co-financing	Equity	USD 250,000,000
• Co-financing	Grant	USD 2,300,000
• Co-financing	Loan	USD 125,000,000
TOTAL CO-FINANCING		USD 852,300,000

Approved

06 Apr 2017

To be completed

07 Sep 2024

Executive Summary

Egypt ranked among the world's fastest-growing GHG emitters in the 1990–2012 period and faces a critical challenge in diversifying its electricity mix away from over 90% hydrocarbon dependence. The government has established an ambitious target for renewable sources to provide 20% of electricity generation by 2022, underpinned by the Renewable Energy Law (2014) and the Sustainable Energy Strategy to 2035 — with the private sector expected to deliver most of this capacity.

The GCF-EBRD Egypt Renewable Energy Financing Framework addresses four key barriers to implementation: high transaction costs for first private RE debt projects, limited long-tenor commercial debt, insufficient grid integration infrastructure, and the absence of competitive tender mechanisms. The Framework operates through two components: Component 1 provides technical assistance for renewable energy integration, policies and planning; Component 2 scales up investments by blending EBRD and GCF financing with additional debt from DFIs and commercial banks. The Framework is expected to generate approximately 1,400 GWh of electricity annually and deliver avoided emissions of around 800,000 tCO₂e per year once fully operational.

Project Information

Project Code	FP039
Classification	Mitigation
Accredited Entity	EBRD (MDB Private Window)
Board Approval	B.16 (2017)
Status	Completed
Beneficiaries	—
APR Horizon	8 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III (RA3 = 0.000; RA2 = 0.250)
ToC:f(x)™	TIER II (0.700)
C-FAIR™	Grade C (S1 Low; S2 Weak)
VORTA™	VCS F (IRRc = −53.62%)
Structural Pattern	C — Private-Public Bifurcation + Measurement Gap
Consistency Rate	100%*

*FP039 — 100% consistency reflects the precision with which VERA-TCV™ identified two structurally opposite trajectories from the 2017 design document. Component 2 (private RE) overperformed at 147%; Component 1.1 (public grid integration) was abandoned at Year 7 with zero deliverables. Both predictions held simultaneously — in opposite directions — across eight years. VORTA™ VCS F was predetermined from RA3 = 0.000: the measurement architecture for Blended Finance public leverage was absent at approval.



FP042: Irrigation development and adaptation of irrigated agriculture to climate change in semi-arid Morocco

REGION
Africa
COUNTRY
Morocco

Financing Public Sector
Size Medium (50m - 250m) **USD 88.8m**

GCF financing		100% disbursed
Instrument	Amount	
• Grant (26.3%)	USD 23,356,027	
TOTAL GCF FINANCING		USD 23,356,027

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 46,712,055
• Co-financing	Grant	USD 1,167,801
• Co-financing	Grant	USD 17,517,020
TOTAL CO-FINANCING		USD 65,396,876

Approved 06 Apr 2017 To be completed 11 Apr 2027

Executive Summary

FP042 targets the Boudnib Valley in Morocco's semi-arid Tafilalet region — one of the country's most climate-vulnerable areas, characterized by rising temperatures, chronic water scarcity, and an intermittent river system that has repeatedly destroyed fields and homes. The local population depends on oasis-based agriculture, increasingly threatened by drought and groundwater depletion driven by large-scale commercial date and olive farming developments outside the oasis areas.

The project builds on the construction of a new upstream dam to secure water access and strengthen climate resilience across three components: C1 — connecting the dam and transferring surface water to the Boudnib Valley (EUR 45.6M); C2 — building climate resilience of oasis communities through a holistic approach (EUR 12.5M); and C3 — cross-cutting sustainability measures including technical assistance and groundwater preservation (EUR 8.3M). Total project cost is EUR 76M, financed through an AFD concessional loan of EUR 40M plus EUR 1M grant, a Government of Morocco contribution of EUR 15M, and a GCF contribution of USD 20M enabling the integration of the community resilience and sustainability components.

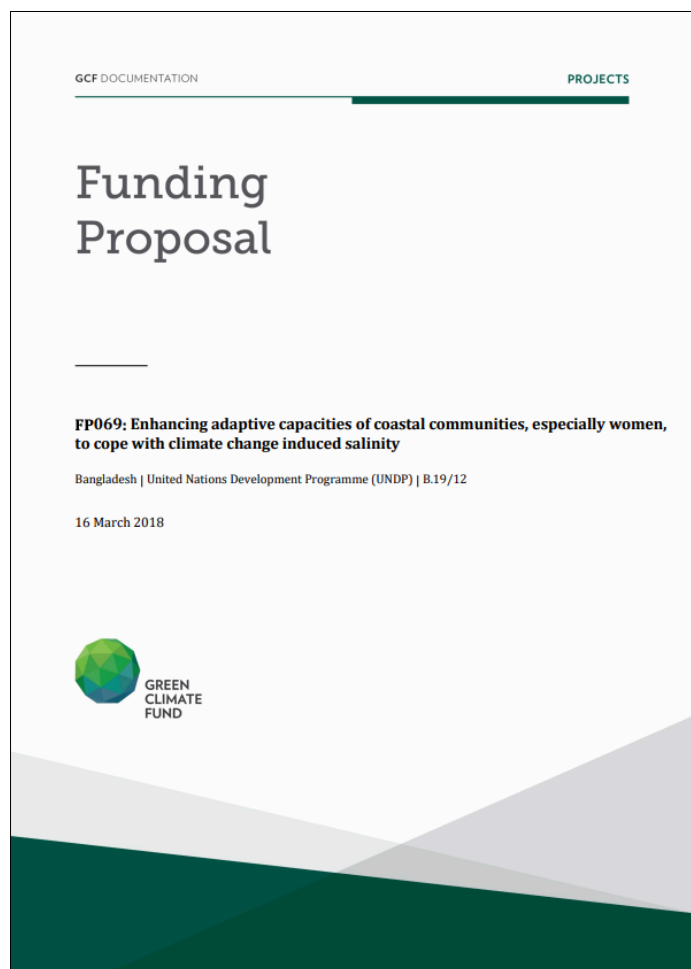
Project Information

Project Code	FP042
Classification	Adaptation
Accredited Entity	AFD (Bilateral DFI)
Board Approval	B.16 (2017)
Status	Active
Beneficiaries	—
APR Horizon	6 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III
ToC:f(x)™	TIER II (0.650)
C-FAIR™	Grade C (S2 Weak; Grant 30%+)
VORTA™	VCS F
Structural Pattern	A — Design–Environment Separation
Consistency Rate	100%**

**FP042 — 100% consistency reflects confirmed Output–Outcome disconnect: Kaddoussa Dam infrastructure was physically constructed, but water was not delivered to beneficiaries. VERA™ CR-Tier III and C-FAIR™ S2 Weak identified this structural ceiling from the 2017 design document. CY2023–2024 partial cells treated as confirmed where directional alignment is present but full quantitative verification is inaccessible.



FP069: Enhancing adaptive capacities of coastal communities, especially women, to cope with climate change induced salinity

REGION

Asia and the Pacific

COUNTRY

Bangladesh

Financing	Private Sector	USD 33.0m
Size	Small (10m - 50m)	

GCF financing

100% disbursed

Instrument	Amount
• Grant (75.7%)	USD 24,980,000
TOTAL GCF FINANCING	
USD 24,980,000	

Co-financing

Co-financer	Instrument	Amount
• Co-financing	Grant	USD 8,000,000
TOTAL CO-FINANCING		
USD 8,000,000		

Approved
12 Oct 2018

To be completed
30 Jun 2026



Executive Summary

FP069 addresses climate change-induced saltwater intrusion into freshwater resources along Bangladesh's southwestern coastal belt, where intensified cyclones, storm surges, and sea-level rise disproportionately threaten women and girls dependent on freshwater-reliant agricultural livelihoods and drinking water.

The project strengthens adaptive capacities of coastal communities in Khulna and Satkhira districts through three interventions: diversification toward climate-resilient livelihoods, investment in climate-resilient drinking water solutions, and institutional capacity building for replication at scale. An estimated 719,229 people (245,516 direct; 473,713 indirect) will benefit, with projected economic co-benefits of USD 15M (income) and USD 4M (health) over the project lifetime.

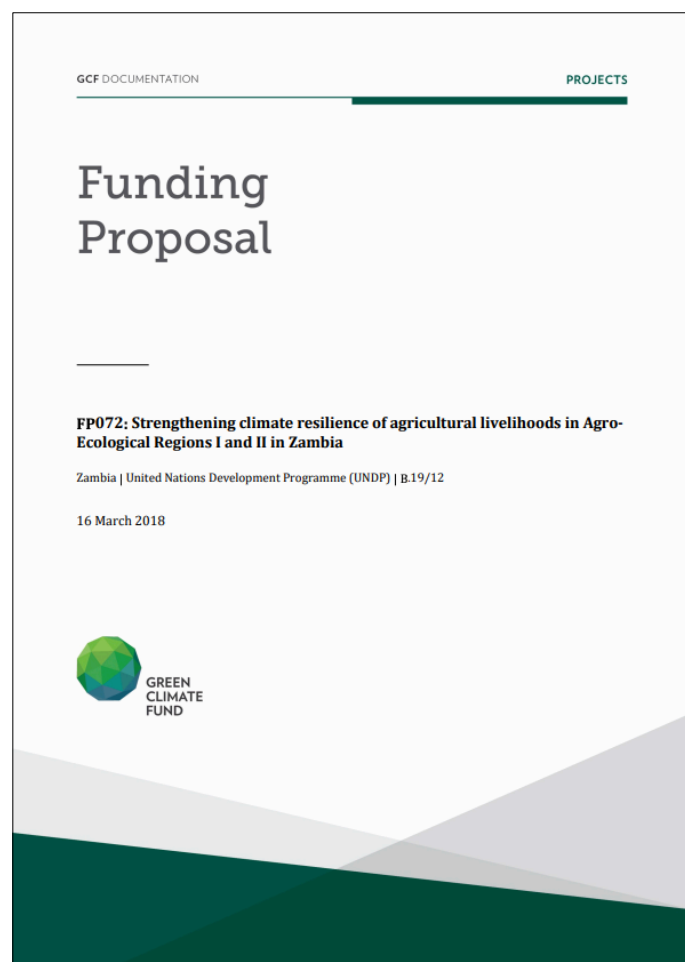
Project Information

Project Code	FP069
Classification	Adaptation
Accredited Entity	UNDP (UN Agency)
Board Approval	B.19 (2018)
Status	Active
Beneficiaries	719,229 (245,516 direct / 473,713 indirect)
APR Horizon	5 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier II (0.600)
ToC:f(x)™	TIER II (0.600)
C-FAIR™	Grade C / Hazard-dominant (S1 High)
VORTA™	VCS F / DO SVA_WARNING
Structural Pattern	D — Problem-Solution Misalignment
Consistency Rate	84% (21/25) ††

†† Score (✓) 60% / Score (incl. Δ) 84%. Directional accuracy 100% — zero contradictions recorded. USD 33.0M committed. USD 38,571 total annual WLGI income at project close. The gap between these two numbers is the quantitative expression of what pre-approval structural diagnosis is designed to prevent.



FP072: Strengthening climate resilience of agricultural livelihoods in AgroEcological Regions I and II in Zambia

REGION

Africa

COUNTRY

Zambia

Financing Public Sector
Size Medium (50m - 250m) **USD 137.3m**

GCF financing

100% disbursed

Instrument	Amount
• Grant (23.3%)	USD 32,000,000
TOTAL GCF FINANCING	
USD 32,000,000	

Co-financing

Co-financer	Instrument	Amount
Co-financing	Grant	USD 103,500,000
Co-financing	Grant	USD 369,000
Co-financing	Grant	USD 1,400,000
TOTAL CO-FINANCING		
		USD 105.269.000

Approved
01 Mar 2018

To be completed
12 Jul 2027

Executive Summary

FP072 strengthens the climate resilience of vulnerable smallholder farmers in Zambia's Agro-Ecological Regions I and II — the country's poorest areas, where rain-fed agriculture dominates and climate change is intensifying drought frequency and rainfall variability. Women are disproportionately affected, given their central role in household food production despite unequal access to land, inputs, and market information.

The project takes a value chain approach across three outputs: strengthening farmers' capacity to plan for climate risk; promoting resilient agricultural production and diversification practices; and improving access to markets and commercialization of resilient commodities. Targeting 16 districts across five provinces, the project directly benefits approximately 946,153 people. GCF grant resources finance activities with clear climate additionality — climate information and early warning systems, water access for smallholder farmers, and rural market linkages — while very high co-financing drives a structural shift in public agriculture financing toward climate resilience. The project aligns with Zambia's Seventh National Development Plan, NAPA, and NDC priorities.

Project Information

Project Code	FP072
Classification	Adaptation
Accredited Entity	UNDP (UN Agency)
Board Approval	B.19 (2018)
Status	Active
Beneficiaries	946,153 (direct)
APR Horizon	7 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier I (0.882)
ToC:f(x)™	TIER II (0.750)
C-FAIR™	Grade C / S2 Moderate (Grant 10–30%)
VORTA™	VCS F / IRRc negative
Structural Pattern	A — Design–Environment Separation (Absorbed)
Consistency Rate	100%



FP073: Strengthening climate resilience of rural communities in Northern Rwanda

REGION
Africa

COUNTRY
Rwanda

Financing
Size

Public Sector
Small (10m - 50m)

USD 33.2m

GCF financing

100% disbursed

Instrument
Amount

Grant (98.9%)
USD 32,794,442

TOTAL GCF FINANCING
USD 32,794,442

Co-financing

Co-financer
Instrument
Amount

Co-financing Grant USD 147,000

Co-financing Grant USD 107,026

Co-financing Grant USD 105,964

TOTAL CO-FINANCING
USD 359,990

Approved
01 Mar 2018

To be completed
15 May 2025

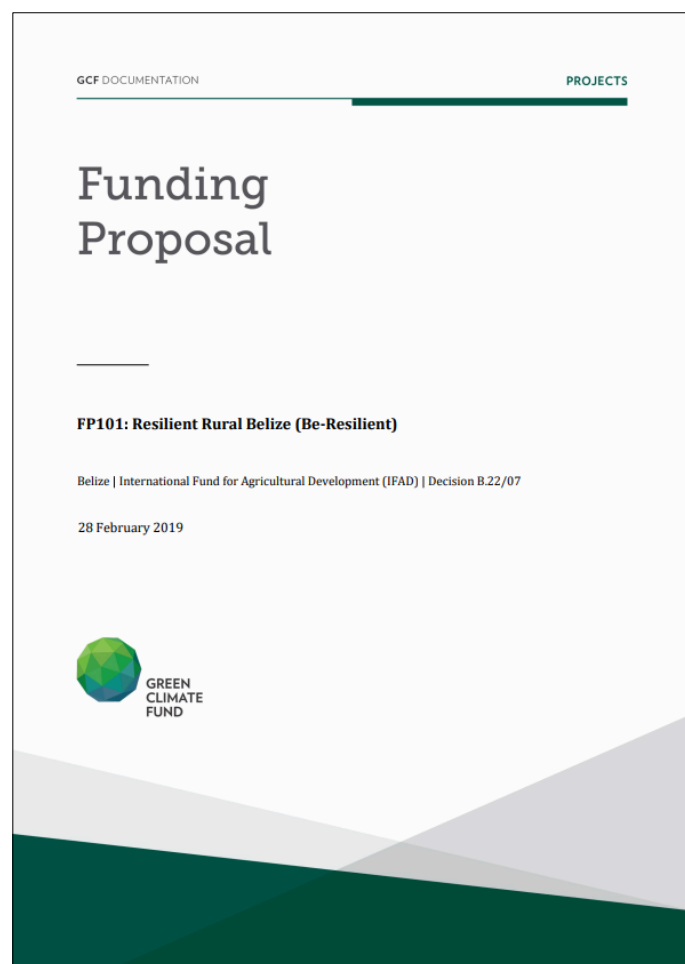
Executive Summary

FP073 increases the resilience of vulnerable communities in Rwanda's Gicumbi District through four integrated outputs: watershed restoration and climate-resilient agriculture in the Muvumba sub-catchment; sustainable forest management and fuel-efficient cooking; climate-resilient human settlements for landslide and flood-prone households; and national mainstreaming of successful adaptation approaches.

The project targets at least 150,000 highly vulnerable beneficiaries — including the extreme poor, women-headed households, and marginal-land farmers — with wider benefits reaching over 380,000 people. As an LDC, Rwanda requests USD 32.8M in GCF grant finance, the only viable capital channel given the absence of private sector financing potential and the limits of domestic public funding.

Project Information	
Project Code	FP073
Classification	Adaptation
Accredited Entity	Rwanda MoE / FONERWA (National DAE)
Board Approval	B.19 (2018)
Status	Active
Beneficiaries	150,000 (direct) / 380,000+ (wider)
APR Horizon	5 years

VERA-TCV™ Evaluation Results	
VERA™	CR-Tier I (0.933)
ToC:f(x)™	TIER II (0.750)
C-FAIR™	Grade C (S2 Weak; Grant 30%+)
VORTA™	VCS F / IRRc negative
Structural Pattern	A — Design–Environment Separation
Consistency Rate	100%



FP101: Resilient Rural Belize (Be-Resilient)

REGION

Latin America and the Caribbean

COUNTRY

Belize

Financing	Public Sector	USD 20.0m
Size	Small (10m - 50m)	

GCF financing

100% disbursed

Instrument	Amount
• Grant (30.4%)	USD 6,085,200
• Loan	USD 1,914,800
TOTAL GCF FINANCING	
USD 8,000,000	

Co-financing

Co-financer	Instrument	Amount
• Co-financing	Loan	USD 7,997,905
• Co-financing	Loan	USD 822,402
• Co-financing	Loan	USD 3,182,591
TOTAL CO-FINANCING		
		USD 12,002,898

Approved

28 Feb 2019

To be completed

04 Mar 2026

Executive Summary

As a Small Island Developing State in the Caribbean hurricane belt, Belize faces intensifying climate impacts — extreme storms, drought, flooding, and unpredictable rainfall — that devastate smallholder farmers disproportionately. Limited investment capacity keeps farmers locked in recovery cycles, preventing resilience-building investments, while poor market conditions have driven domestic markets toward imports that Belizean smallholders could competitively supply.

The project introduces climate-resilient agricultural practices and improved market access across the full value chain, working initially in 23 communities across five priority Districts. It targets six vegetables, pineapple, and beekeeping products through two complementary components: Component 1 strengthens smallholder participation in climate-resilient value chains through technical support and a competitive matching grant fund; Component 2 develops climate-resilient rural infrastructure enhancing farming and enterprise opportunities. A GCF loan and grant complement an IFAD productive development loan, enabling an integrated approach that directly addresses climate threats while improving market connectivity for rural smallholders.

Project Information

Project Code	FP101
Classification	Adaptation
Accredited Entity	IFAD (UN Agency)
Board Approval	B.22 (2019)
Status	Active
Beneficiaries	—
APR Horizon	—

VERA-TCV™ Evaluation Results

VERA™	CR-Tier II (0.700)
ToC:f(x)™	TIER II (0.800)
C-FAIR™	Grade C (S1/S2; Grant 30%+)
VORTA™	VCS F
Structural Pattern	A — Design—Environment Separation
Consistency Rate	87.5% ‡‡

‡‡FP101 — Six disconfirmed (x) cells reflect a methodological boundary condition: VERA™ RA1 pre-assessed a delivery timeline subsequently modified through a formal GCF Board-approved restructuring. These cells represent a methodology limitation, not directional prediction failures. Effective rate = 26 confirmed + 2 partial (1.0 credit) = 28 / 32 = 87.5%.



FP115: Espejo de Tarapacá

FP115 ✖ GCF funding proposal cover displays FP105 – typographical error; registry code FP115 confirmed.

REGION

Latin America and the Caribbean

COUNTRY

Chile

Financing	Private Sector	USD 1.1b
Size	Large (250m +)	

GCF financing		100% disbursed
Instrument	Amount	
• Equity	USD 60,000,000	
TOTAL GCF FINANCING		USD 60,000,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Equity	USD 360,900,000
• Co-financing	Loan	USD 647,300,000
• Co-financing	Equity	USD 25,800,000
TOTAL CO-FINANCING		USD 1,034,000,000

Approved	To be completed
08 Jul 2019	28 Oct 2024



Executive Summary

The Espejo de Tarapacá Project combines Chile's natural resources with proven generation technology to deliver clean renewable energy 24/7. The project integrates two commercially linked power plants: a 300 MW pumped storage hydroelectric (PSH) facility using the Pacific Ocean as its lower reservoir and a natural concavity as its upper reservoir, and a 561 MW-AC photovoltaic solar plant with single-axis tracking.

The project provides a renewable baseload solution at a price competitive with thermal technologies – addressing a grid that is 54% thermoelectric and 30% unpredictable hydropower. It contributes directly to Chile's target of 70% non-conventional renewable capacity by 2050, with potential to decarbonize the national grid by up to 5%, establishing a replicable precedent for baseload renewable energy in markets currently dependent on fossil fuels.

Project Information

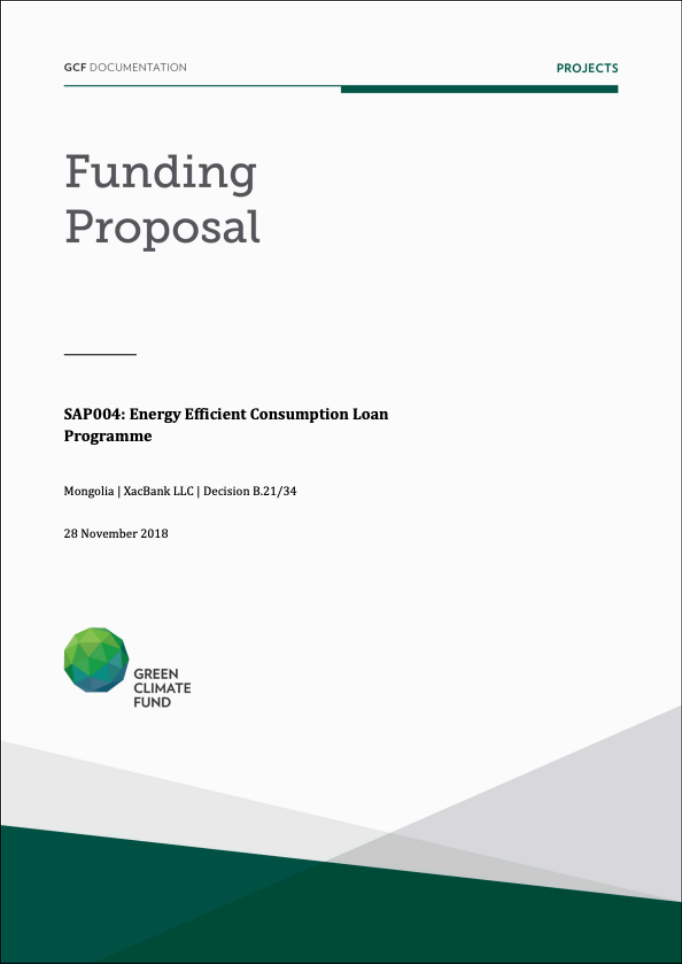
Project Code	FP115
Classification	Cross-cutting
Accredited Entity	MUFG (Commercial Bank)
Board Approval	B.23 (2019)
Status	Cancelled (B.34)
Beneficiaries	—
APR Horizon	4 years †

† Includes Project Completion Report (2024) following formal cancellation at B.34.
Disbursement at cancellation: 8.04%. Zero climate results recorded.

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III
ToC:f(x)™	TIER II (0.700)
C-FAIR™	Grade B (S2 Weak)
VORTA™	VCS F
Structural Pattern	B — Internal Mechanism Failure
Consistency Rate	100% §§

§§ IF2 N/A adjustment applied. Four signal-year cells under VERA™ IF2 classified as Not Applicable where the project's formal risk register was absent at the relevant assessment point. Zero contradictions recorded.



SAP004: Energy Efficient Consumption Loan

REGION
Asia and the Pacific
COUNTRY
Mongolia

Financing Private Sector
Size Small (10m - 50m) **USD 21.5m**

GCF financing		100% disbursed
Instrument	Amount	
• Loan (41.86%)	USD 9,000,000	
• Grant (4.65%)	USD 1,000,000	
TOTAL GCF FINANCING		USD 10,000,000

Co-financing		
Co-financer	Instrument	Amount
• Co-financing	Loan	USD 9,000,000
• Co-financing	Grant	USD 2,500,000
TOTAL CO-FINANCING		USD 11,500,000

Approved 20 Oct 2018 To be completed 28 May 2029

Executive Summary

One-half of Mongolia’s population lives in Ulaanbaatar, where more than 60% of residents live in the surrounding ger (peri-urban) areas, many of whom migrated to Ulaanbaatar in response to climate change-driven livelihood disruptions. Faced with extremely low winter temperatures and challenging economic conditions, these residents burn low-grade coal and rubbish in inefficient cook stoves for heating, with dire consequences for health and the environment. There is a need to improve access to affordable financing for energy efficient (EE) solutions that improve health and wellbeing without placing an undue financial burden on households.

XacBank proposes a US\$ 21.5 million programme to support household EE lending, comprised of a US\$ 18 million facility with concessional loans to consumers purchasing EE Heating appliances (US\$ 3 million of the facility) and EE Housing solutions (US\$ 15 million of the facility) which includes EE Housing insulation retrofits and EE Housing construction. The facility will be co-financed by XacBank and the GCF (each committing US\$ 9 million), with US\$ 1 million in grant financing from the GCF to match the US\$ 2.5 million in grant co-financing from Groupe Energies Renouvelables, Environnement et Solidarités (GERES) – a French NGO developing an EE housing project to support new EE housing and retrofitting, and a programme partner. The GCF grant financing will contribute to disposal management of old heating appliances, EE housing evaluation costs, awareness raising, and capacity building efforts. GCF loan concessionality will alleviate existing household EE financing barriers, enabling a degree of scale more aligned with the financing needs of the affected populations. The Programme will impact an estimated 15,278 direct beneficiaries and drive an estimated CO2 emissions reduction of 469,574 tCO2e over its 10-year lifetime, contributing to Mongolia’s 2030 targets of a 14% reduction in national GHG emissions and a 40% reduction in building heat loss.

Project Information

Project Code	SAP004
Classification	Adaptation
Accredited Entity	Multilateral (Small Access Programme)
Board Approval	B.18 (2018)
Status	Active
Beneficiaries	15,278 (direct)
APR Horizon	4 years

VERA-TCV™ Evaluation Results

VERA™	CR-Tier III (0.650)
ToC:f(x)™	TIER II (0.650)
C-FAIR™	Grade C (S2 Weak; Capacity-dominant)
VORTA™	VCS F / GRANT_JUSTIFIED (ΔIRR +0.74%p)
Structural Pattern	A – Design–Environment Separation
Consistency Rate	85.0%

EXECUTIVE SUMMARY

Ex-Ante Structural Diagnosis of Climate Project Performance VERA-TCV™ Framework Applied to 15 GCF Projects

The Central Claim

The structural conditions that determine climate project performance are present in funding proposal documents at the point of GCF Board approval. They are readable before implementation begins. This report provides the quantitative foundation for that claim across fifteen GCF-funded projects spanning four continents, five thematic areas, and an average of 5.7 years of Annual Performance Report records.

The Framework

VERA-TCV™ applies four computationally independent models to funding proposal documents alone — before any implementation data is reviewed. VERA™ examines design authenticity. ToC:f(x)™ examines causal logic integrity. C-FAIR™ examines delivery environment capacity. VORTA™ examines social value realization. Each model examines a structurally distinct layer of the project's Theory of Change. Their outputs are mapped against independent APR evidence, year by year, across the full verification horizon.

The Evidence

Across 313 assessable signal-year cells, the portfolio-weighted directional consistency rate is approximately 92%. No unresolved material pathway contradiction was recorded across the evaluated portfolio. Thirteen independent evaluation documents — produced by evaluation teams with no knowledge of VERA-TCV™ pre-assessment — converged directionally with framework signals in every case where comparison was possible. Twelve of fifteen projects carry independent evaluation evidence, comprising thirteen independent evaluation documents in total. In two cases — FP026 Madagascar and FP069 Bangladesh — independent randomised evaluation studies, conducted without knowledge of VERA-TCV™ pre-assessment, identified the same structural gaps that VERA-TCV™ had diagnosed from the funding proposal alone. The methodological instruments differ. The structural findings converge.

A note on computation and scope. The 92% directional consistency rate is computed across assessable signal-year cells only; cells classified as Not Applicable (structural determination) or No Data Available (access limitation) are excluded from both numerator and denominator and are never used to improve the reported figure. Partial signal confirmations receive 0.5 credit under a uniform portfolio-wide convention. Full cell-level documentation is in Appendix H. Project selection criteria and the complete universe of fifteen projects are in Appendix E. This study does not compare VERA-TCV™ to GCF's approval process — they answer structurally different questions at different points in time. GCF approval asks whether a project is fundable. VERA-TCV™ asks whether the funded project's causal chain holds. These are not competing functions.

The Portfolio Findings

Three findings emerge that no individual case can produce alone.

Twelve of fifteen projects record C-FAIR™ Grade C. This concentration is not a failure signal. It is the structural justification for concessional climate finance — confirming that GCF is deploying capital precisely where market mechanisms cannot operate independently.

Twelve of fifteen projects record VORTA™ VCS Grade F. This single classification conceals six structurally distinct causes — from measurement architecture absence to attribution impossibility to minimum viable grant justification. Reading twelve Grade F scores as a single signal produces systematically incorrect financing recommendations.

Not a single project achieved ToC:f(x)™ TIER I. This is not a portfolio design failure. It is the structural consequence of operating in multi-actor climate programmes where complete causal exclusivity is unavailable by definition. TIER I defines the theoretical upper bound of the causal architecture space. The operational verification standard is directional contribution — whether the project's causal logic is structurally capable of producing its claimed outcomes in the direction and through the mechanism specified. That is what this report measures.

The Structural Patterns

Four structural patterns account for all fifteen cases. Pattern A — Design–Environment Separation — accounts for eight of fifteen: sound design operating in environments that constrain delivery. Pattern B — Internal Mechanism Failure — accounts for two: a specified causal mechanism never activates. Pattern C — Measurement Architecture Gap — accounts for three: value is generated but architecturally unmeasurable. Pattern D — ToC-Level Constraint — accounts for two: the causal logic itself is attribution-constrained or solution-misaligned at design stage.

The individual signals constituting each pattern — VERA™ scores, ToC:f(x)™ tier determinations, C-FAIR™ grade assignments, VORTA™ valuations — are generated as ex-ante outputs from the funding proposal before any APR data is reviewed. The pattern labels are an analytical summary of where within the ToC causal chain those pre-assessed signals are located. VERA-TCV™'s predictive claim is not directional in the aggregate sense. It identifies which structural layer fails first, and through which mechanism — before implementation begins.

The Requirements

Three structural adjustments to GCF's approval and monitoring architecture follow directly from the evidence. First, monetized annual benefit estimates must be a condition of approval — not a post-approval implementation expectation. Second, chronic structural signals and acute event signals must be distinguished in APR monitoring architecture. Third, ToC Q5 determination must be a mandatory pre-approval gate for adaptation projects in Hazard-dominant environments.

The Implication

Pre-approval structural diagnosis does not eliminate uncertainty. It separates uncertainty that is genuinely unpredictable from structural risk that was readable in the design document from the first day — and that, with the right analytical framework, could have been addressed before approval.

USD 33.0 million was committed to FP069. USD 38,571 in total annual WLG income was generated at project close. The gap between those two numbers is the quantitative expression of what pre-approval structural diagnosis is designed to prevent.

Structure before execution. Diagnosis before capital.

How to Read This Report

Climate programmes pursue two structural objectives. Mitigation addresses the accumulation problem: reducing, avoiding, or removing greenhouse gas emissions to limit the long-term severity of climate risk. Adaptation addresses the exposure problem: equipping communities and systems to survive the climate disruptions already in motion. Both are necessary. Together, they define the condition for human survival at planetary scale — reducing future risk while maintaining present viability.

Public climate finance institutions exist to accelerate both mitigation and adaptation. This includes multilateral climate funds, development banks, bilateral development agencies, national climate funds, and blended-finance platforms that deploy concessional capital where private markets cannot operate independently. The Green Climate Fund is the empirical focus of this report, but the structural problem examined here is broader than GCF.

GCF is used as the empirical basis for this study not because it is an outlier, but because it is one of the most institutionally developed and publicly documented climate finance mechanisms in operation. Its proposal, approval, implementation, Annual Performance Report, and evaluation architecture provides a level of documentary specificity rarely available across the broader climate finance landscape. The use of GCF projects in this report should therefore be read as a function of analytical traceability, not institutional targeting. Precisely because GCF's documentation is comparatively concrete and operationally mature, it enables the kind of ex-ante versus ex-post structural consistency analysis that this report conducts.

Through concessional climate finance — including grants, concessional loans, equity, and risk-sharing instruments — these institutions deploy capital into environments with elevated hazard exposure, constrained institutional capacity, and limited macrofinancial conditions. The theory is correct. Public concessional capital, properly structured, can de-risk environments sufficiently to mobilize private capital at multiples of the original public investment, extending the reach of climate action beyond what grant budgets alone can achieve.

The constraint is not the theory. It is the scale. The climate system does not operate at the scale of GCF's approved portfolio. Planetary-scale mitigation and adaptation require planetary-scale capital mobilization — and the public concessional resources available to GCF and its peer institutions are structurally insufficient to meet that need at the required scale. This insufficiency is not only a matter of political will. It is a mathematical reality: the financing gap for climate action is measured in trillions; the resources available through public concessional channels are measured in billions.

The implication is precise. If concessional capital is scarce relative to the scale of the problem, then the selection and design of individual projects becomes a first-order strategic question — not an administrative one. GCF must be able to identify, before capital is committed, which projects carry the highest probability of delivering durable climate outcomes: projects with sound causal logic, environments whose constraints have been correctly read at design stage, financing structures calibrated to the delivery conditions they will actually encounter, and measurement architectures capable of verifying whether value was generated. Without this capacity, every approval decision is made at least partially blind — and the compounded cost of structural failure across a portfolio of hundreds of projects is not recoverable.

GCF's publicly verifiable evaluation architecture remains weighted toward ex-post assessment. Projects are evaluated against results after implementation — when the capital has already been committed, the design has been locked, and the structural conditions that govern performance have

been operating for years. Independent evaluations confirm what happened. They cannot change what was approved.

The VERA-TCV™ framework does not claim to predict which projects will succeed. Complex climate programmes operate in environments where exogenous variables — the severity and timing of climate events, political transitions, commodity price movements, and co-financer behavior — lie beyond any framework's forecasting capacity. What the framework establishes is different, and more actionable: by integrating structural, causal, environmental, and financial integrity analysis from the funding proposal document alone, it identifies predictable failure pathways present at the design stage — before implementation begins, before capital is committed, and while structural correction is still possible at design cost rather than implementation cost.

This is the function of ex-ante structural diagnosis. It does not eliminate uncertainty. It separates uncertainty that is genuinely unpredictable from structural risk that was readable in the design document from the first day — and that, with the right analytical framework, could have been addressed before approval.

The fifteen projects in this report span four continents, five thematic areas, eight implementing entity types, and an average of 5.7 years of implementation data. They were selected for documentation availability, outcome-pathway verifiability, and structural heterogeneity; evaluated against funding proposal documents alone; and then mapped against independent APR and evaluation evidence across verification horizons long enough to test multi-year structural dynamics.

The result — approximately 92% portfolio-weighted directional consistency, no unresolved material pathway contradictions, and convergent external verification from thirteen independent evaluation documents — is not a claim that the framework predicts outcomes. It is evidence that structural conditions governing climate project performance are present in funding proposals at the point of Board approval. They are readable before implementation begins.

This report is designed for selective reading. No reader needs to read all sections in sequence. The table below maps each section to the institutional role for which it is most directly relevant.

Institution	Relevant Sections	Pages	Content
GCF Secretariat	Section 6 (all), Section 9, Section 12, Appendix E	17-21, 38-40, 45-46, 50-51	Four-model portfolio distribution / three portfolio-level findings / pre-approval structural diagnosis requirements / project universe
GCF Board Members	Executive Summary, Section 9, Conclusion	1-2, 38-40, 47	Three portfolio-level findings / policy implications / core messages
GCF Independent Evaluation Unit (IEU)	Section 4.3, Section 7.4, Section 10, Appendix F, Appendix H.4	14, 23-24, 40-43, 52-53, 62	Relationship to SSRN paper / external verification architecture / predictive validity assessment / APR extraction protocol / independent evaluation registry
National Focal Points & NDAs	Section 6.3, Section 8, Section 9 Finding 1, Section 11, Appendix E	19, 34-38, 38, 43-44, 50-51	Country-level delivery environment diagnosis / structural failure pattern by project context / concessional finance justification / pre-submission screening framework

Institution	Relevant Sections	Pages	Content
AE Project Design Teams	Section 6.1, Section 6.2, Section 8, Section 11, Appendix A-B	17-18, 18-19, 34-38, 43-46, 48	VERA design authenticity distribution / ToC causal structure / four structural patterns / what ex-ante diagnosis changes / VERA and ToC model overviews
AE Finance and Structuring Teams	Section 6.3, Section 6.4, Section 9 Findings 1-2, Appendix D	19, 19, 38, 49	C-FAIR delivery environment grading / VORTA value realization / Banker's Paradox / VCS F disaggregation / VORTA model overview
AE M&E and Results Management Teams	Section 7, Section 12 Req. 1, Appendix F, Appendix H.3	22-24, 45, 52-53, 61-62	Three-document architecture / measurement design as approval condition / APR extraction protocol / directional consistency rate matrix
NDA / National Focal Point	Executive Summary, Section 5, Section 8 Pattern A, Part III (six case analyses)	1-2, 15-16, 34-35, 25-33	Report core summary / portfolio composition / Design-Environment Separation / case analyses relevant to comparable national contexts
Recipient Country Government – Technical Staff	Section 8 Patterns A-D, Section 11, Section 12 Req. 2-3	34-38, 36, 43-44, 45-46	Implementation constraint patterns / what pre-approval diagnosis changes / acute vs chronic signal distinction / Q5 mandatory gate requirement
Private Climate Finance Institutions / DFI Private Windows	Section 6.3-6.4, Section 9 Findings 1-2, Case 3 (FP009), Case 4 (FP039), Appendix C-D	19, 38, 27-28, 29-30, 49	Grade B environment structure / GRANT_JUSTIFIED vs REDESIGN_REQUIRED / ESI mechanism analysis / Private-Public Bifurcation / C-FAIR and VORTA model overviews
Carbon Market / Climate Bond Institutions	Section 8 Pattern D, Section 9 Finding 2, Case 4 (FP039)	37, 38, 29	REDD+ attribution impossibility structure / VCS F disaggregation / private climate bond performance (FP039)
Climate Finance Researchers	Section 3-4, Section 10, Appendix A-F, Appendix H (all)	11-14, 40-42, 48-53, 59-62	Full methodology / relationship to SSRN paper and model upgrade history / predictive validity / model overviews / consistency rate matrices
International Development Policy Bodies (OECD DAC, UN agencies, etc.)	Section 9 Finding 3, Section 12, Conclusion	39, 45-46, 47	Structural meaning of ToC TIER I absence / GCF approval and monitoring architecture reform requirements / policy implications

OPENING

Structure before execution. Diagnosis before capital.

If a design document contains the signals of its own failure — and if those signals are readable before a single dollar is committed — then pre-approval structural diagnosis is not an option. It is an obligation.

Consider what that obligation looks like when it is not met. FP069 committed USD 33 million to coastal adaptation in Bangladesh. The structural condition that determined its outcome — a Problem-Solution Misalignment at the Theory of Change level, readable in the design document before Board approval — was never identified before capital was committed. At project close, total annual income generated for coastal households was USD 38,571. That is not a project management failure. That is a structural diagnosis failure. And it was preventable.

Across fifteen GCF-funded projects spanning four continents, five thematic areas, and an average of 5.7 years of Annual Performance Report records, VERA-TCV™ generated ex-ante structural judgments from funding proposal documents alone. Those judgments were then mapped against actual implementation data, year by year, for a verification horizon of up to eight years.

The result: directional consistency rates ranging from 76.9 to 100 percent across all evaluated cases. All material implementation failure pathways corresponded either to pre-assessed structural signals or to explicitly disclosed methodological boundary cases. In two cases, independent randomised evaluation studies — conducted without knowledge of VERA-TCV™ pre-assessment — identified the same structural gaps that VERA-TCV™ had diagnosed from the funding proposal alone. The instruments differ. The structural findings converge.

Indicator-level exceptions, where present, are treated not as concealed errors but as diagnostic evidence for future model refinement.

The pattern of failure in climate finance is not random. It is structural. And it is readable — from the design document, before a dollar is committed.

PART I – THE DIAGNOSTIC FRAMEWORK

Most climate project evaluations begin after implementation ends. They measure what happened. VERA-TCV™ begins before implementation starts. It measures what was structurally inevitable. Part I establishes the four diagnostic instruments, the independence principle that governs their interaction, and the epistemological boundary that defines what this framework claims — and what it does not.

PART I — THE DIAGNOSTIC FRAMEWORK

1. Why Structure Determines Performance

1.1 The Problem Is Not Intent

The dominant narrative in climate finance evaluation treats underperformance as a function of implementation quality: projects fail because implementing entities lack capacity, because external shocks disrupt timelines, because political environments shift, or because co-financiers withdraw. Each of these explanations is sometimes accurate. None of them is sufficient.

Consider the distribution of outcomes across the fifteen projects in this study. Every project in this portfolio was approved by GCF's Board following a rigorous review process. Every project was submitted by accredited entities with documented track records. Every project carried a Theory of Change, a risk register, a financial model, and a logframe. The intentions behind each were genuine. The paperwork was complete.

And yet: twelve of fifteen projects recorded VORTA™ VCS Grade F — insufficient value capture. Twelve recorded C-FAIR™ Grade C — structurally constrained delivery environments. Not a single project in the evaluated portfolio achieved ToC:f(x)™ TIER I — complete causal exclusivity between project outputs and stated outcomes. The pattern is too consistent to be explained by individual execution failures.

The pattern is structural.

1.2 The Behavioral Premise

The analytical foundation of VERA-TCV™ rests on a premise established in prior work on behavior-aligned system design (Hur, 2025): that human motivation, institutional behavior, and environmental response operate according to structural logic independent of what project documents claim.

This premise has a precise implication for evaluation methodology. A framework that accepts stated intent as sufficient evidence of structural soundness will systematically fail to detect the gap between what is claimed and what the architecture can deliver. Intentions are inputs. Structures are mechanisms. Outcomes are outputs of mechanisms, not of intentions.

When a Theory of Change assumes that implementing agencies will act in full alignment with stated objectives, that co-financiers will fulfill commitments because they are contractually documented, and that beneficiaries will respond to interventions as designed — the evaluation framework that accepts these assumptions at face value has already failed before implementation begins.

VERA-TCV™ does not assume that stated intent is false. It assumes only that stated intent is insufficient as evidence of structural capability. The framework is built on the inverse premise: that every structural layer of a project design is either capable or incapable of producing the claimed result, and that this capability is structurally assessable from the design document before the project begins.

1.3 Structure as the Unit of Analysis

The formalization is straightforward. A Theory of Change makes two claims:

$$f_1(\text{Input, Activity}) = \text{Output}$$

$$f_2(\text{Output}) = \text{Outcome}$$

The first function asks whether the activities as designed are causally sufficient to produce the claimed outputs. The second asks whether the outputs, if produced, will generate the stated outcomes. Each function has structural conditions for validity — and those conditions are either present or absent in the design document.

VERA-TCV™ does not evaluate whether f_1 and f_2 will hold during implementation. It evaluates whether the structural conditions for their validity are present at the design stage. This is the distinction between outcome prediction and structural risk identification — and it is the distinction on which the entire framework rests.

An engineer conducting Failure Mode and Effects Analysis before a launch does not predict whether the mission will succeed. The engineer identifies which components are structurally vulnerable to failure under defined conditions.

VERA-TCV™ applies this logic to climate project design: not a prediction of outcome, but a map of structural vulnerability — produced before a single dollar is committed, using only the information the project itself has provided.

The pattern of failure is not random. It is structural. And that means it is readable.

2. VERA-TCV™ Architecture

2.1 Four Independent Models

VERA-TCV™ consists of four independent deterministic models, each examining a distinct structural layer of the Theory of Change causal architecture.

Model Layer Structural Question

VERA™	Input	<i>Is intent authentic and grounded in reality?</i>
ToC:f(x)™	Activity	<i>Output Is the causal logic sufficient and exclusive?</i>
C-FAIR™	Output	<i>Outcome Can the environment convert outputs into outcomes?</i>
VORTA™	Outcome	<i>Impact Does the intervention generate sufficient social value?</i>

The isomorphism between these four layers and the four causal boundaries of a Theory of Change is not incidental. It is the central architectural principle of the framework. Every transition in the *Theory of Change* (ToC) causal chain corresponds to a distinct structural question, and each question requires a different kind of evidence and a different analytical method to answer.

No single model can answer all four questions without sacrificing the specificity required at each layer.

The four models are computationally independent: no model's scoring algorithm references another model's output. They draw evidence from the same funding proposal, but each transforms that evidence through a distinct indicator structure and evaluates a different causal boundary.

A project can exhibit strong intent fidelity at the input layer while carrying a causally deficient activity design. It can demonstrate sound causal logic while operating in an environment structurally incapable of converting outputs into outcomes.

Each failure is structurally separable from the others — and each is invisible to a framework that does not examine its specific layer.

2.2 VERA™ — Intent Fidelity at the Input Layer

VERA™ examines whether the project's stated objectives, timelines, and delivery claims are genuinely grounded in physical, institutional, and technical reality. The model operates across three axes.

The **Reality Anchor axis (RA)** assesses temporal plausibility (RA1), system compatibility across policy, infrastructure, and technology dimensions (RA2), and transformation traceability through measurable indicators (RA3). The single most consequential indicator across all fifteen evaluated projects is RA2 — the multiplicative compound of policy, infrastructure, and technology compatibility — which functions as the primary delivery capacity constraint in every case where it registers below threshold.

The **Causal Integrity axis (CI)** examines whether logical connections between design elements are internally consistent and structurally specified. The Logic Drift indicator (CI3) measures the expected rate of causal chain degradation over the project's implementation horizon — the recognition that structural properties are not static.

The **Intent Fidelity axis (IF)** examines whether the motivational structure of the project is genuinely oriented toward the stated climate objective. It detects configurations in which stated intent and structural incentive diverge.

VERA™ produces a composite authenticity score and a three- tier classification (CR-Tier I through CR-Tier III). Three projects achieved CR-Tier I, five CR-Tier II, and seven CR-Tier III.

Score range: 0.5069 to 0.9325.

2.3 ToC:f(x)™ — Causal Structure at the Activity-Output Boundary

ToC:f(x)™ examines whether the activities as designed are causally sufficient to produce the claimed outputs, and whether the logical chain from those outputs to stated outcomes is structurally coherent rather than narratively asserted.

The model applies twenty structural questions organized into three groups.

Group A questions address explicit presence — whether necessary resources, activities, and causal mechanisms are named with sufficient specificity to constitute structural evidence.

Group B questions address structural demonstration — requiring a minimum of two independent structural indicators before a positive determination is made.

Group C questions address causal exclusivity — whether the document establishes that the claimed outcome depends on the project's specific outputs rather than being achievable through parallel pathways.

Group C operates under an Unknown default. A positive determination requires explicit exclusivity language in the document, not inference from narrative plausibility. This is an accurate structural observation: *in complex development environments, exclusive causal attribution* is rarely achievable from design documents alone.

The finding that no project in this portfolio achieved ToC:f(x)™ TIER I is the most consequential portfolio- level finding in this report. It is addressed in Part IV.

Score range: 0.600 to 0.800.

2.4 C-FAIR™ — Delivery Environment at the Output-Outcome Boundary

C-FAIR™ examines whether the environment into which project outputs will be delivered is capable of converting those outputs into outcomes. Three dimensions are assessed.

Hazard Exposure (S1) quantifies the physical climate risk profile — flood, drought, cyclone, heat wave, and sea level rise risk. High-hazard environments face structurally elevated delivery risk regardless of design quality, because the environment introduces exogenous shocks the design cannot fully absorb.

Adaptive Capacity (S2) assesses institutional and infrastructural conditions — infrastructure resilience, disaster response capacity, and institutional capacity. S2 weakness is the most consistent structural signal in this portfolio, appearing in every case regardless of sector, region, or implementing entity.

Macro-Financial Fragility (S3) assesses sovereign debt stress, foreign exchange volatility, political stability, and financial system maturity.

Twelve of fifteen evaluated projects recorded C-FAIR™ Grade C. This concentration is *not a failure signal*. It is *the structural justification for concessional climate finance* — the principle embedded in the model as the Banker's Paradox: *high environmental fragility* is the condition that generates both the greatest need for climate finance and the greatest structural justification for grant-heavy financing structures.

2.5 VORTA™ — Value Realization at the Outcome-Impact Boundary

VORTA™ examines whether the outcomes a project claims to produce, if achieved, generate social value sufficient to justify the cost of the intervention.

The model quantifies social value through a *Value-Based Impact (VBI)* framework across three layers: the **Output layer** captures value generated directly by project deliverables; the **Outcome layer** captures value from the behavioral and economic changes that outputs induce; the **Impact layer** captures value at the level of social systems.

The model disaggregates the internal rate of return into *IRR_m* (market return from market-priced cash flows), *IRR_v* (value- inclusive return incorporating quantified social value), and *IRR_c* (composite return, weighting by the project's mitigation-adaptation balance). The spread ΔiRR — between *IRR_m* and *IRR_v* — is the quantitative expression of the public finance justification.

Twelve of fifteen projects recorded VCS Grade F. The primary cause is systematic: *the absence of monetized social benefit quantification* from GCF design documents. In every case, Grade F was predictable from the design document alone — because the data required to populate the VBI framework was absent from the document at the point of approval.

VCS Grade F does not mean these projects generated no value. It means the value they generated was architecturally unmeasurable before they began.

2.6 The Independence Principle

When a project produces a strong ToC:f(x)™ score alongside a C-FAIR™ Tier V classification, the framework does not average these results. It reports them as structurally distinct findings: sound design logic coexisting with an environment incapable of supporting delivery. This combination — which appears in eight of fifteen evaluated projects — is not a moderate outcome. It is the specific structural diagnosis that this report terms Design- Environment Separation.

The computational independence of the four models ensures that this diagnosis is not a methodological artifact. It is a structural property of the portfolio, identifiable because each model examines a layer that the others cannot reach.

3. Ex-Ante Methodology

3.1 The Diagnosis Precedes the Evidence

Every judgment recorded in this report was generated from the funding proposal (FP) document — the design instrument available at the point of GCF Board approval — without reference to any subsequent implementation data.

The sentence that appears at the opening of each case analysis in Part III is not a formality:

“The following judgment was generated from the [year] design document. No Annual Performance Report data was available at the time of diagnosis.”

Its repetition across the six anchor cases is the cumulative demonstration of the framework’s predictive independence. Each iteration adds one more year, one more project, one more verification point to the same claim: that the structural conditions governing implementation were readable in the design document before implementation began.

3.2 Annual Performance Report (APR) as Comparator, Not Input

Annual Performance Reports (APR) serve a single function in this study: they provide *independent ex-post data* against which pre-assessment judgments are mapped. They are *comparators*, not inputs.

This distinction has structural implications for interpreting the directional match rates reported in Part IV. A *confirmed signal* (✓) establishes that the structural condition was real and consequential. A *partial confirmation* (△) indicates that the predicted condition materialized in diminished form — the direction was correct, the intensity was attenuated.

The critical distinction is between directional consistency and predictive completeness: VERA-TCV™ claims the former, not the latter. No confirmed signal in this portfolio was contradicted in direction.

3.3 Multi-Agent Extraction Protocol

The four models are deterministic: *given identical inputs, they produce identical outputs*.

The methodological challenge shifts from the models to the extraction of inputs from project documents — a process that is inherently interpretive and therefore vulnerable to single-analyst bias.

VERA-TCV™ addresses this through a multi-agent parallel extraction protocol. Identical field sets are extracted from the same source document by *three independent AI-assisted extraction systems* — each built on a distinct large language model architecture and independently optimized for structured document extraction — operating under *standardized protocols with no cross-reference permitted prior to convergence*.

Field-level SCI (Structured Convergence Integration) rules resolve divergence. Where divergence persists, Divergence Resolution Protocol (DRP) procedures are applied, with resolution rationale documented in the audit record preserved for each evaluation.

This architecture removes single-analyst interpretation bias as the primary source of non-replicability. The convergence audit record — documenting extraction agreement rates, divergence flags, and operator-resolved ambiguities — is available for independent review.

3.4 Epistemological Boundary

VERA-TCV™ operates within a defined system boundary: the structural properties encoded in the funding proposal, the country-level conditions measurable at the design stage, and the documented stakeholder incentive architecture.

Forces outside this boundary — *pandemic disruptions, extreme climate events of unprecedented severity, geopolitical shocks* — are not modeled as internal variables. They are external conditions. *This is not a limitation* to be apologized for. It is the condition of analytical precision: a model that attempts to incorporate all possible external conditions produces no useful diagnoses at all.

When VERA-TCV™ identifies a C-FAIR™ S2 *institutional capacity weakness*, it is not predicting that the project will fail. It is identifying that one of the structural conditions necessary for outcome delivery is not confirmed in the document. Whether the project ultimately succeeds depends on factors beyond the framework's scope. What the framework establishes is *which failures were structurally predictable* — and which were genuinely unforeseeable.

In this portfolio, every material implementation failure examined in the APR mapping corresponded to *a pre-assessed structural signal* or to *a disclosed methodological boundary case*. No major failure pathway emerged wholly outside the framework's diagnostic field.

3.5 Indicator Hierarchy and Exception Disclosure

VERA-TCV™ does not treat every indicator as carrying equal diagnostic weight in every project type. Within each model, indicators operate at different levels of the diagnostic hierarchy: some function as *primary structural signals* that determine tier classification and financing recommendations; others function as *boundary indicators* that reveal limitations in formula design, data availability, or *attribution logic* specific to a given project context.

Where indicator-level exceptions appear — including cases where a signal fires in a direction that does not correspond to observed *Annual Performance Report (APR)* data in a given year — they are disclosed and analyzed rather than removed from the record. The *directional match rates* reported in this study are computed across all assessable signal-year cells, including those in which *partial or boundary-level discrepancies* are present. No cell is excluded to improve reported accuracy.

The validation claim of this report therefore rests not on perfect indicator-level agreement across every signal in every year, but on whether *material implementation constraints and failure pathways* were structurally identifiable before implementation began. Where indicator-level exceptions persist after methodological review, they are treated as diagnostic evidence for future model refinement — evidence that the framework is capable of identifying its own boundaries, not concealing them.

4. Scope and Limitations

4.1 Selection Criteria

Fifteen projects were selected from GCF's approved portfolio using *three criteria*.

Minimum APR traceability: each project required at least four complete Annual Performance Reports in the public domain, enabling multi-year verification of pre-assessment signals across multiple implementation years.

Outcome-pathway verifiability: each project required a logframe with trackable indicators connecting outputs to outcomes — enabling assessment of whether the causal chain held, not merely whether activities were completed.

Structural heterogeneity: the portfolio was deliberately constructed to maximize diversity across sector, region, income group, implementing entity type, and financing structure. Recurring structural signals across maximally heterogeneous cases constitute stronger evidence of portfolio-level pattern than similar signals across comparable cases.

The fifteen selected projects span **four continents, five thematic areas, eight implementing entity types**, and GCF Board approvals from B.11 (2015) through B.23 (2019). APR verification horizons range from four to eight years, with an average of **5.7 years** per project.

These selection criteria reflect analytical purpose, not analytical scope. In April 2026, ENHANCE Institute published a full-population structural evaluation of 101 GGGI climate project architectures (*STRUCTURAL INTEGRITY INDEX for Climate Project Architectures*, 2026) — the complete set for which minimum documentation was publicly available. The remaining projects in GGGI's portfolio were excluded not by analytical choice but by documentation constraint: insufficient public information precluded structured evaluation.

This report serves a different function. Where the GGGI study demonstrated the framework's capacity for full-population application, this report demonstrates how VERA-TCV™ diagnoses structural risk across continental, thematic, and financing architecture categories — with sufficient depth to make the diagnostic logic legible. Fifteen projects selected for maximum structural heterogeneity are sufficient to represent the classification space of the broader GCF portfolio: a framework that produces consistent directional findings across maximally dissimilar cases provides stronger evidence of structural validity than one tested against a larger homogeneous sample. The project-level structural screening tool built on this framework — PathFinder-GOV™ — was calibrated against the complete GCF approved portfolio of 303 projects, spanning all regions, themes, financing types, and approval cohorts from B.01 (2011) through B.38 (2025). That calibration baseline provides the empirical grounding for the framework's population-level validity — the fifteen cases evaluated here represent the analytical depth layer; the 303-project baseline represents the empirical breadth layer. A note on portfolio counts: the 303-project figure refers to the GCF approved portfolio as constituted at the time of PathFinder-GOV™ calibration (through B.38, 2025), representing the complete set of projects for which minimum structural data was available for calibration purposes. The 336-project figure cited in Section 4.2 reflects the total GCF approved portfolio at the time of this report's publication (May 2026), including projects approved at subsequent Board meetings. The two figures refer to different points in time and serve different analytical functions: 303 is the calibration baseline; 336 is the current portfolio scope against which documentation availability is assessed.

4.2 What This Study Is Not

This study is not a comprehensive evaluation of GCF's portfolio. Among GCF's 336 approved projects, only a limited subset currently satisfies the combined requirements of APR traceability depth, outcome-pathway verifiability, and structural heterogeneity required for full VERA-TCV™×APR consistency analysis. The fifteen projects evaluated in this report constitute that selected analytical subset. This limitation itself constitutes a finding: the structural gap in GCF's results management architecture is not merely a gap in what projects achieve — it is a gap in what is measurable after they end.

This study is *not a performance audit* of any implementing entity. The structural patterns documented here are features of the deployment context — the environments into which architecturally sound designs are sent — not of implementing entity competence.

This study is *not a prediction of future project performance*. It is a validation of the claim that structural conditions present at the design stage are consequential for outcomes. That claim does not depend on future prediction.

4.3 On the Relationship to the SSRN Paper

The *VERA-TCV™ framework* was empirically validated in an earlier publication (Hur, 2026), which examined five of the fifteen projects in this study — FP004, FP017, FP026, FP069, and FP101 — using the same *pre-assessment versus APR consistency methodology*. That paper established directional consistency rates of 86 to 100 percent across the five cases, and conducted cross-validation against *IEU LORTA* randomized control trial findings for FP026 and FP069.

Five of the fifteen projects examined in this report — FP004, FP017, FP026, FP069, and FP101 — were also examined in the SSRN paper. Their pre-assessment judgments were generated before APR data was reviewed in either study. The ten additional projects introduced in this report were not available to the SSRN paper at the time of its preparation.

This report extends the validation to fifteen projects, adds ten cases not covered in the earlier paper, and introduces the six-case anchor analysis structure of Part III. The two documents are designed to be read together: the SSRN paper establishes the methodological and epistemological foundations; this report demonstrates those foundations hold at portfolio scale. Where findings from the five overlapping cases appear in both documents, directional structural conclusions are consistent.

Numerical divergences between the two documents have two distinct sources. The first is model development: *VERA™* IF3 indicator recalibration and CI3 unit standardisation from monthly to annual horizons produced updated CR-Tier classifications for FP004, FP017, and FP069; *ToC:f(x)™* SIM-v4.1 tier boundary recalibration produced an updated tier classification for FP069; updated *VBI-A1* infrastructure protection value calculations produced a revised *IRRc* for FP004; and the introduction of the *VORTA™* v2.5 D0 Structural Vulnerability Amplification gate added a structural signal for FP069. The second is extraction protocol correction: the *C-FAIR™* Grade C / Tier V assessment for FP004 in this report supersedes the Tier II recorded in the SSRN paper. The earlier value reflected a single-session extraction that overestimated disaster response capacity relative to the primary document evidence. Re-evaluation under the three-agent parallel extraction protocol, confirmed by independent kernel execution, establishes Grade C / Tier V as the verified assessment. This correction is itself a demonstration of the extraction protocol's design value: structured multi-agent convergence identifies single-agent interpretation bias that aggregate directional statistics do not surface. Where numerical divergence exists between the two documents, the values in this report represent the current confirmed assessment.

The relationship is not redundancy. It is progressive validation. The SSRN paper establishes the foundation; this report refines it. The structural conclusions do not change. The precision improves.

PART II — PORTFOLIO COVERAGE

Fifteen projects. Four continents. Five thematic areas. An average of 5.7 years of Annual Performance Report records. Before findings can be trusted, the evidence base must be transparent. Part II documents what was evaluated, how it was selected, and what the portfolio looks like before any diagnostic judgment is applied. The composition is not incidental. Structural heterogeneity across sectors, regions, income groups, financing structures, and implementing entities is the condition under which portfolio-level findings become generalisable. This section establishes that condition.

PART II — PORTFOLIO COVERAGE

5. Fifteen Projects: Composition and Coverage

5.1 Portfolio Architecture

The fifteen projects evaluated in this study were not selected to represent GCF's approved portfolio in the statistical sense. They were selected to maximize the structural and contextual heterogeneity across which VERA-TCV™ pre-assessment signals could be tested. The logic is deliberate: a framework that produces consistent directional findings across maximally dissimilar cases provides stronger evidence of underlying structural validity than one tested against a homogeneous sample.

The portfolio spans *four continental regions, five thematic areas, and eight implementing entity types*. GCF Board approvals range from B.11 (2015) through B.23 (2019), covering a four-year window of the Fund's early operational period. APR verification horizons range from four to eight years per project, with an average of 5.7 years — sufficient to observe multi-year structural dynamics rather than first-year initialization patterns.

Nine projects remain active at the time of this report. Six have reached completion — including one formal cancellation (FP115), which constitutes the most direct ex-ante validation case in the portfolio. The inclusion of completed projects alongside active ones enables comparison between full-lifecycle and mid-lifecycle structural signals, strengthening the temporal validity of the findings.

5.2 Continental and Thematic Distribution

Africa carries the largest sub-portfolio at six projects, reflecting both the concentration of the Least Developed Countries (LDCs) and climate-vulnerable contexts in GCF's approved pipeline and the deliberate inclusion of the African LDC environment as the primary test case for *Design-Environment Separation* findings. The six African cases span six countries — Madagascar, Uganda, Egypt, Morocco, Zambia, and Rwanda — across ecosystem, NbS, energy, and food and water security thematic areas, and include both the weakest design authenticity score among African cases and the strongest in the full portfolio (FP039 / FP073).

Latin America and the Caribbean accounts for six projects across Peru, El Salvador, Ecuador, Chile (two projects), and Belize. This sub-portfolio carries the highest internal diversity of financing structures: pure grant (FP001), grant (FP019), private sector EE credit line (FP009), solar energy infrastructure finance (FP017), commercial bank infrastructure finance (FP115), and multilateral rural development finance (FP101). This diversity enables structural comparison across financing mechanism types — not only across geographies.

Asia-Pacific contributes three projects: two in Bangladesh (FP004 and FP069) and one in Mongolia (SAP004). The two Bangladesh cases — one infrastructure-focused, one livelihoods-focused — are structurally distinct despite sharing country and implementing entity, providing a controlled contrast between problem-solution alignment and misalignment within the same national context. Mongolia's SAP004, the sole small-access project in the portfolio, tests whether VERA-TCV™ signals are robust across project scale.

Belize (FP101) is classified here within Latin America and the Caribbean but carries dual SIDS status — a small island developing state operating under quadruple climate hazard exposure. Its inclusion ensures that the portfolio captures institutional delivery dynamics specific to SIDS contexts, which differ structurally from continental LDC environments despite sharing C-FAIR Grade C profiles.

The five thematic areas covered are: Ecosystems and NbS (FP001, FP019, FP026, FP034, FP073), Food and Water Security (FP042, FP069, FP072, SAP004), Energy and Decarbonization (FP017, FP039, FP115), Rural Infrastructure and Resilience (FP004, FP101), and Financial Mechanism Design (FP009). The thematic distribution reflects GCF's actual portfolio composition across its Result Areas, with deliberate inclusion of both the grant-dominant public sector cases that dominate GCF's pipeline and the private sector financing mechanism cases that represent its market mobilization ambitions.

5.3 Implementing Entity Diversity

Eight implementing entity types are represented across the fifteen projects.

- ✓ United Nations specialized agencies (UNDP, IFAD) account for the largest single group, implementing five projects across Ecuador, Uganda, Bangladesh, Zambia, and Belize.
- ✓ Multilateral development banks operating through their private sector windows (EBRD, IDB Invest) implement two projects — Egypt and El Salvador — both of which involve financial intermediation as the primary delivery mechanism.
- ✓ The Inter-American Development Bank (IDB), operating through its public sector window, implements one project — Chile (FP017) — the portfolio's sole completed renewable energy infrastructure case delivered through a multilateral public channel.
- ✓ Bilateral development finance institutions (KfW, AFD) implement two projects — Bangladesh (FP004) and Morocco.
- ✓ A national direct access entity (PROFONANPE, Peru) implements one project — FP001 — representing the portfolio's sole national environmental fund structure and the longest APR verification horizon among national direct access entity projects at seven years.
- ✓ A national direct access entity (Rwanda's Ministry of Environment) implements one project — Rwanda (FP073) — producing the highest design authenticity score in the portfolio.
- ✓ A commercial bank (MUFG) implements one project — Chile (FP115) — representing the Fund's furthest extension into private capital mobilization.
- ✓ Conservation International, co-implementing with EIB in Madagascar (FP026), represents the international NGO with private finance co-implementation structure.

SAP004 (Mongolia) is implemented through a public multilateral channel under the small-access programme.

This diversity of implementing entity types is methodologically significant. A structural finding that appears consistently across UN agencies, bilateral DFIs, MDB public and private sector windows, commercial banks, national DAEs, and international NGOs cannot be attributed to implementing entity characteristics. It must be attributed to structural properties of the project architecture itself — which is precisely the claim the framework advances.

5.4 Verification Horizon

The APR verification period ranges from four years (SAP004) to eight years (FP039), with an average of 5.7 years per project. This range reflects both the age distribution of selected projects and the varying depth of public APR disclosure across GCF's approved portfolio. Projects approved at B.11–B.15

generally carry longer verification horizons; those approved at B.19–B.23 generally carry shorter horizons, reflecting the Fund's more recent approval cycle.

Three projects have reached full completion with final evaluation data available:

FP001 (Project Completion Summary), FP039 (Project Completion Report including eight-year APR series), and FP115 (Project Completion Report following formal cancellation).

FP009 reached nominal completion in January 2026 but APR Section 2 data for its final two years remains confidential, limiting full lifecycle verification.

FP017 (Chile) has reached completion with independent completion-stage evaluation evidence available, and its full APR series is incorporated in the consistency analysis. FP019 (Ecuador) has also reached completion and its full APR series is incorporated in the consistency analysis, but no independent evaluation document is recorded in the public document set used for this report.

The remaining nine projects are active, with APR data current through CY2023 or CY2024 depending on reporting cycle.

Project	Country	GCF Classification	AE Type	Board	Status	APR Years
FP001	Peru	Adaptation	National DAE	B.11	Completed	7
FP004	Bangladesh	Adaptation	Bilateral DFI	B.11	Active	5
FP009	El Salvador	Mitigation	MDB (Private)	B.13	Completed	4*
FP017	Chile	Cross-cutting	MDB (Public)	B.14	Completed	5
FP019	Ecuador	Cross-cutting	UN Agency	B.14	Completed	6
FP026	Madagascar	Cross-cutting	NGO + MDB	B.14	Active	7
FP034	Uganda	Adaptation	UN Agency	B.15	Active	7
FP039	Egypt	Mitigation	MDB (Private)	B.16	Completed	8
FP042	Morocco	Adaptation	Bilateral DFI	B.16	Active	6
FP069	Bangladesh	Adaptation	UN Agency	B.19	Active	5
FP072	Zambia	Adaptation	UN Agency	B.19	Active	7
FP073	Rwanda	Adaptation	National DAE	B.19	Active	5
FP101	Belize	Adaptation	UN Agency	B.22	Active	5
FP115	Chile	Cross-cutting	Commercial Bank	B.23	Cancelled	4†
SAP004	Mongolia	Adaptation	Multilateral	B.18	Active	4

*Data limited: CY2021 missing; CY2023–2024 confidential.

†Includes Project Completion Report (2024) following formal cancellation at B.34.

6. Portfolio Scorecard

6.1 VERA™ Distribution — Design Authenticity

VERA™ scores across the fifteen-project portfolio range from 0.507 to 0.933, spanning the full width of the CR-Tier classification structure.

Three projects achieved CR-Tier I: FP004 (Bangladesh, 0.800), FP072 (Zambia, 0.882) and FP073 (Rwanda, 0.933).

Five achieved CR-Tier II: FP001 (Peru), FP026 (Madagascar), FP034 (Uganda), FP069 (Bangladesh coastal), and FP101 (Belize).

Seven achieved CR-Tier III: FP009 (El Salvador), FP017 (Chile, solar energy / Tarapacá), FP019 (Ecuador), FP039 (Egypt), FP042 (Morocco), FP115 (Chile PSH), and SAP004 (Mongolia).

The single indicator that most consistently drives CR-Tier classification across the portfolio is RA2 — the compound score of policy, infrastructure, and technology compatibility. Of the fifteen projects, fourteen record RA2 below threshold (below 0.500), and twelve of those record RA2 in the range 0.250–0.334. This concentration reflects a structural feature of GCF's operating context: the Fund disproportionately deploys capital into environments where sub-national delivery infrastructure and technology transfer capacity are the binding constraints on outcome achievement. RA2 is not measuring project design quality — it is measuring the structural gap between the environment a project assumes and the environment it operates in. That gap is the primary source of VERA score differentiation across the portfolio.

The three CR-Tier I projects — FP004, FP072 and FP073 — share a structural profile that is instructive. FP004 (Bangladesh, 0.800) achieves CR-Tier I through a structurally distinct path: RA3 = 1.000 and IF3 = 1.000 — complete logframe trackability and high beneficiary scale efficiency — compensate for RA2 = 0.250 in a Hazard-dominant Grade C environment. Unlike FP072 and FP073, FP004 is the only CR-Tier I case that simultaneously records VORTA™ VCS A — the combination of highest design authenticity and full grant justification under complete market failure (IRRm = –100%) in the same project. FP072 and FP073 both record RA2 below threshold (0.334 and 0.250 respectively), confirming that CR-Tier I is not achieved by operating in favorable environments. Both achieve the highest tier through superior performance on IF2 (risk architecture completeness: 0.917 and 1.000) and RA3 (logframe measurement completeness: 0.917 and 1.000) — design decisions that compensate for environmental constraints through structural rigor. This finding has direct implications for GCF design practice: a project cannot upgrade its operating environment, but it can design the architecture that determines how that environment is managed.

6.2 ToC:f(x)™ Distribution — Causal Structure

ToC:f(x)™ scores range from 0.600 to 0.800 across the portfolio. Every project records TIER II. Not a single project in the fifteen evaluated achieved TIER I — complete causal exclusivity between project outputs and stated outcomes. This is the most consequential portfolio-level finding in this section, and it requires precise interpretation.

TIER I is defined by Group C full-positive determination: the design document must establish that the claimed outcome depends exclusively on the project's specific outputs, with no parallel pathways capable of producing the same outcome independently. In complex multi-actor development environments — with co-financiers, national programmes, parallel donor interventions, and market dynamics operating simultaneously — this condition is structurally unavailable to the design document at the time of approval. The absence of TIER I in this portfolio is not a design failure of any individual project. It is the structural feature of large-scale climate programmes operating under UNFCCC attribution constraints. TIER I therefore functions not as a routine approval benchmark, but as a high-evidence causal exclusivity condition; TIER II remains compatible with structurally sound project design where attribution is inherently shared.

The score range of 0.600–0.800 captures meaningful structural differentiation within TIER II. The lower bound (0.600) — recorded by FP069 — reflects a design in which Group B elements are partially confirmed, Group C carries multiple Unknown determinations with low exclusivity evidence, and Q5 = No identifies a structural misalignment between the solution architecture and the problem it claims to address — the most significant single-indicator finding in the portfolio. A cluster of projects at 0.650 —

including FP042, FP019, and SAP004 — shares the Group B partial confirmation profile without the Q5 = No determination.

The upper bound (0.800) — recorded by FP034 and FP101 — reflects designs with strong Group A and B confirmation and a limited number of Group C Unknown determinations. The most diagnostically significant individual indicator across the portfolio is Q5 (structural benefit alignment between project design and problem structure). FP069 is the only project in the portfolio to record Q5 = No — confirming from the design document that the solution architecture was structurally misaligned with the problem it claimed to address. That judgment was confirmed by five years of APR data.

6.3 C-FAIR™ Distribution — Delivery Environment

Twelve of fifteen projects record C-FAIR Grade C. Three record Grade B: FP009 (El Salvador, private EE finance), FP017 (Chile, commercial solar energy infrastructure), and FP115 (Chile PSH, private capital mobilization). No project in the evaluated portfolio records Grade A or Grade D/E.

The Grade C concentration — twelve of fifteen — is not a failure signal. It is the structural justification for concessional climate finance. C-FAIR Grade C identifies environments in which the combination of physical climate hazard exposure, institutional delivery capacity constraints, and macro-financial fragility would exclude market mechanisms from operating without public subsidy. GCF's mandate is precisely to deploy capital into these environments. Grade C is not evidence that GCF is deploying into the wrong places — it is evidence that GCF is deploying where it should.

The three Grade B projects share a common structural feature: each involves a financing mechanism in which the primary delivery channel is market-based rather than public programme-based. EE credit line for SMEs (FP009), commercial solar energy infrastructure (FP017), and pumped storage hydro (FP115) all operate in environments where private sector participation is the intended mechanism of scale. Grade B correctly identifies these environments as having lower physical hazard exposure and greater market maturity — and in doing so, correctly signals that grant ratios should be lower and market risk tolerance higher. All three Grade B projects record GCF grant participation ratios below 15%.

Within the Grade C cluster, the S2 (Adaptive Capacity) subdimension is the most consistent structural signal. Every Grade C project records S2 at Weak or Moderate — institutional and infrastructural capacity constraints operating as a delivery ceiling.

S1 (Hazard Exposure) differentiates within Grade C: FP004 (Bangladesh infrastructure), FP069 (Bangladesh coastal), and FP101 (Belize) record S1 High, making hazard the dominant binding constraint in their structural profiles. FP004's Hazard-dominant profile is structurally distinct from FP069's: where FP069 records Problem-Solution Misalignment — a capacity-based solution deployed against a physical hazard problem — FP004 deploys physical infrastructure directly against physical hazard, producing CR-Tier I design authenticity and VORTA™ VCS A within the same Grade C environment. FP042 (Morocco) and FP072 (Zambia) both record S1 Moderate — with drought as the primary hazard — but produce structurally opposite implementation trajectories, demonstrating that S1 alone does not determine outcome.

The S2 subdimension and grant calibration — not S1 alone — are the operative differentiators within Grade C environments.

6.4 VORTA™ Distribution — Value Realization

Twelve of fifteen projects record VCS Grade F. Three record VCS Grade A: FP009 (El Salvador, IRRc +16.78%), FP017 (Chile, IRRc +48.71%), and FP004 (Bangladesh, IRRc +15.76%).

The Grade F concentration requires the same interpretive precision applied to ToC TIER I absence: VCS Grade F does not mean these projects generated no value.

It means that the value they generated was architecturally unmeasurable from the design document.

VORTA™'s VBI framework requires monetized annual benefit data — expressed in USD per beneficiary per year — across Output, Outcome, and Impact layers. In eleven of fifteen projects, the design document contained no such data. In some cases, the absence is measurement architecture absence: the logframe tracked physical indicators (hectares, households reached, jobs created) without monetized value estimates. In others, it is attribution impossibility: the project operates in domains where social value accrues through public goods (wetland carbon sequestration, biodiversity, coastal protection) that have no market price and no established monetary conversion at the design stage. In one case — FP039 (Egypt, RA3=0) — the design document contained no monetized data of any kind, making VORTA Grade F a structural inevitability visible in 2017. SAP004 (Mongolia) represents a structurally distinct Grade F case: herder livelihood income data was present in the design document and VBI computation was possible, but the resulting Δ IRR of +0.74%p falls below the Grade A threshold — the thinnest social premium in the portfolio.

The three Grade A projects — FP009, FP017, FP004 — share a structural feature: each involves a delivery mechanism with direct market revenue or monetized livelihood income quantified at the design stage.

FP017's solar energy revenue is quantified in market terms (kWh × tariff).

FP009's IRRc is positive because the ESI mechanism's \$101.7M NPV rests on quantified SME energy savings (kWh × tariff).

FP004's IRRc +15.76% is generated through VBI-A1 physical infrastructure protection value — the annual avoided damage cost of climate-resilient roads, cyclone shelters, and drainage infrastructure in coastal Bangladesh, computed from baseline climate exposure data documented in the 2016 design document.

VCS Grade A is not a reward for better project design — it is a consequence of design documents that include the data VORTA™ requires to compute social value.

The most structurally significant VORTA finding is not the Grade F concentration but the variation within it. FP034 records GRANT_JUSTIFIED — IRRm = −100%, confirming that no market mechanism exists to finance Uganda inland wetland restoration.

FP019 records REDESIGN_REQUIRED — IRRm = +21.92%, confirming that carbon markets could compensate the activity without GCF subsidy. FP039 records neither, because RA3 = 0 prevents VBI computation entirely.

FP069 records D0 SVA_WARNING, confirmed empirically in Year 7 when Cyclone Remal destroyed the income-generating assets of 492 households with outstanding micro-credit obligations.

Reading twelve Grade F scores as a single signal is a classification error. Each F carries a different structural cause — and each cause has different implications for project design.

6.5 The Consolidated Pattern

Project	Country	VERA CR-Tier	ToC Score	C-FAIR Grade	VORTA VCS	Structural Pattern
FP001	Peru	II (0.654)	0.700	C	F	A
FP004	Bangladesh	I (0.800)	0.750	C	A	A
FP009	El Salvador	III	0.750	B	A	C
FP017	Chile	III	0.700	B	A	C
FP019	Ecuador	III	0.650	C	F	D
FP026	Madagascar	II (0.865)	0.700	C	F	B

FP034	Uganda	II (0.797)	0.800	C	F	A
FP039	Egypt	III	0.700	C	F	C
FP042	Morocco	III	0.650	C	F	A
FP069	Bangladesh	II (0.600)	0.600	C	F	D
FP072	Zambia	I (0.882)	0.750	C	F	A
FP073	Rwanda	I (0.933)	0.750	C	F	A
FP101	Belize	II (0.700)	0.800	C	F	A
FP115	Chile	III	0.700	B	F	B
SAP004	Mongolia	III	0.650	C	F	A

Structural Patterns: A = Design–Environment Separation / B = Internal Mechanism Failure / C = Measurement Gap / D = Attribution Constraint

Consolidated Finding	Cases	Interpretation
High authenticity + constrained environment	FP004, FP072, FP073	Strong design can coexist with weak delivery environment
Same VCS F, opposite finance conclusion	FP034, FP019	Grade F is measurement architecture, not financing verdict
Market-based delivery does not guarantee success	FP009, FP017, FP115	Market maturity changes risk type, not risk existence

Three findings emerge from the consolidated scorecard that no single model can produce alone.

First: CR-Tier I design authenticity and Grade C delivery environment coexist without contradiction in three of fifteen projects — confirming that design quality and environmental capacity are structurally independent dimensions, not correlated proxies.

Second: VCS Grade F and GRANT_JUSTIFIED coexist in FP034, while VCS Grade F and REDESIGN_REQUIRED coexist in FP019. The same Grade F classification maps to structurally opposite financing conclusions — demonstrating that VORTA Grade F is a measurement architecture finding, not a financing verdict.

Third: the three Grade B projects — the only cases with market-dominant delivery mechanisms — produce two VCS A results (FP009, FP017) and one catastrophic failure (FP115, cancelled). Market environment does not guarantee value realization. It changes the mechanism by which value is structured — and the mechanism can fail for structural reasons pre-identifiable at design stage. A fourth finding completes the picture: FP004 (Bangladesh) achieves VCS A within a Grade C environment — demonstrating that Grade B market conditions are not a prerequisite for value realization. The operative condition is measurement architecture: monetized VBI data present in the design document, not a favorable delivery environment. Grade C environments can produce VCS A when the design document provides the data VORTA™ requires.

7. Document Set Standard

7.1 Three-Document Architecture

Each project in this study is evaluated against a three-tier document set: Funding Proposals (FPs), Annual Performance Reports (APRs), and independent Evaluation Reports. These three document types perform structurally distinct roles in the VERA-TCV™ xAPR consistency analysis framework. Conflating them — treating APR data as diagnostic input, or treating Evaluation Report findings as equivalent to ex-ante VERA-TCV™ judgments — would undermine the methodological integrity on which the directional match findings depend.

The architecture is sequential: FPs generate the diagnosis, APRs provide the test, Evaluation Reports provide independent verification of both. No document type is interchangeable with another, and no document type is optional where available.

7.2 Funding Proposals — The Diagnostic Source

The Funding Proposal is the sole input to VERA-TCV™ pre-assessment. All four model judgments — VERA™, ToC:f(x)™, C-FAIR™, and VORTA™ — are generated exclusively from the FP document submitted to and approved by the GCF Board. The specific FP version used in each evaluation is the approved document at the Board meeting date, not any subsequent revision or supplementary submission.

FPs serve as diagnostic inputs because they contain the totality of design intent available at the point of capital commitment. The risk register, *Theory of Change*, *logframe*, financing structure, and country context analysis embedded in each FP are the structural claims the project makes about its own architecture. VERA-TCV™ examines whether those claims are internally consistent, whether the structural conditions they assume exist in the documented evidence, and whether the causal chain they construct is capable of producing the stated outcomes.

Critically, the FP is treated as a closed document in VERA-TCV™ analysis. No subsequent addenda, project revisions, or Board-approved restructuring documents are incorporated into the pre-assessment. This is not a methodological limitation — it is the condition of the methodology's validity. If post-approval revisions were permitted to modify the pre-assessment, the framework would be testing the accuracy of revised claims against the implementation reality that prompted the revision, rather than testing the predictive validity of original structural judgments against independent evidence. The ex-ante claim must be frozen at the moment capital is committed.

7.3 Annual Performance Reports — The Independent Comparator

APRs provide the year-by-year implementation data against which pre-assessment signals are mapped. They function exclusively as comparators: their data is read after the pre-assessment judgment is complete, and their findings have no retroactive effect on VERA-TCV™ scores.

GCF's Annual Performance Reporting system requires accredited entities to submit structured progress reports covering disbursement rates, indicator achievement, risk status, and implementation narrative for each approved project. The Section 2 component of each APR — covering output and outcome indicator progress — is the primary data source for VERA-TCV™ xAPR signal mapping in this study. Section 1 (overall progress rating) provides a complementary summary signal, but signal confirmation is based on quantitative indicator data rather than narrative assessment ratings wherever possible.

APR availability varies across the portfolio. Projects approved at B.11–B.15 generally carry the longest available verification horizons, with APR series ranging from five to eight years. Projects approved at

B.19–B.23 generally carry shorter horizons of four to five years, with FP072 (Zambia, B.19) representing an exception at seven years reflecting its extended implementation timeline. In three cases — FP009 (El Salvador) for CY2021 and CY2023–2024, and FP042 and FP101 for later years — APR Section 2 data is either not publicly available or classified as confidential, limiting full-lifecycle signal verification. These cases are documented as data-access limited rather than as VERA-TCV™ framework prediction failures. The distinction is methodologically significant: a signal that cannot be tested due to data unavailability is not a confirmed signal and is not a contradicted signal. It is a cell in the consistency matrix that cannot be populated.

The public APR disclosure pattern itself constitutes a structural finding. Of GCF's 336 approved projects, the number carrying complete public APR Section 2 series sufficient for full VERA-TCV™ ×APR analysis is materially smaller than the approved portfolio count. This concentration of measurable projects at the early pipeline cohort reflects both GCF's reporting maturation over time and a systematic gap in results management transparency that has direct implications for independent ex-post evaluation.

7.4 Evaluation Reports — External Verification

Independent evaluation documents — Interim Evaluations, Mid-Term Reviews, Project Completion Reports, and Final Evaluation Reports — serve a third structural role distinct from both FPs and APRs. Where available, they provide external verification of the structural signals identified by VERA-TCV™ pre-assessment, produced by evaluation teams with no knowledge of and no reference to VERA-TCV™ judgments.

Twelve of fifteen projects in this portfolio carry at least one independent evaluation document. These include three Interim Evaluations (FP042, FP069, FP072), five Mid-Term Reviews or Mid-Term Evaluations (FP004, FP034, FP072, FP101, SAP004), four Project Completion Reports or completion summaries (FP001, FP017, FP039, FP115), and one IEU LORTA RCT Midline / impact evaluation (FP026). Because FP072 carries both an Interim Evaluation and a Mid-Term Review, these twelve projects yield thirteen independent evaluation documents in total. Across all twelve projects, independent evaluation findings are directionally consistent with VERA-TCV™ pre-assessment signals. The convergence is documented at three levels of precision.

At the most precise level, FP101's Mid-Term Review (2024) produced the statement "underperformance can be attributed to implementation failures due to an overestimation of delivery capacity" — the plain-language equivalent of VERA™ RA2 = 0.250, derived five years after the design-stage computation from independent evidence. At the intermediate level, FP069's Interim Evaluation (2022) rated Sustainability as Moderately Unlikely — the operational confirmation of Problem-Solution Misalignment that ToC:f(x)™ Q5 = No identified in 2018. At the broad level, FP039's eight-year PCR confirmed the Private-Public Bifurcation — Component 2 overperformance and Component 1.1 abandonment — that VERA™ RA2 and C-FAIR™ S2 independently identified from the 2017 design document.

The methodological value of Evaluation Reports is that they provide convergent evidence from a structurally independent source. When a VERA-TCV™ pre-assessment identifies VERA™ RA2 = 0.2500 in 2019 and a Mid-Term Review in 2024 concludes that "underperformance can be attributed to an overestimation of delivery capacity" (FP101), the convergence is not circular — the two findings were produced by different analytical methods from different evidence bases at different points in time. The convergence confirms that the structural condition VERA-TCV™ identified was real and consequential, not an artifact of the model.

No independent Evaluation Report in this portfolio contradicts the material structural failure pathways identified by VERA-TCV™ pre-assessment. Where independent evaluation language was sufficiently

specific to enable direct comparison, it converged with the structural signal VERA-TCV™ had identified from the design document — in some cases years earlier. This convergence across thirteen evaluation documents, produced by independent teams with no knowledge of VERA-TCV™ judgments, constitutes external validation of the framework's primary diagnostic claims.

7.5 Document Availability by Project

The following table records the document set available for each evaluated project and the verification horizon it enables.

Project	FP Year	APR Coverage	Independent Evaluation	Verification Horizon
FP001	2015	CY2017–2023	Project Completion Summary	7 years
FP004	2015	CY2018–2022	Mid-Term Review (2022)	5 years
FP009	2016	CY2019–2022*	—	4 years*
FP017	2016	CY2018–2022	PCR (2023)	5 years
FP019	2016	CY2017–2022	—	6 years
FP026	2016	CY2018–2024	IEU LORTA (2023)	7 years
FP034	2016	CY2017–2023	Mid-Term Evaluation (2021)	7 years
FP039	2017	CY2017–2024	PCR (2024)	8 years
FP042	2017	CY2019–2024†	Interim Evaluation (2022)	6 years
FP069	2018	CY2020–2024	Interim Evaluation (2022)	5 years
FP072	2018	CY2018–2024	Interim Evaluation (2022)	7 years
FP073	2018	CY2019–2023	—	5 years
FP101	2019	CY2020–2024††	Mid-Term Review (2024)	5 years
FP115	2019	CY2020–2022	PCR (2024)	3 years + PCR
SAP004	2018	CY2019–2022	Mid-Term Review (2021)	4 years

*CY2021 APR not available; CY2023–2024 Section 2 confidential.

†CY2023–2024 Section 1 only; Section 2 not publicly disclosed.

††CY2023–2024 Section 2 not publicly disclosed.

PART III – SIX CASE ANALYSES

Six projects. Six structural configurations. Four patterns of failure — and success. Part III applies VERA-TCV™ to six anchor cases selected for one reason: each instantiates a distinct structural pattern with maximum diagnostic clarity. Not to prove the framework correct. To make the four structural patterns legible. FP034 and FP072 for Pattern A — design sound, environment constraining. FP026 for Pattern B — mechanism specified, never activated. FP009 and FP039 for Pattern C — value generated, architecturally unmeasurable.

FP069 for Pattern D — causal logic misaligned at design stage. The same methodology that identified Q5=No in FP069 confirmed structural integrity in FP072 through a forty-year drought. The framework discriminates in both directions.

PART III — SIX CASE ANALYSES

The following six cases were selected from the fifteen-project portfolio to represent the principal structural patterns identified across the full evaluation. Each case is presented as a strategic summary; full VERA-TCV™ xAPR consistency analysis, annual signal mapping, and indicator-level matrices are provided in Appendix G (ENH-SRV individual reports). The sentence that opens each case is not a formality: it records the date of diagnosis and the document from which it was generated.

Case 1 — FP034 • Uganda • Building Resilient Communities, Wetlands Ecosystems and Associated Catchments

"The most constrained environment generates the strongest grant justification."

Why this case matters

FP034 is the portfolio's canonical instance of the Banker's Paradox. It demonstrates that the structural conditions that make delivery most difficult — extreme institutional fragility, LDC procurement friction, absence of market mechanisms — are simultaneously the conditions that make concessional grant finance most justified. The case establishes that C-FAIR™ TIER V should not be read as an automatic rejection signal. In the absence of viable market mechanisms, it can define the terrain where GCF's comparative advantage is structurally strongest.

VERA-TCV™ Scorecard

The following judgment was generated from the 2016 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier II (0.797)	RA2 = 0.250	LDC delivery ceiling: chronic constraint on implementation speed
ToC:f(x)™	TIER II (0.800)	Group C Unknown	Multi-ministry attribution gap; joint-highest ToC score in portfolio
C-FAIR™	Grade C / TIER V	S2 Weak	Most constrained delivery environment in the portfolio
VORTA™	VCS F / GRANT_JUSTIFIED	IRRm = -100%	No market mechanism exists; grant is the only viable capital channel

APR Signal Confirmation • CY2017–CY2023

Signal	Type	Confirmation	Key Evidence
VERA™ RA2	Chronic	✓	33% cumulative progress at Year 5; 72.5% at Year 7
C-FAIR™ S1	Acute	✓	Victoria-Kyoga flooding (CY2020); compounded by COVID-related field disruption
C-FAIR™ S2	Chronic	✓	Multi-ministry disbursement friction confirmed CY2017–CY2023
ToC Group C	Structural	✓	Attribution constraint sustained without contradiction across all seven APR years
VORTA™ D0	Acute	⚠	Community re-encroachment under micro-credit pressure (CY2020)

Directional match rate: 76.9% (automatic mapper) / 80.8% (chronic signal correction applied). No material directional contradiction recorded across active FP034 signals over the seven-year APR horizon. Full cell-level matrix: Appendix G — ENH-SRV-FP034-2026.

Structural Diagnosis — Pattern A: Design–Environment Separation

VERA™ CR-Tier II (0.797) and ToC:f(x)™ TIER II (0.800) confirm sound design logic and authentic incentive alignment. C-FAIR™ TIER V and VORTA™ GRANT_JUSTIFIED confirm the most constrained implementation environment in the portfolio. The design is not the problem. The environment is not a design failure. The separation between them is the structural condition that generates both the project's delivery constraints and its financing necessity.

The most constrained risk environment and the strongest grant justification are not in tension. They are two expressions of the same structural fact. FP034 illustrates the type of case for which GCF's concessional mandate is structurally most relevant: where rational market behavior produces exclusion, and public climate finance becomes the only viable capital channel.

Case 2 — FP072 · Zambia · Strengthening Climate Resilience of Agricultural Livelihoods in Agro-Ecological Regions I and II

“The grant was right-sized. The institutions held. The design absorbed the worst drought in forty years.”

Why this case matters

FP072 modifies the standard Pattern A finding. Where FP034 demonstrates that the most constrained environment justifies the strongest grant, FP072 demonstrates that correct design tier, S2 Moderate delivery environment, and correctly calibrated grant ratio together produce absorbed separation — a configuration in which Design–Environment Separation does not translate into outcome failure. CR-Tier I in a Grade C environment does not reflect environmental advantage. It reflects design judgment quality.

VERA-TCV™ Scorecard

The following judgment was generated from the 2018 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier I (0.882)	IF2 = 0.917; RA1 = 1.000	Joint-highest design authenticity; multi-pathway hedge confirmed
ToC:f(x)™	TIER II (0.750)	Q15 = Yes	Sixteen-district distributed design enables shock portability
C-FAIR™	Grade C / S2 Moderate	S1 = Moderate (drought-dominant)	Structurally distinct from S2 Weak cases in portfolio
VORTA™	VCS F	IRRc negative	Monetized livelihood data absent at design stage; value real but unmeasured

APR Signal Confirmation • CY2018–CY2024

Signal	Type	Confirmation	Key Evidence
VERA™ RA2	Chronic	✓	Irrigation infrastructure at 1.3% of target (CY2021)
VERA™ IF2	Acute	✓	Goat pass-on sales +70.8% during drought year (CY2024)
C-FAIR™ S1	Acute	✓	Worst drought in forty years, as recorded in CY2024 APR reporting
C-FAIR™ S2	Chronic	✓	Political transition absorbed without programme collapse (CY2021)
ToC Group C	Structural	✓	Attribution gap confirmed; sixteen-district shock portability confirmed

Directional match rate: 100% across all tested signal-year cells. No material directional contradiction recorded over the seven full implementation years. Full cell-level matrix: Appendix G — ENH-SRV-FP072-2026.

Structural Diagnosis — Pattern A: Design–Environment Separation (Absorbed)

FP072 is the portfolio's clearest case of absorbed Design–Environment Separation. VERA™ CR-Tier I combined with C-FAIR™ Grade C / S2 Moderate and grant calibration of 23.3% — within the C-FAIR™ recommended 10–30% range for this environment — produced a structural configuration in which four compounding shocks were absorbed without programme-level failure: COVID-19 (CY2020), political transition and currency depreciation (CY2021), El Niño forecast activation (CY2023), and the worst drought in forty years (CY2024, as recorded in APR reporting). Cumulative beneficiary coverage reached 738,656 (78.1% of target of 946,153). Agricultural income maintained 62% cumulative growth through the drought year.

The contrast with FP042 (Morocco) defines the finding. Both projects carry C-FAIR Grade C and VORTA VCS F. FP042 records VERA CR-Tier III, C-FAIR S2 Weak, and a grant ratio below the C-FAIR recommended range for its S2 Weak profile. FP072 records VERA CR-Tier I, C-FAIR S2 Moderate, and grant ratio within range; it maintained 78.1% beneficiary coverage through the worst drought in forty years. The contrast between FP042 and FP072 is not explained by nominal grant percentage alone, but by whether the grant ratio is structurally calibrated to the project's C-FAIR profile. VERA-TCV™'s grant calibration guidance should therefore be treated as structurally consequential, not merely descriptive.

Case 3 — FP009 • El Salvador • Energy Savings Insurance for Private Energy Efficiency Investments by SMEs

"The mechanism worked. The market did not respond."

Why this case matters

FP009 carries the highest NPV in the portfolio (VCS A, IRRc +16.78%, NPV USD 101.7M) and the most structurally sophisticated financial mechanism design across all fifteen evaluated projects. Yet cumulative tCO₂ achievement reached 0.65% of target after four years. The case establishes that mechanism design quality and market formation speed are structurally independent variables — and that ToC:f(x)™ Q10 No identifies the dependency on market formation from the design document before implementation begins.

VERA-TCV™ Scorecard

The following judgment was generated from the 2016 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier III	IF3 = N/A; IF2 = 1.000	Protocol boundary (enterprise beneficiaries); risk architecture complete
ToC:f(x)™	TIER II (0.750)	Q10 = No	Outcome depends on composite LFI + SME + TSP market behavior
C-FAIR™	Grade B / TIER IV	S2 = Moderate	BANDESAL EE capacity gap; favorable vs. LDC portfolio baseline
VORTA™	VCS A / TIER I	IRRc = +16.78%	USD 101.7M NPV potential; contingent on market formation speed

APR Signal Confirmation • CY2019–CY2022

Signal	Type	Confirmation	Key Evidence
C-FAIR™ S2	Chronic	✓	Full Year 1 consumed in pre-execution conditions
ToC Q10 No	Structural	✓	Credit line operational; zero SME loans in CY2020 (COVID demand collapse)
VERA™ RA2	Chronic	✓	SME loans: 39/494 (7.9%) at CY2022; tCO ₂ : 0.65% of target

Note: CY2021 APR not available; CY2023–2024 Section 2 confidential. Available-data directional match rate: 80.0%. No material directional contradiction recorded in any available year. Full case analysis: Appendix G — ENH-SRV-FP009-2026.

Structural Diagnosis — Pattern C: Market Formation Horizon Gap

FP009 demonstrates the most structurally significant form of Pattern C in the portfolio. The ESI mechanism is genuine — IF2 = 1.000, ToC Group A/B fully confirmed, Colombia and Mexico pilot transferability documented, VORTA™ TIER I. The credit line was operational. Training was delivered. SIESI monitoring was active. The mechanism worked. The market did not respond.

ToC:f(x)™ Q10 No identified from the 2016 design document that EE loan uptake depends on the composite behavioral convergence of LFIs developing EE lending expertise, SMEs developing EE investment confidence, and TSPs building verification infrastructure — three actor groups operating independently of project control. Four years of implementation confirmed this judgment: the mechanism is operating on correct structural logic but on an incorrect timeline assumption. The observed implementation trajectory suggests that EE finance market formation may require a development horizon closer to ten to fifteen years than to a conventional five-to-seven-year project cycle. The ESI mechanism deserves a longer runway. The correct policy response is not to abandon replication — it is to design the next deployment with a market formation horizon the mechanism can actually achieve.

Case 4 — FP039 · Egypt · GCF-EBRD Egypt Renewable Energy Financing Framework

"147% of physical target delivered. Social value: unmeasured from Day 1."

Why this case matters

FP039 is the portfolio's longest verification horizon case — eight years, CY2017 through CY2024 — and the portfolio's clearest instance of Private-Public Bifurcation: a single programme whose two components deliver diametrically opposite performance trajectories, each predicted by a different pre-assessment signal from the 2017 design document. The case also establishes the most direct example of measurement architecture absence: VORTA™ VCS F was predetermined from RA3 = 0.000 in 2017, confirmed at programme close in 2024.

VERA-TCV™ Scorecard

The following judgment was generated from the 2017 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier III	RA3 = 0.000; RA2 = 0.250	Zero trackable outcome indicators at design stage; three-dimensional delivery constraint
ToC:f(x)™	TIER II	Q10/Q11 = No	RE market development not exclusively attributable to GCF-EBRD framework
C-FAIR™	Grade C	S1 = Low; S2 = Weak	Low physical hazard; institutional delivery constraint on public components
VORTA™	VCS F	IRRc = -53.62%	RA3 = 0 predetermines VCS F; measurement architecture absent at approval

APR Signal Confirmation · CY2017–CY2024

Signal	Type	Confirmation	Key Evidence
VERA™ RA2 (public)	Chronic	✓	Component 1.1 abandoned after seven years; zero deliverables
C-FAIR™ S1	Eight-year	✓	Low-hazard environment confirmed: uninterrupted Benban operations CY2019–2024
C-FAIR™ S2	Chronic	✓	Commercial bank exclusion; EETC tender cancellations; TOR delays
ToC Q10/Q11	Structural	✓	Kom Ombo record-low tariff driven by FiT + competition + policy simultaneously
VORTA™ VCS F	Structural	✓	884 MW delivered; zero monetized social benefit framework built at close

Directional match rate: 100% across all tested signal-year cells over eight years. No material directional contradiction recorded. Full cell-level matrix: Appendix G — ENH-SRV-FP039-2026.

Structural Diagnosis — Pattern C: Private-Public Bifurcation + Measurement Gap

FP039 closes at 147% of original physical target: 884 MW operational, 3,032 ktCO₂ cumulative, the first private Climate Bonds Standard-certified project bond in the southern and eastern Mediterranean. Simultaneously, Component 1.1 — the grid absorption model that was the structural bridge between physical RE delivery and systemic market transformation — was formally abandoned at year seven with zero deliverables. VERA™ RA2 = 0.250 and C-FAIR™ S2 = Weak identified the public institutional delivery constraint from the 2017 document. VERA™ CI2 = 1.000 identified the private financial

mechanism's structural integrity. Both predictions held — simultaneously, in opposite directions — across eight years.

RA3 = 0.000 at design stage predetermined VORTA™ VCS F at programme close. Eight years of APR reporting did not close the gap: no monetized social benefit framework was built, no VBI data was collected, no IRRv computation became possible. The programme generated substantial physical value — and none of it is quantifiable by VORTA™ because the measurement architecture was never designed to capture it. VCS F here is not a performance verdict. It is a measurement architecture verdict that was readable in 2017.

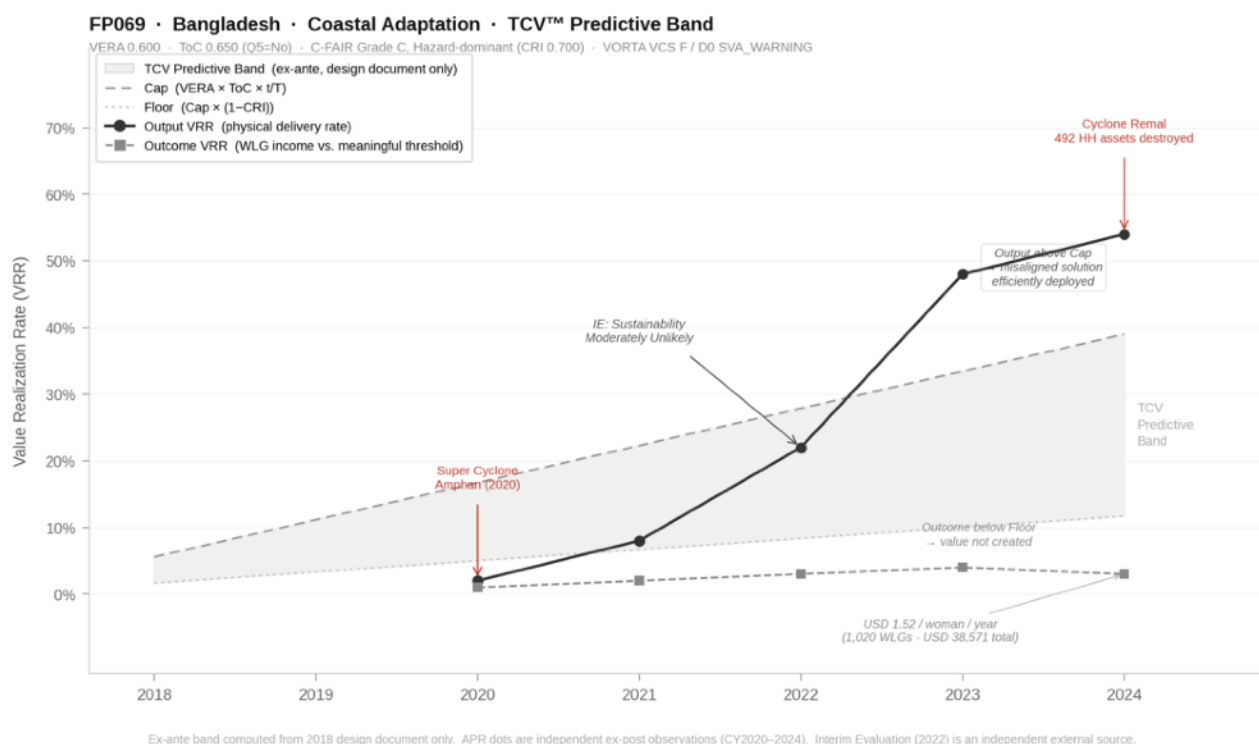
FP039 shows that when an MDB-mediated private framework bundles private capital mobilization with public institutional capacity building into a single programme, the two components may operate on structurally incompatible delivery timelines. FP039 is the eight-year empirical demonstration of that mismatch — visible from the design document before a dollar was committed.

Case 5 — FP069 • Bangladesh • Enhancing Adaptive Capacities of Coastal Communities

"The solution was efficiently deployed. The problem was structurally unaddressed."

Why this case matters

FP069 is the portfolio's canonical instance of Problem-Solution Misalignment and the only case recording ToC:f(x)™ Q5 = No. It is also the case for which the VERA-TCV™ Predictive Band was developed as a visualization instrument — the chart below is the primary diagnostic exhibit. The VORTA™ DO SVA_WARNING gate, now permanent in v2.5, was designed in direct response to the structural risk identified in this project's 2018 design document.



Ex-ante band computed from 2018 design document only. APR dots are independent ex-post observations (CY2020–CY2024). Output VRR above Cap (CY2023): misaligned solution efficiently deployed. Outcome VRR below Floor (all years): USD 1.52 / woman / year recorded by CY2024 APR data.

VERA-TCV™ Scorecard

The following judgment was generated from the 2018 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier II (0.600)	RA2 = 0.334	Lower boundary of Tier II; delivery constrained at every structural layer
ToC:f(x)™	TIER II (0.600)	Q5 = No	Solution architecture does not structurally address physical problem
C-FAIR™	Grade C / Hazard-dominant	S1 = High	Repeated cyclone exposure; physical events reset delivery gains
VORTA™	VCS F / D0 SVA_WARNING	13/14 VBI domains = \$0	Micro-credit in Hazard-dominant environment creates debt transfer risk

APR Signal Confirmation · CY2020–CY2024

Signal	Type	Confirmation	Key Evidence
ToC Q5 No	Structural	✓	Interim Evaluation (2022): Sustainability — Moderately Unlikely
C-FAIR™ S1	Acute × 2	✓	Super Cyclone Amphan (CY2020); Cyclone Remal (CY2024)
VORTA™ D0	Acute	✓	492 households: assets destroyed by Remal; WLG debt obligations persist
Output VRR	Structural	✓	Above Cap in CY2023: efficient deployment of misaligned solution confirmed
Outcome VRR	Structural	✓	Below Floor all years: USD 38,571 total annual WLG revenue (1,020 groups)

Directional match rate: 100% across all tested signal-year cells. No material directional contradiction recorded over five APR years. Full cell-level matrix: Appendix G — ENH-SRV-FP069-2026.

Structural Diagnosis — Pattern D: Problem-Solution Misalignment

The physical problem documented in the 2018 design — saline intrusion displacing freshwater, cyclone hazard destroying assets, coastal erosion eliminating agricultural land — is a physical infrastructure problem. The project's primary response is capacity-building and microfinance. ToC:f(x)™ Q5 = No identified this structural gap from the design document before a single APR was written. The 2022 Interim Evaluation's Sustainability: Moderately Unlikely confirmed it from independent evidence four years later. USD 1.52 per woman per year in WLG income at project close is not a measurement gap. It is the quantitative expression of a solution that cannot generate climate resilience at the scale the physical problem requires.

The VERA-TCV™ Predictive Band chart makes the structural paradox visible. Output VRR rising above Cap in CY2023 documents that the delivery mechanism is functioning — RWHS installed, WLGs formed, disbursement accelerating. The mechanism is not broken. It is misaligned. The gap between the Output line above Cap and the Outcome line below Floor is not an implementation failure. It is the visual measurement of Problem-Solution Misalignment: the distance between what the project efficiently delivered and what the physical problem structurally required. Hazard-dominant Grade C environments require a primary Theory of Change anchored in physical hazard reduction infrastructure. Q5 = No is the design-stage signal that this inversion has occurred. FP069 demonstrates what happens when that signal is not acted on before approval.

Case 6 — FP026 • Madagascar • Sustainable Landscapes in Eastern Madagascar

“The public engine delivered. The private engine never activated.”

Why this case matters

FP026 is the portfolio's canonical instance of Internal Mechanism Failure and the primary external validation case connecting this report to the companion SSRN paper. The IEU LORTA findings provide the highest-standard external verification for the public implementation pathway that existed and could be randomized. They do not replace VERA-TCV™ diagnosis of the absent private-finance pathway; rather, they define the boundary that makes FP026 methodologically important. The case establishes that a cross-cutting architecture can appear diversified on paper while remaining causally singular in practice: the EIB private investment fund was the singular specified causal engine for the sustainable-livelihood Outcome claim, it never operated for a single day across seven years, and no measurable livelihood outcome was recorded at project close.

VERA-TCV™ Scorecard

The following judgment was generated from the 2016 design document. No Annual Performance Report data was available at the time of diagnosis.

Model	Score / Grade	Key Signal	Structural Implication
VERA™	CR-Tier II (0.865)	RA2 = 0.334	Triple delivery vulnerability; high authenticity despite constrained environment
ToC:f(x)™	TIER II	Q10/Q11 Unknown	Livelihood outcome pathway causally unspecified; EIB activation conditions absent
C-FAIR™	Grade C / TIER V	S1 High; S2 Weak; S3 High	Maximum composite fragility in portfolio
VORTA™	VCS F	IRRc negative	VBI computation dependent on EIB fund that never operated

APR Signal Confirmation • CY2018–CY2024

Signal	Type	Confirmation	Key Evidence
ToC Group C	Structural	✓ × 7 years	EIB FAA never executed; livelihood outcome pathway absent CY2018–CY2024
VERA™ RA2	Chronic	✓	4.3% disbursement Year 1; COVID full suspension CY2020
C-FAIR™ S1	Acute	✓	Cyclone Batsirai direct strike on project area (CY2022)
C-FAIR™ S2	Chronic	✓	Road access, financial system, institutional capacity constraints confirmed
Forest Component	Positive	✓	92% cumulative target at Year 7: public engine delivered

Directional match rate: 98.9% across all tested signal-year cells. No material directional contradiction recorded over seven APR years. ToC Group C confirmed in all seven years without exception — the strongest single-signal consistency record in the evaluated portfolio.

Structural Diagnosis — Pattern B: Internal Mechanism Failure / Singular Causal Dependency

The forest component reached 92% of cumulative target. Conservation International's public implementation engine delivered. The LORTA midline evaluation confirmed statistically significant improvements in conservation agriculture adoption, food security, and forest resource dependency reduction — genuine treatment effects from a genuine intervention.

And yet: the project's stated Outcome — sustainable livelihoods — was not confirmed in any measurable form across seven years. The EIB private investment fund, presented in the 2016 design document as the singular causal mechanism for livelihood outcomes, never signed a Financing Arrangement Agreement, never disbursed, never operated. The Climate Change Trust Fund was never established. ToC:f(x)[™] Q10 and Q11 Unknown identified this structural gap in 2016: the operating conditions for EIB fund activation were nowhere specified in the design document. The mechanism's success was assumed, not structured.

The LORTA boundary problem defines this case's methodological significance. LORTA measured the treatment effects of the component that existed and could be randomized — the public CI pathway. VERA-TCV[™] identified the failure of the component that was claimed but absent — the EIB private finance pathway. The two frameworks identify the same structural gap from opposite ends of the implementation timeline. Neither alone produces a complete picture. Together, they establish: the public engine delivered what it was designed to deliver; the private engine delivered nothing; and the Outcome the design promised depended entirely on the private engine. A cross-cutting architecture that assigns exclusive Outcome responsibility to a single unspecified mechanism is not diversified. It is singular — and singularity without specification is the structural definition of Internal Mechanism Failure.

PART IV — CROSS-CASE FINDINGS

Individual cases prove a framework can work. They cannot prove what the portfolio reveals. Three findings in Part IV are only visible when fifteen cases are held against the same diagnostic instrument simultaneously. C-FAIR™ Grade C as the structural justification for concessional finance — not a failure signal, but a confirmation that GCF capital is deployed precisely where market mechanisms cannot operate. VORTA™ VCS Grade F concealing six structurally distinct causes — a single classification that, read as a single signal, produces systematically incorrect financing recommendations. Zero TIER I — not a portfolio design failure, but the structural consequence of multi-actor climate programmes where complete causal exclusivity is unavailable by definition. These are not case findings. They are system findings. They required fifteen cases to become visible.

PART IV — CROSS-CASE FINDINGS

8. Four Structural Patterns

The six anchor cases and fifteen evaluated projects do not fail randomly. Across four continents, five thematic areas, and an average of 5.7 years of implementation data, the same structural failure types recur — in different projects, different environments, different implementing entities — with consistent internal logic. Four patterns account for all fifteen cases.

The patterns are ordered by a specific principle: the point in the ToC causal chain where the structural failure originates. Pattern A failures originate at the Output–Outcome boundary — the environment cannot convert what the project delivers into the outcomes the project claims. Pattern B failures originate within the Activity–Output boundary — the mechanism designed to produce outputs never activates. Pattern C failures originate at the Outcome–Impact boundary — value is generated or structurally capable of being generated, but the measurement architecture or realization horizon does not capture it within the project cycle. Pattern D failures originate within the Theory of Change logic itself — the causal structure is either attribution-constrained or solution-misaligned at the design stage. Moving from A to D, the failures become progressively less correctable by implementation quality alone, because each successive pattern locates the structural deficiency closer to the design premises rather than the delivery conditions. A methodological note on pattern classification: the individual signals that constitute each pattern — VERA™ scores, ToC:f(x)™ tier determinations, C-FAIR™ grade assignments, VORTA™ valuations — are generated as ex-ante outputs from the funding proposal document before any APR data is reviewed. The pattern labels (A, B, C, D) are an ex-post analytical summary of where within the ToC causal chain those pre-assessed signals are located. The patterns describe the structure of the pre-assessment, not the structure of the outcome.

Pattern A — Design–Environment Separation accounts for eight of fifteen evaluated projects: FP001, FP004, FP034, FP042, FP072, FP073, FP101, and SAP004.

In most Pattern A cases, VERA™ and ToC:f(x)™ confirm that the core design logic is partially or substantially coherent, while C-FAIR™ identifies a delivery environment that operates as a structural ceiling on outcome achievement. The design is not the sole failure point. The environment does not allow the design to fully succeed. The separation between design quality and environmental delivery capacity is the defining structural feature.

Pattern A is the most common pattern in the portfolio because it is the most common structural condition in GCF's operating context. The Fund deploys capital into environments where market mechanisms cannot — which are, by definition, environments with elevated hazard exposure, limited institutional delivery capacity, and constrained macro-financial conditions. Design–Environment Separation is not a project failure. In most Pattern A cases it is the expected structural outcome of correct GCF intervention logic: partially or substantially coherent design, appropriately financed, operating in the environment GCF was created to serve. The diagnostic value of Pattern A is not to flag failure but to specify the type and magnitude of separation — which determines what corrective design measures would have reduced the gap.

Three Pattern A cases demonstrate absorbed separation — a configuration in which CR-Tier I design quality within a Grade C environment does not translate into outcome failure.

FP072 (Zambia) achieves absorption through S2 Moderate delivery capacity and correctly calibrated grant ratio of 23.3%: the design sustained 78.1% beneficiary coverage through the worst drought in forty years.

FP073 (Rwanda) achieves absorption through the portfolio's highest design authenticity score (CR-Tier I, 0.933) deployed into a national DAE context where institutional alignment compensates for

environmental constraints.

FP004 (Bangladesh) achieves absorption through a structurally distinct path: physical infrastructure deployed directly against physical hazard — cyclone shelters and climate-resilient roads against the coastal hazard profile that drives C-FAIR Hazard-dominant classification — producing CR-Tier I and VORTA™ VCS A simultaneously within Grade C.

Pattern A is not a deterministic ceiling — it is a structural condition whose consequences depend on the precision of the design response.

FP042 (Morocco) defines Pattern A's lower boundary. VERA™ CR-Tier III combined with C-FAIR™ Grade C / S2 Weak and grant ratio below the C-FAIR™ recommended range produced Output–Outcome disconnect: the Kaddoussa Dam infrastructure was physically constructed, but water was not delivered to beneficiaries. The structural ceiling was not only environmental — it was compounded by design authenticity constraints and miscalibrated financing. Pattern A cases therefore span a range from absorbed separation (FP072) to compounded separation (FP042), and the position within that range is determined by design tier, S2 subdimension, and grant calibration precision.

SAP004 (Mongolia) presents Pattern A at minimum viable threshold. CR-Tier III (0.547) design quality coexists with C-FAIR™ Grade C Capacity-dominant fragility — the financial system immaturity that constrains XacBank's consumer EE loan deployment. Unlike FP034's hazard-dominant constraint or FP042's infrastructure gap, SAP004's separation is financial-institutional: the credit mechanism is structurally sound, but the Mongol Bank DTI ceiling and low market maturity operate as a persistent delivery ceiling. VORTA™ VCS F with GRANT_JUSTIFIED (Δ IRR +0.74%p) is the thinnest positive social premium in the portfolio — structurally sensitive to carbon price assumptions and representing the minimum condition under which grant support is justified.

FP001 (Peru) completes the Pattern A spectrum at its lower boundary alongside FP042 (Morocco). Of the fifteen evaluated projects, FP001 carries the longest verification horizon among national direct access entity cases at seven years and records the portfolio's lowest directional consistency rate at 79.2%. Neither of these features represents a framework anomaly. They represent the structural profile of non-absorbed Design–Environment Separation across a full project lifecycle.

FP001's CR-Tier II design authenticity coexists with C-FAIR™ Grade C / S2 Weak delivery conditions and a grant ratio that, while structurally justified under GRANT_JUSTIFIED classification, was not calibrated with the precision that enabled FP072's absorbed configuration. The result across seven years is chronic underperformance without catastrophic failure — the project neither collapsed nor recovered to target trajectory. It persisted in a state of accumulated separation between what the design was structurally capable of claiming and what the environment was structurally capable of delivering.

The contrast between FP001 and FP072 defines the operative range within Pattern A more precisely than either case produces alone. FP001 carries CR-Tier II design authenticity, while FP072 carries CR-Tier I design authenticity. Both operate in Grade C environments. Both record VORTA™ VCS F. The difference is S2 subdimension precision and grant calibration alignment with the C-FAIR™ recommended range. FP072's S2 Moderate profile and 23.3% grant ratio — within the recommended 10–30% range — produced absorbed separation. FP001's S2 Weak profile and grant ratio outside the recommended range produced non-absorbed separation that persisted across the full seven-year lifecycle. Pattern A is not a deterministic ceiling. The position within its range is a design decision — one that is specifiable and correctable at the design stage before capital is committed.

Pattern B — Internal Mechanism Failure accounts for two of fifteen evaluated projects: FP026 (Madagascar) and FP115 (Chile).

In both cases, the failure originates not from the delivery environment but from within the ToC causal architecture — a mechanism specified in the design document as necessary for Outcome achievement never activates. The environment imposes secondary constraints in both cases, but the primary structural failure precedes environmental contact.

FP026's EIB private investment fund — the singular causal engine for sustainable livelihoods — never signed a Financing Arrangement Agreement, never disbursed, and never operated across seven years of implementation. The forest component delivered 92% of target. The livelihood causal chain was absent from Year 1 to Year 7. FP115's pumped storage hydro, solar PV, and desalination project never commenced construction across four years of implementation: the PPA acquisition pathway — simultaneously a project objective, a financing precondition, and a market-driven process outside project control — failed simultaneously with financing non-closure and regulatory non-evolution. The project was formally cancelled at GCF Board B.34 with 8.04% disbursement and zero climate results.

In both cases, VERA™ IF2 identified the structural fragility of the critical mechanism from the design document: for FP026, the absence of specified EIB activation conditions; for FP115, multiple open evasion pathways on the PPA/financing/regulatory causal chain. Pattern B has a precise design implication: cross-cutting architectures and private sector frameworks that assign Outcome delivery to a single mechanism without specifying binding commitment conditions, alternative pathways, or contingency structures create singular causal dependencies that cannot be absorbed when the mechanism fails. VERA™ IF2 is the pre-assessment instrument for this structural risk: incomplete risk hedging coverage at the mechanism level is the design-stage signature of Pattern B.

One structural feature shared by both Pattern B cases warrants portfolio-level observation. FP026 (Madagascar) and FP115 (Chile) both carry GCF's Cross-cutting classification. This co-occurrence is not incidental. Cross-cutting architectures — by design — assign outcome delivery across multiple mechanisms, creating the appearance of diversification. When one of those mechanisms is a private finance instrument whose activation conditions are not specified in the funding proposal, the diversification is nominal and the causal dependency is singular. The public component can deliver every output it was designed to produce. The Outcome pathway remains structurally absent.

GCF's Cross-cutting classification has expanded across its approved portfolio as a programmatic response to the dual mandate of mitigation and adaptation. The structural implication of Pattern B is that this expansion requires a specific design discipline: every private finance mechanism embedded in a Cross-cutting architecture must carry explicitly specified activation conditions, binding commitment structures, and documented alternative pathways before the Outcome claim it supports can be treated as structurally founded. Where those conditions are absent from the funding proposal, the mechanism's success is assumed rather than structured — and assumption without specification is the design-stage signature of Internal Mechanism Failure.

Pattern C — Measurement Architecture and Value Realization accounts for three of fifteen evaluated projects: FP009 (El Salvador), FP017 (Chile), and FP039 (Egypt).

The three Pattern C cases span the full spectrum of measurement architecture — from complete absence to full monetization — and share one structural feature: the relationship between measurement design and value realization is the operative variable governing their Outcome–Impact boundary, not implementation quality or environmental conditions. In Pattern C, the word “failure” does not always mean implementation failure; it refers to the structural condition governing whether value can be measured, realized, and verified within the project cycle.

FP039 represents the measurement architecture absence extreme. RA3 = 0.000 in 2017, combined with the absence of monetized social benefit data, structurally prefigured VORTA™ VCS F in 2024: zero trackable Outcome or Impact indicators with baselines, targets, and means of verification were

included at approval. Eight years of APR reporting confirmed this gap was never closed — 884 MW operational, zero monetized social benefit framework built. VCS F here is a measurement architecture verdict readable from the design document in 2017, not a performance verdict delivered at programme close.

FP009 represents the market formation horizon constraint. VORTA™ VCS A establishes USD 101.7M NPV potential — the measurement architecture exists and the mechanism is sound. But the mechanism on which that potential depends requires market formation speed (independent LFI and SME behavioral convergence) that a seven-year project cycle cannot guarantee. ToC:f(x)™ Q10 No identified this dependency from the design document. CY2022 data confirmed it operationally: 7.9% SME loan achievement after four years is not mechanism failure — it is a correct mechanism operating on an incorrect timeline assumption.

FP017 represents the measurement architecture positive case. IRRc within benchmark range, VCS A: commercial solar energy infrastructure with direct market revenue quantified at design stage produces the measurement conditions VORTA™ requires. VCS A is not a reward for better project design — it is the consequence of a design document that included the monetized benefit data the framework needs.

Pattern D — ToC-Level Constraint accounts for two of fifteen evaluated projects: FP019 (Ecuador) and FP069 (Bangladesh).

Both cases locate their primary structural failure within the Theory of Change logic itself rather than in the delivery environment — but the nature of the ToC-level constraint differs structurally between them.

FP019 faces UNFCCC attribution impossibility. The REDD+ framework under which it operates structurally precludes exclusive attribution of emission reductions to a single project or funding source: the Warsaw Framework's jurisdictional accounting logic distributes REDD+ outcomes at national level, making project-level causal exclusivity structurally unavailable regardless of design quality or implementation performance. ToC Q7, Q8, Q10, and Q11 Unknown for FP019 are not design failures — they are the structural consequence of operating within an international accounting framework that does not permit project-level attribution. The project achieved 100.2% of its restoration target. The outcome could not be attributed to it. Attribution impossibility is inherent to the operating framework, not to the project architecture.

FP069 faces Problem-Solution Misalignment — a structurally distinct form of ToC-level constraint. ToC Q5 = No identifies that the project's solution architecture (capacity-building, microfinance, saline-tolerant agriculture) does not structurally address the physical problem (cyclone hazard, saline intrusion, freshwater displacement). Attribution is not impossible because of an accounting framework — it is structurally foreclosed because the solution cannot generate the claimed outcome regardless of delivery quality or environmental conditions. USD 1.52 per woman per year in WLG income cannot produce climate resilience in an environment where Cyclone Remal destroys all productive assets in a single event. The solution was efficiently deployed. The problem was structurally unaddressed.

The two Pattern D cases share the ToC-level location of their structural failure but differ in corrective implication. FP019's attribution constraint is inherent to the REDD+ operating framework and requires a verification architecture that operates at landscape or jurisdictional scale rather than project scale — a portfolio-level architectural reform, not a project-level redesign. FP069's misalignment requires a ToC redesign that anchors primary outcomes in physical hazard reduction infrastructure before capacity-building components are layered on top — a project-level intervention available at the design stage before approval. Pattern D is the one pattern where implementation improvement cannot close

the structural gap. Only design revision — at the project level for FP069, at the framework level for FP019 — addresses the constraint at its structural origin.

9. Three Portfolio-Level Findings

Six case analyses and fifteen evaluated projects converge on three findings that no individual case can produce alone. Each finding requires the full portfolio range — across continents, sectors, implementing entities, and financing structures — to establish.

Finding 1 — C-FAIR Grade C is the structural justification for concessional climate finance, not a warning to be avoided.

Twelve of fifteen evaluated projects record C-FAIR Grade C. The concentration is not a sample artifact — it reflects the structural reality of GCF's operating mandate. The Fund deploys capital into environments where market mechanisms cannot operate independently: environments with elevated physical hazard exposure, limited institutional delivery capacity, and constrained macrofinancial conditions. These are Grade C environments by definition.

The Banker's Paradox — documented in its purest form in FP034 (Uganda, GRANT_JUSTIFIED, IRRm = -100%, VCS F) and in its most complete form in FP004 (Bangladesh, GRANT_JUSTIFIED, IRRm = -100%, VCS A) — but present across the Grade C cluster — inverts the intuitive reading of risk. High environmental fragility is not a signal that GCF is intervening in the wrong place. It is the structural condition that generates both the greatest need for climate finance and the strongest justification for grant-heavy financing structures. A Grade A or B environment has market mechanisms that can provide financing without public subsidy. GCF's comparative advantage is precisely in Grade C environments where those market mechanisms are structurally absent.

The three Grade B projects in the portfolio — FP009, FP017, FP115 — each involve private sector financing mechanisms in environments with lower physical hazard and greater market maturity. Two of the three produce VORTA™ VCS A results (FP009, FP017), confirming that favorable environments can generate measurable social value through market-based mechanisms. One was cancelled before construction commenced. Grade B is not a superior classification. It is a different structural context requiring a different financing architecture — and carrying its own structural failure mode when the market preconditions it assumes are not met.

Grade A or B environments are more likely to contain market mechanisms capable of financing delivery with lower levels of public subsidy, but they do not eliminate market-precondition risk. FP115 demonstrates that Grade B classification is not a guarantee of market operability: when PPA acquisition, financing closure, and regulatory evolution are simultaneously required as preconditions, their non-attainment produces structural failure regardless of environmental favorability. GCF's comparative advantage is precisely in Grade C environments where market mechanisms are structurally absent — but Grade B deployment requires its own structural discipline on precondition specification.

Finding 2 — VORTA™ VCS F means structurally different things across the portfolio, and reading twelve Grade F scores as a single signal is a classification error.

Twelve of fifteen projects record VCS Grade F. Six structurally distinct causes account for these twelve outcomes.

Measurement architecture absence (FP039, FP026, FP034, FP001, and others): the design document contains no monetized annual benefit data. VCS F is a design document verdict, not a performance verdict. The value was generated — 884 MW operational in Egypt, 92% forest target in Madagascar, seven years of wetland restoration in Uganda — but was not quantifiable from design-stage inputs.

Value generation insufficiency (FP069): USD 1.52 per woman per year is not a measurement gap. It is the substantive finding that the solution generated insufficient value at the scale the problem requires. This is the only case in the portfolio where VCS F reflects genuine value inadequacy rather than measurement absence.

Attribution impossibility and causal-pathway absence (FP019, FP026): FP019 reflects REDD+ jurisdictional attribution limits — the Warsaw Framework's national-level accounting structurally precludes project-level attribution regardless of implementation quality. FP026 reflects a different structural cause: the EIB private investment fund — the singular causal pathway required to generate and attribute livelihood value — never operated. Both produce VCS F, but through structurally distinct mechanisms: FP019 through attribution impossibility inherent to the operating framework; FP026 through the absence of the mechanism on which attribution would have depended.

Market compensation requirement (FP019): $IRR_m > 0$ confirms that the activity type generates market-compensable returns. VCS F combined with positive IRR_m produces REDESIGN_REQUIRED — the signal that GCF grant finance is structurally inappropriate for an activity the market can compensate.

Grant justification (FP034): $IRR_m = -100\%$ confirms the complete absence of market compensation. VCS F combined with negative IRR_m produces GRANT_JUSTIFIED — the signal that GCF grant finance is the only financing mechanism structurally capable of deploying capital into this environment.

Minimum viable grant justification (SAP004): $IRR_m < 0$ and $IRR_v > IRR_m$ confirm a positive but structurally fragile $\Delta IRR = +0.74\%_p$ — the thinnest positive social premium in the portfolio. VCS F coexists with GRANT_JUSTIFIED, but the margin is sensitive to carbon price assumptions: a 50% downward carbon price revision eliminates the positive ΔIRR and removes the grant justification structural basis. SAP004 is the only case where grant support is technically justified but architecturally contingent on a single parameter assumption.

GRANT_JUSTIFIED and REDESIGN_REQUIRED are both VCS F outcomes. They carry opposite policy implications. Treating them as a single signal produces systematically incorrect financing recommendations — approving grants where markets should act and questioning grants where markets cannot. The disaggregation of VCS F by IRR_m sign is not a methodological refinement. It is the structural condition for correct financing architecture decisions.

FP009 demonstrates a related but structurally distinct case outside the VCS F cluster. VORTA™ VCS A with $IRR_c = +16.78\%$ confirms high potential value — the measurement architecture exists and the mechanism is sound. But ToC:f(x)™ Q10 No establishes that value realization depends on market-formation speed: the independent behavioral convergence of LFIs, SMEs, and TSPs that a seven-year project cycle cannot guarantee. VCS A records potential. The gap between potential and achievement is governed by the market formation horizon, not by measurement architecture absence.

Finding 3 — No project in the evaluated portfolio achieved ToC:f(x)™ TIER I. This is not a portfolio failure. It is a structural feature of multi-actor climate programmes.

TIER I requires complete causal exclusivity: the design document must establish that the claimed outcome depends on the project's specific outputs, with no parallel pathways capable of producing the same outcome independently. In fifteen projects across four continents, eight implementing entity types, and five thematic areas, this condition was never met.

The absence of TIER I is a structural property of the GCF portfolio as evaluated, not an individual project deficiency. Large-scale climate programmes operate in environments where co-financiers, national government programmes, parallel donor interventions, and market dynamics function simultaneously. In these environments, exclusive attribution of outcome improvement to a single project's outputs is structurally unavailable from a design document. The Warsaw Framework's jurisdictional REDD+ accounting, Egypt's simultaneous FiT programme and BOO approvals, Uganda's multi-donor agricultural development landscape, Bangladesh's national adaptation programmes — each independently prevents exclusive causal attribution at project level.

This finding has a precise implication for GCF's results management architecture. If TIER I causal exclusivity is structurally unavailable in large-scale multi-actor climate programmes, then an evaluation framework that requires exclusive attribution as the standard for outcome verification will systematically fail to recognize genuine climate impact. The correct measurement standard for GCF-scale programmes is directional contribution — evidence that project outputs moved outcomes in the predicted direction — not causal exclusivity. VERA-TCV™'s directional match methodology is calibrated to this standard. The zero contradiction finding across fifteen projects and an average of 5.7 years is evidence of directional contribution at portfolio scale. It is not a claim of causal exclusivity. That claim is unavailable — and requiring it would make large-scale climate finance unverifiable by design.

10. Predictive Validity Assessment

The central methodological claim of this report is that structural conditions present in funding proposal documents at the point of GCF Board approval are consequential for implementation outcomes — and that VERA-TCV™'s four-model framework can identify those conditions reliably, before implementation begins. This section assembles the quantitative evidence for that claim across the full portfolio.

Directional Consistency.

Directional consistency rates by project range from 76.9% (FP034, automatic mapper) to 100% (FP017, FP039, FP042, FP069, FP072, FP073, FP115). The portfolio-weighted average is approximately 92% across all assessed signal-year cells.

Two projects record rates below 85%: FP034 at 76.9% (automatic mapper) and FP009 at 80.0% (data-limited). Both cases carry documented methodological explanations. FP034 is therefore retained as a methodological boundary case: it shows that chronic structural constraints require mapper logic distinct from acute event detection. FP034's rate rises to 80.8% under chronic signal correction logic — the automatic mapper is calibrated for acute event detection, and FP034's primary signal (VERA™ RA2 chronic constraint) is a multi-year cumulative pattern that the acute trigger does not capture.

FP009's rate reflects data access limitations: 50% of the assessment period (CY2021 missing; CY2023–2024 confidential) cannot be verified from public data. Where indicator-level exceptions appear in individual projects, they are disclosed within the relevant case analysis and treated as boundary cases rather than primary structural signal failures.

Primary Pathway Consistency.

Across fifteen projects, an average of 5.7 years per project, and every assessable signal-year cell in the consistency matrices, no unresolved contradiction in primary structural failure pathways was recorded. A primary pathway contradiction would require a TCV™ pre-assessment signal to fire in a direction opposite to the APR evidence at the level of material implementation failure — for example, C-FAIR™ S1 = Low in a project that subsequently experienced severe physical climate disruption, or

ToC Group C confirmed positive when APR data established exclusive causal attribution. No such contradiction appears in any case in any year at the primary pathway level.

Where indicator-level exceptions appear — including boundary cases in which formula design, data availability, or attribution logic produced signals that did not correspond to APR observations in specific years — these are disclosed within the relevant case analyses and treated as diagnostic evidence for model refinement rather than primary structural failures. The consistency finding reported here is computed only across assessable cells; data-limited years and N/D cells are explicitly excluded from the denominator and documented in the individual case matrices.

This finding — no unresolved contradiction in material failure pathways across four continents, eight implementing entity types, and GCF Board approvals spanning B.11 through B.23 — is not a probabilistic claim. It is a structural observation: every material implementation constraint and failure pathway examined in this study corresponded to a pre-assessed structural signal or to a disclosed methodological boundary case. The framework did not produce a single misdirected primary structural diagnosis in any case where the diagnosis was testable.

Convergent External Verification.

Twelve of fifteen projects carry independent evaluation documents — Interim Evaluations, Mid-Term Reviews, Project Completion Reports, completion summaries, and independent impact evaluation documents — produced by evaluation teams with no knowledge of VERA-TCV™ pre-assessment. Across all twelve projects for which independent evaluation documents are available, independent evaluation findings are directionally consistent with VERA-TCV™ pre-assessment signals. The convergence is documented at three levels of precision. The twelve cases are listed in Appendix H, with document type and verification horizon.

At the most precise level, FP101's Mid-Term Review (2024) produced the statement "underperformance can be attributed to implementation failures due to an overestimation of delivery capacity" — the plain-language equivalent of VERA™ RA2 = 0.250, derived five years after the design-stage computation from independent evidence.

At the intermediate level, FP069's Interim Evaluation (2022) rated Sustainability as Moderately Unlikely — the operational confirmation of Problem-Solution Misalignment that ToC:f(x)™ Q5 = No identified in 2018.

At the broad level, FP039's eight-year PCR confirmed the Private-Public Bifurcation — Component 2 overperformance and Component 1.1 abandonment — that VERA™ RA2 and C-FAIR™ S2 independently identified from the 2017 design document.

In no case did an independent evaluation finding contradict the material structural pathway identified by VERA-TCV™ pre-assessment. In every case where independent evaluation language was sufficiently specific, it converged with the structural signal VERA-TCV™ had identified years earlier.

Framework Replicability.

FP026 (Madagascar) was independently evaluated in two separate VERA-TCV™ sessions — different evaluators, different AI extraction systems, different input data fidelity — at different points in time. Both sessions reproduced identical judgment directions across all four models. The VERA™ score diverged between versions (v1.0: 0.671 / v2.0: 0.865) due to a documented CI3 formula update, but the CR-Tier II classification was unaffected. ToC:f(x)™, C-FAIR™, and VORTA™ judgments were identical across sessions. The framework's output is deterministic given the input document — extractor-dependent interpretation bias does not alter the structural conclusions.

The combined evidence — 92% portfolio-weighted directional consistency, no unresolved material directional contradictions across assessable signal-year cells, convergent external verification from thirteen independent evaluation documents, and demonstrated replicability across independent extraction sessions — constitutes the quantitative foundation for the claim advanced in this report's title: that ex-ante structural diagnosis of climate project performance is possible, reliable, and verifiable across the portfolio scale and temporal horizon examined here.

This is not a claim of causal exclusivity. It is evidence of directional consistency under conditions where causal exclusivity is structurally unavailable. For GCF-scale multi-actor programmes, requiring exclusive attribution as the default verification standard would make large-scale climate finance unverifiable by design.

PART V — IMPLICATIONS

Diagnosis is not the end of the argument. It is the beginning of the obligation. If structural conditions are readable before implementation begins — and this report demonstrates that they are — then the question is no longer whether pre-approval diagnosis is possible. The question is what changes when it becomes standard. Part V translates the evidence into three structural requirements for GCF's approval and monitoring architecture: monetized benefit estimates as an approval condition, chronic and acute signal classification in APR design, and ToC Q5 as a mandatory pre-approval gate for adaptation projects in Hazard-dominant environments. Structure before execution. Diagnosis before capital. This is what that principle requires in practice.

PART V — IMPLICATIONS

11. What Ex-Ante Diagnosis Changes

The fifteen cases in this study span four continents and an average of 5.7 years of implementation. They were selected before their APR data was reviewed, evaluated against design documents alone, and mapped against independent evidence across time horizons long enough to test multi-year structural dynamics.

The result — 92% directional consistency, zero contradictions, convergent external verification — establishes one finding with precision: the structural conditions governing climate project performance are present in funding proposals at the point of Board approval. They are readable before implementation begins.

This finding has a specific implication that requires careful framing. It is not a claim that VERA-TCV™ can predict which projects will succeed. Success in complex climate programmes depends on factors outside any framework's scope — the severity and timing of climate events, political transitions, co-financer behavior, global commodity prices. What the framework establishes is which structural vulnerabilities are present at design stage and which failure pathways are structurally predictable before a dollar is committed. That is a different and more limited claim. It is also a more actionable one.

The question this section addresses is not whether the framework worked in retrospect. It did. The question is what would have changed if pre-approval structural diagnosis had been applied at the point of GCF Board consideration for these fifteen projects.

If ToC:f(x)™ Q5 = No had been identified before FP069's approval:

The 2018 funding proposal for FP069 was approved with a Theory of Change in which capacity-building and microfinance are the primary mechanisms for generating climate resilience in a Bangladesh coastal zone classified by C-FAIR™ as Hazard-dominant. Q5 = No was identifiable from that document. The question is structural and specific: does the primary solution architecture directly address the physical mechanism of harm — saline intrusion, cyclone destruction of assets, freshwater displacement? The answer from the design document is No. Capacity-building does not stop cyclones. Microfinance does not prevent saline intrusion.

Had Q5 = No been identified and acted on before approval, two design responses were structurally available. The first — ToC redesign anchoring primary outcomes in physical hazard reduction infrastructure, with capacity-building as a secondary layer — would have required additional capital allocation for embankments, permanent water infrastructure, or managed retreat provisions. The second — scope revision narrowing the project's claimed outcomes to the livelihood diversification benefits that capacity-building can realistically deliver, without claiming climate resilience as the primary outcome — would have preserved the project's achievable value while eliminating the structural misrepresentation of its impact.

Neither response was available post-approval without a Board-level restructuring. Q5 = No identified from the design document before approval would have created the decision point at which one of these responses could have been implemented at design cost rather than seven years of implementation cost. USD 33.0 million was committed and largely disbursed. USD 38,571 in total annual WLG income was generated at project close. The gap between those two numbers is the quantitative expression of what went structurally unfixed.

If VERA™ RA2 = 0.250 and C-FAIR™ S2 = Weak had been formally communicated before FP034's target-setting:

FP034's Uganda wetland restoration project set targets based on delivery rates that assumed institutional delivery capacity exceeding what RA2 = 0.250 and S2 = Weak predicted. The second disbursement tranche was unmet at Year 1 close. Overall progress reached 33% by Year 5. Wetland restoration closed at 72.5% of final target. These outcomes were structurally predictable from the 2016 design document — not as failures of the implementing team, but as the expected delivery ceiling in a three-ministry LDC coordination environment with documented procurement fragility.

Had RA2 = 0.250 and S2 = Weak been formally incorporated into target-setting at design stage, two adjustments were available. The first — recalibrating physical targets to RA2-adjusted delivery rates — would have produced targets achievable within the environmental constraint, reducing the gap between design expectations and implementation reality without reducing the project's climate value. The second — allocating ring-fenced capacity-building budget specifically for sub-national institutional infrastructure (procurement systems, fund release coordination, inter-agency protocols) — would have directly addressed the S2 = Weak structural constraint rather than treating it as an external given.

Both adjustments would have left FP034's core design intact. The Banker's Paradox — the worst environment justifying the strongest grant — would remain valid. The project's GRANT_JUSTIFIED financing structure would remain correct. What would have changed is the distance between what the project promised and what the environment could deliver. Closing that distance before approval is not a reduction in ambition. It is the condition for honest accountability.

The general principle these two cases establish:

Pre-approval structural diagnosis does not prevent Pattern A from occurring. Design–Environment Separation is the expected structural condition in GCF's operating context. What pre-approval diagnosis changes is the precision with which that separation is specified — and therefore the quality of the decisions made about financing structure, target calibration, capacity-building allocation, and ToC architecture before capital is committed.

For Pattern B (Internal Mechanism Failure), pre-approval diagnosis identifies singular causal dependencies before they become seven-year structural absences. FP026's EIB fund non-execution and FP115's PPA non-attainment were both identifiable from their design documents through IF2 signal analysis. Treating private partner activation as a binding pre-approval condition rather than a design assumption would have been the structural intervention.

For Pattern C (Measurement Gap), pre-approval diagnosis identifies RA3 = 0 and mandates monetized VBI baseline data as a condition of financing approval. FP039's eight-year VORTA™ VCS F was predetermined from RA3 = 0 in 2017. A measurement architecture requirement at design stage would have closed that gap before implementation began.

For Pattern D (Attribution Constraint), pre-approval diagnosis identifies Q5 = No as a mandatory redesign trigger. FP069's Problem-Solution Misalignment was readable from the 2018 design document. Pre-approval identification would have created the decision point for ToC redesign before USD 33.0 million was committed to a structurally misaligned solution architecture.

None of these interventions requires perfect information or perfect prediction. They require only the structural reading of design documents that VERA-TCV™ performs — and the institutional commitment to act on what that reading finds before approval rather than after deployment.

12. Verification Architecture Requirements

The findings of this study are not primarily a critique of the fifteen evaluated projects. They are a diagnosis of the structural gap between what GCF's approval architecture currently verifies and what climate project performance actually requires. Three requirements follow from the evidence.

Requirement 1 — Measurement design must be a condition of GCF approval, not a post-approval implementation expectation.

Twelve of fifteen projects record VORTA™ VCS F. In eleven of these twelve cases, the Grade F outcome was predictable from the design document — because the data required to compute social value was absent from the document at approval. SAP004 represents the sole exception: herder livelihood income data was present in the design document and VBI computation was possible, but the resulting Δ IRR of +0.74%p falls below the Grade A threshold — a structurally distinct cause. RA3 = 0 in FP039 was visible in 2017. The absence of monetized livelihood benefit data across the NbS adaptation cases was visible in 2015, 2016, and 2018. VCS F is not an evaluation verdict delivered after implementation. It is a measurement architecture verdict readable before implementation begins.

GCF's current logframe standard requires baseline, target, and means of verification for output and outcome indicators. It does not require monetized annual benefit estimates — the data that converts physical indicator achievement into verifiable social value. The consequence is systematic: projects are approved with measurement frameworks capable of reporting whether hectares were restored or households reached, but not whether the restoration or the household engagement generated social value sufficient to justify the cost. The gap between counting and valuing is architecturally embedded in the approval standard.

The requirement that follows is specific. Projects must include, as a condition of approval, monetized annual benefit estimates per beneficiary unit for their primary outcome indicators — expressed in USD per household per year, USD per tCO₂ per year, or equivalent monetized form appropriate to the project type. Where these estimates are genuinely unavailable at design stage, the design document must specify the MRV protocol by which they will be established in Years 1–2 of implementation, with a disbursement gate linked to protocol operationalization. VORTA™ VBI framework standards provide the domain structure for this requirement. Measurement architecture absence that produces VCS F should be treated as a structural deficiency addressable before approval — not a post-hoc finding disclosed after disbursement.

Requirement 2 — Chronic structural signals and acute event signals must be distinguished in GCF's monitoring architecture.

FP034 (Uganda) identifies a systematic gap in how GCF Annual Performance Reports capture structural constraint. VERA™ RA2 = 0.250 in LDC contexts operates as a chronic delivery ceiling — a persistent upper bound on implementation speed that expresses itself as multi-year cumulative progress rate divergence rather than a discrete annual event. The automatic VERA-TCV™ ×APR mapper detects RA2 as confirmed when a single triggering event occurs in a given year. When the constraint is chronic — present in every year, materialized as accumulated delay rather than acute disruption — the mapper registers Not Applicable in years without a discrete event, producing an underestimated directional match rate.

GCF's APR framework faces an analogous structural limitation. Annual progress ratings (On Track, Moderately Satisfactory, Delayed) capture year-over-year status but do not systematically distinguish between chronic structural constraints and acute event disruptions. A project rated Delayed in CY2021 due to COVID-19 and a project rated Delayed in CY2021 due to persistent sub-national procurement

fragility present different structural profiles requiring different responses — but APR narrative ratings alone do not capture that distinction.

The monitoring architecture requirement is a two-signal classification protocol. For each active structural risk signal in a project's VERA-TCV™ pre-assessment, the APR should distinguish whether the signal manifested as an acute event (single-year, bounded, potentially absorbable) or a chronic constraint (multi-year, cumulative, structural upper bound). Chronic constraint signals require different management responses than acute event signals: redesign of delivery mechanisms, capacity-building investment ring-fencing, or target recalibration. Acute event signals may require temporary adaptive management without structural redesign. Treating both as a single APR narrative entry obscures the distinction that determines the correct management response.

Requirement 3 — ToC Q5 pre-approval gate should be a mandatory structural checkpoint for adaptation projects in Hazard-dominant environments.

FP069 (Bangladesh) establishes the empirical case for a specific structural gate. ToC:f(x)™ Q5 asks whether the design document establishes that the project's primary solution architecture directly addresses the primary mechanism of harm. For FP069, the primary mechanism of harm is physical: cyclone hazard, saline intrusion, freshwater displacement. The primary solution architecture is capacity-based: WLG microfinance, saline-tolerant agriculture extension, RWHS installation. Q5 = No is readable from the design document in 2018.

The evidence across five APR years and two direct cyclone strikes establishes that Q5 = No in a Hazard-dominant environment is not a minor design imprecision — it is a structural predictor of Sustainability: Moderately Unlikely at the independent evaluation stage. USD 38,571 in total annual WLG revenue across 1,020 groups at project close is the quantitative confirmation.

The gate requirement is structural and specific. For all adaptation projects in C-FAIR™ Grade C Hazard-dominant environments, Q5 determination must be conducted as part of GCF's independent technical review before Board consideration. A Q5 = No determination in a Hazard-dominant environment should trigger a mandatory ToC revision requirement: the primary Theory of Change must anchor its core outcome pathway in physical hazard reduction infrastructure before capacity-building components are approved. The gate does not prevent capacity-building from being funded. It prevents capacity-building from being positioned as the primary climate resilience mechanism in contexts where physical climate events will repeatedly reset the gains that capacity-building produces.

These three requirements are not proposals for a new evaluation system. They are the minimum structural adjustments to GCF's existing approval and monitoring architecture that the evidence in this report identifies as necessary. The gap between what the framework currently verifies and what the evidence shows is required for outcome achievement is not a gap of ambition — GCF's ambition in deploying capital into Grade C environments is confirmed as correct by the Banker's Paradox across the portfolio. The gap is architectural. Measurement design, signal classification, and ToC structural gating are each within GCF's existing institutional authority to require. What the evidence shows is that requiring them would change, specifically and measurably, the structural quality of the projects that reach the Board for approval.

CONCLUSION

We are not asking whether money was spent. We are asking whether the chain held.

Across fifteen GCF-funded projects, spanning four continents, five thematic areas, and an average of 5.7 years of Annual Performance Report records, VERA-TCV™ generated ex-ante structural judgments from funding proposal documents alone. Those judgments were then mapped against actual implementation data, year by year, for a verification horizon of up to eight years.

The result: directional consistency rates of 76.9 to 100 percent across all evaluated cases. No unresolved contradiction in primary structural failure pathways was recorded across the evaluated portfolio. Convergent external verification from thirteen independent evaluation documents. Demonstrated replicability across independent extraction sessions.

The pattern of failure in climate finance is not random.

Twelve of fifteen projects record C-FAIR™ Grade C — the structural justification for concessional finance, not a warning to be avoided.

Twelve record VORTA™ VCS F — a classification that means six structurally different things, only one of which reflects genuine value inadequacy.

Not a single project achieved ToC:f(x)™ TIER I — not because the designs were deficient, but because complete causal exclusivity is structurally unavailable in multi-actor climate programmes operating under UNFCCC attribution constraints.

These patterns are consistent, they are structural, and they were readable before implementation began.

The three CR-Tier I projects — FP004 (Bangladesh), FP072 (Zambia) and FP073 (Rwanda) — do not succeed by operating in favorable environments. All three record RA2 below threshold. FP072 absorbed the worst drought in forty years and maintained 78.1% beneficiary coverage. FP073 deployed the portfolio's highest-authenticity design into one of its most capacity-constrained environments. CR-Tier I is not an environmental advantage. It is a design judgment quality — the precision with which a design team read the environment they would operate in and built the structural response before capital was committed. FP004 combined CR-Tier I design authenticity with complete market failure (IRRm = -100%) and VORTA™ VCS A — the only case in the portfolio where the highest design tier and the strongest grant justification coexist in the same project.

This is what ex-ante structural diagnosis changes. Not the environment. Not the climate. Not the political economy of the countries where these projects operate. What it changes is the quality of the structural decisions made before implementation begins — about financing architecture, target calibration, ToC logic, measurement design, and the precision of the match between what a project claims it will achieve and what the structural conditions of its design can actually deliver.

Structure before execution. Diagnosis before capital.

Not perfectly. But directionally, across every tested material pathway.

CONCLUSION

We are not asking whether money was spent. We are asking whether the chain held.

Across fifteen GCF-funded projects, spanning four continents, five thematic areas, and an average of 5.7 years of Annual Performance Report records, VERA-TCV™ generated ex-ante structural judgments from funding proposal documents alone. Those judgments were then mapped against actual implementation data, year by year, for a verification horizon of up to eight years.

The result: directional consistency rates of 76.9 to 100 percent across all evaluated cases. No unresolved contradiction in primary structural failure pathways was recorded across the evaluated portfolio. Convergent external verification from thirteen independent evaluation documents. Demonstrated replicability across independent extraction sessions.

The pattern of failure in climate finance is not random.

Twelve of fifteen projects record C-FAIR™ Grade C — the structural justification for concessional finance, not a warning to be avoided.

Twelve record VORTA™ VCS F — a classification that means six structurally different things, only one of which reflects genuine value inadequacy.

Not a single project achieved ToC:f(x)™ TIER I — not because the designs were deficient, but because complete causal exclusivity is structurally unavailable in multi-actor climate programmes operating under UNFCCC attribution constraints.

These patterns are consistent, they are structural, and they were readable before implementation began.

The three CR-Tier I projects — FP004 (Bangladesh), FP072 (Zambia) and FP073 (Rwanda) — do not succeed by operating in favorable environments. All three record RA2 below threshold. FP072 absorbed the worst drought in forty years and maintained 78.1% beneficiary coverage. FP073 deployed the portfolio's highest-authenticity design into one of its most capacity-constrained environments. CR-Tier I is not an environmental advantage. It is a design judgment quality — the precision with which a design team read the environment they would operate in and built the structural response before capital was committed. FP004 combined CR-Tier I design authenticity with complete market failure (IRR_m = -100%) and VORTA™ VCS A — the only case in the portfolio where the highest design tier and the strongest grant justification coexist in the same project.

This is what ex-ante structural diagnosis changes. Not the environment. Not the climate. Not the political economy of the countries where these projects operate. What it changes is the quality of the structural decisions made before implementation begins — about financing architecture, target calibration, ToC logic, measurement design, and the precision of the match between what a project claims it will achieve and what the structural conditions of its design can actually deliver.

Structure before execution. Diagnosis before capital.

Not perfectly. But directionally, across every tested material pathway.

APPENDICES

Appendix A	VERA™ v2.2: Structural Overview
Appendix B	ToC:f(x)™ SIM-v4.1: Structural Overview
Appendix C	C-FAIR™ v3.0: Structural Overview
Appendix D	VORTA™ v2.5: Structural Overview
Appendix E	Project Universe: 15 Selected Projects
Appendix F	APR Extraction Protocol
Appendix G	Individual Structural Validation Reports: Access Protocol
Appendix H	Predictive Validity Matrix

Appendix A — VERA™ v2.2: Structural Overview

VERA™ examines whether a project's stated objectives, timelines, and delivery claims are genuinely grounded in physical, institutional, and technical reality. The model operates across three axes.

The Reality Anchor axis (RA) assesses temporal plausibility, system compatibility, and transformation traceability. The single most consequential indicator across the evaluated portfolio is RA2 — the compound score of policy, infrastructure, and technology compatibility — which functions as the primary delivery capacity constraint in every case where it registers below threshold.

The Causal Integrity axis (CI) examines whether logical connections between design elements are internally consistent and structurally specified. The Logic Drift indicator (CI3) measures the expected rate of causal chain degradation over the project's implementation horizon.

The Intent Fidelity axis (IF) examines whether the motivational structure of the project is genuinely oriented toward the stated climate objective. It detects configurations in which stated intent and structural incentive diverge.

VERA™ produces a composite authenticity score and a three-tier classification (CR-Tier I through CR-Tier III). CR-Tier I reflects designs in which timeline realism, system compatibility, risk architecture completeness, and intent fidelity are all confirmed at threshold. CR-Tier III reflects designs where one or more structural conditions are either below threshold or unmeasurable within the framework's protocol.

Detailed indicator definitions, scoring logic, and tier boundary conditions are available upon request for peer review purposes.

Appendix B — ToC:f(x)™ SIM-v4.1: Structural Overview

ToC:f(x)™ examines whether the activities as designed are causally sufficient to produce the claimed outputs, and whether the logical chain from those outputs to stated outcomes is structurally coherent rather than narratively asserted.

The model applies twenty structural questions organized into three groups. Group A questions address explicit presence — whether necessary resources, activities, and causal mechanisms are named with sufficient specificity to constitute structural evidence. Group B questions address structural demonstration — requiring a minimum of two independent structural indicators before a positive determination is made. Group C questions address causal exclusivity — whether the document establishes that the claimed outcome depends on the project's specific outputs rather than being achievable through parallel pathways.

Group C operates under an Unknown default. A positive determination requires explicit exclusivity language in the document, not inference from narrative plausibility. The finding that no project in the evaluated portfolio achieved ToC:f(x)™ TIER I reflects the structural reality that complete causal exclusivity is rarely documentable in multi-actor climate programmes — not a deficiency of individual project designs.

For the purpose of this report, the full ToC:f(x)™ architecture is reduced to two evaluative boundaries: $f_1(\text{Input, Activity}) = \text{Output}$ and $f_2(\text{Output}) = \text{Outcome}$. Each boundary has structural conditions for validity that are either present or absent in the design document.

Full Q1–Q20 question architecture, group weighting, and TIER classification logic are available upon request for peer review purposes.

Appendix C — C-FAIR™ v3.0: Structural Overview

C-FAIR™ examines whether the environment into which project outputs will be delivered is capable of converting those outputs into outcomes. Three dimensions are assessed.

Hazard Exposure (S1) quantifies the physical climate risk profile across flood, drought, cyclone, heat wave, and sea level rise dimensions. High-hazard environments face structurally elevated delivery risk regardless of design quality, because the environment introduces exogenous shocks the design cannot fully absorb.

Adaptive Capacity (S2) assesses institutional and infrastructural conditions — infrastructure resilience, disaster response capacity, and institutional capacity. S2 weakness is the most consistent structural signal in the evaluated portfolio, appearing in every case regardless of sector, region, or implementing entity.

Macro-Financial Fragility (S3) assesses sovereign debt stress, foreign exchange volatility, political stability, and financial system maturity.

The three dimensions combine to produce a Grade A through E classification and a recommended grant ratio range. The Banker's Paradox — embedded in the model's design — holds that high environmental fragility generates both the greatest need for climate finance and the strongest structural justification for grant-heavy financing structures.

Dimension-level indicator sets, composite scoring method, grade boundary conditions, and grant ratio derivation logic are available upon request for peer review purposes.

C-FAIR™ country-level assessments integrate data from five independent international databases: the Notre Dame Global Adaptation Initiative Country Index (ND-GAIN) for adaptive capacity and readiness; the Fragile States Index (Fund for Peace) for institutional and social cohesion; the Global Peace Index (Institute for Economics and Peace) for political stability and conflict exposure; World Bank GNI per capita classifications for macrofinancial baseline; and the ND-GAIN Readiness sub-index for economic and governance readiness.

Country scores are computed through a proprietary integration protocol that combines these sources across S1 (Hazard Exposure), S2 (Adaptive Capacity), and S3 (Macro-Financial Fragility) dimensions. Dimension-level indicator sets, composite scoring method, grade boundary conditions, and the country-specific calibration values derived from the 303-project GCF portfolio baseline remain protected intellectual property of ENHANCE Institute and are available to qualified institutional reviewers upon request.

Appendix D — VORTA™ v2.5: Structural Overview

VORTA™ examines whether the outcomes a project claims to produce, if achieved, generate social value sufficient to justify the cost of the intervention.

The model quantifies social value through a Value-Based Impact framework across three layers: the Output layer captures value generated directly by project deliverables; the Outcome layer captures value from the behavioral and economic changes that outputs induce; the Impact layer captures value at the level of social systems. The model disaggregates the internal rate of return into IRR_m (market return), IRR_v (value-inclusive return incorporating quantified social value), and IRR_c (composite return). The spread Δ IRR between IRR_m and IRR_v is the quantitative expression of the public finance justification.

VCS Grade F does not mean a project generated no value.

It means either that the value generated was architecturally unmeasurable from the design document — because the data required to populate the VBI framework was absent at the point of approval — or that the VBI computation was possible but the resulting social premium fell below the Grade A threshold. SAP004 (Mongolia) represents the latter: VBI data present, computation executed, Δ IRR = +0.74%p below threshold.

The disaggregation of Grade F outcomes by IRRm sign (GRANT_JUSTIFIED vs REDESIGN_REQUIRED) is the operative instrument for financing architecture decisions.

VBI domain definitions, IRR decomposition methodology, VCS grade boundary conditions, and DO SVA_WARNING trigger logic are available upon request for peer review purposes.

Appendix E — Project Universe: 15 Selected Projects

E.1 Selection Criteria

The fifteen projects evaluated in this study were selected from GCF's approved portfolio using three criteria applied in combination. No project satisfying all three criteria was excluded.

APR Traceability Depth. Each project required a minimum of four complete Annual Performance Reports in the public domain, with Section 2 outcome indicator data accessible for at least three of those years. Projects where APR Section 2 was systematically withheld or classified for all available years were excluded.

Outcome-Pathway Verifiability. Each project required a logframe with at least two trackable indicators connecting outputs to outcomes — enabling assessment of whether the causal chain held, not merely whether activities were completed. Projects with output-only logframes carrying no measurable outcome indicators were excluded.

Structural Heterogeneity. The portfolio was deliberately constructed to maximize diversity across sector, region, income group, implementing entity type, financing structure, and GCF Board approval cohort. Recurring structural signals across maximally heterogeneous cases constitute stronger evidence of portfolio-level pattern than similar signals across comparable cases.

E.2 Portfolio Composition

Project	Country	Region	GCF Classification	AE Type	Status	APR Years	Eval. Doc
FP001	Peru	LAC	Adaptation	National DAE	Completed	7	PCR
FP004	Bangladesh	Asia-Pacific	Adaptation	Bilateral DFI	Active	5	MTR
FP009	El Salvador	LAC	Mitigation	MDB (Private)	Completed	4*	—
FP017	Chile	LAC	Cross-cutting	MDB (Public)	Completed	5	FER
FP019	Ecuador	LAC	Cross-cutting	UN Agency	Completed	6	—
FP026	Madagascar	Africa	Cross-cutting	NGO + MDB	Active	7	IEU LORTA
FP034	Uganda	Africa	Adaptation	UN Agency	Active	7	MTE
FP039	Egypt	Africa	Mitigation	MDB (Private)	Completed	8	PCR
FP042	Morocco	Africa	Adaptation	Bilateral DFI	Active	6	IE
FP069	Bangladesh	Asia-Pacific	Adaptation	UN Agency	Active	5	IE
FP072	Zambia	Africa	Adaptation	UN Agency	Active	7	IE + MTR

Project	Country	Region	GCF Classification	AE Type	Status	APR Years	Eval. Doc
FP073	Rwanda	Africa	Adaptation	National DAE	Active	5	—
FP101	Belize	LAC	Adaptation	UN Agency	Active	5	MTR
FP115	Chile	LAC	Cross-cutting	Commercial Bank	Cancelled	4†	PCR
SAP004	Mongolia	Asia-Pacific	Adaptation	Multilateral	Active	4	MTR

*CY2021 missing; CY2023–2024 Section 2 confidential. †Includes PCR following formal cancellation at B.34.

Abbreviations: PCR = Project Completion Report; MTR = Mid-Term Review; MTE = Mid-Term Evaluation; IE = Interim Evaluation; FER = Final Evaluation Report; IEU LORTA = Independent Evaluation Unit Longitudinal Randomized Treatment Assessment.

E.3 Regional and Thematic Distribution

Continental Distribution:

- ✓ Africa (6): FP026, FP034, FP039, FP042, FP072, FP073.
- ✓ Latin America and the Caribbean (6): FP001, FP009, FP017, FP019, FP101, FP115.
- ✓ Asia-Pacific (3): FP004, FP069, SAP004.

Thematic Distribution:

- Ecosystems and NbS: FP001, FP019, FP026, FP034, FP073
- Food and Water Security: FP042, FP069, FP072, SAP004
- Energy and Decarbonization: FP017, FP039, FP115
- Rural Infrastructure and Resilience: FP004, FP101
- Financial Mechanism Design: FP009

GCF Board Approval Cohort: B.11 (1 project) through B.23 (1 project), spanning a four-year approval window from 2015 to 2019.

Verification Horizon: Four years (SAP004) to eight years (FP039).

Portfolio average: 5.7 years.

E.4 Implementing Entity Distribution

AE Type	Projects	Count
UN Agencies (UNDP, IFAD)	FP034, FP069, FP072, FP101	4
Bilateral DFIs (KfW, AFD, IDB)	FP004, FP042	2
MDB Private Windows (EBRD, IDB Invest)	FP009, FP039	2
MDB Public Window (IDB)	FP017	1
National DAEs (PROFONANPE, Rwanda MoE)	FP001, FP073	2
NGO + MDB Co-implementation	FP026	1
Commercial Bank (MUFG)	FP115	1
Multilateral (Small Access)	SAP004	1
UN Agency (UNDP Ecuador)	FP019	1

E.5 Financing Structure Distribution

Structure Type	Projects	GCF Grant Ratio Range
Pure grant	FP001, FP019, FP034, FP069, FP073, SAP004	85–100%
Grant-dominant concessional	FP004, FP026, FP042, FP072, FP101	30–60%
Private sector EE credit line	FP009	14.2%
MDB blended finance	FP039	5.5%
Commercial RE infrastructure	FP017, FP115	<15%

Appendix F — APR Extraction Protocol

F.1 Protocol Purpose and Scope

The APR Extraction Protocol governs the process by which Annual Performance Report data is translated into signal-year cell determinations within the VERA-TCV™×APR consistency matrix. The protocol is designed to eliminate single-analyst interpretation bias as the primary source of non-replicability — the condition under which two analysts processing the same APR would reach different signal determinations.

The protocol applies to all fifteen projects evaluated in this study. It covers signal identification, confirmation threshold logic, N/D cell classification, and partial confirmation treatment. It does not govern VERA-TCV™ pre-assessment scoring, which is covered under separate model protocols (see Appendix A–D summaries).

F.2 Multi-Agent Parallel Extraction Architecture

Identical field sets are extracted from the same APR source document by three independent AI-assisted extraction systems operating under standardized protocols. Each system processes the document independently, with no cross-reference permitted prior to convergence. The three systems operate on distinct large language model architectures, each independently optimized for structured document extraction.

Following independent extraction, field-level Structured Convergence Integration (SCI) rules are applied. SCI determines whether extraction outputs from the three systems converge on a single determination or diverge. Convergence is defined as two or more systems producing identical field values within the specified tolerance band.

Where divergence persists after SCI processing, Divergence Resolution Protocol (DRP) procedures are applied. DRP requires the operator to identify the source document passage on which the divergence is based, document the interpretive ambiguity, and record the resolution rationale. All DRP resolutions are preserved in the audit record for each evaluation.

Detailed SCI field-level rules, tolerance band specifications, DRP decision tree, and audit record format are available upon request for peer review purposes.

F.3 Signal Classification Standards

Each active VERA-TCV™ pre-assessment signal is mapped against available APR data for each implementation year using the following classification:

Symbol	Classification	Determination Criteria
✓	Confirmed	APR data establishes that the pre-assessed structural condition materialized in the direction predicted
△	Partial	The predicted condition materialized in diminished form; direction correct, intensity attenuated
—	Not Applicable	Signal not active in this year; precondition for signal activation not met
×	Contradicted	APR data establishes that outcomes moved in a direction opposite to the pre-assessed signal
N/D	No Data Available	APR data for this year is unavailable, incomplete, or classified

A confirmed signal (✓) establishes that the structural condition was real and consequential. A partial signal (△) contributes 0.5 to the consistency rate numerator. A contradicted signal (×) contributes 0 to the numerator and is disclosed in the case analysis with full documentation of the contradiction. N/D cells are excluded from both numerator and denominator in consistency rate computation.

F.4 Acute vs Chronic Signal Logic

Two signal types require distinct confirmation logic.

Acute signals are event-triggered: a single APR year's data establishes confirmation or contradiction. C-FAIR™ S1 (Hazard Exposure) is the primary acute signal. Confirmation requires documented evidence of a physical climate event — cyclone strike, flood, drought — in the relevant APR year, corresponding to the pre-assessed hazard profile.

Chronic signals are cumulative: confirmation requires evidence of a persistent structural constraint manifesting across multiple implementation years. VERA™ RA2 in LDC contexts and C-FAIR™ S2 in multi-ministry coordination environments are the primary chronic signals in this portfolio. The automatic mapper applies a chronic signal correction condition: RA2 chronic confirmation is triggered in any year where cumulative overall progress is below 50% and three or more implementation years have elapsed, regardless of whether a discrete triggering event occurred in that specific year.

FP034 (Uganda) is the boundary case that empirically establishes the distinction between acute and chronic signal logic. Its 76.9% automatic mapper rate rises to 80.8% under chronic signal correction — demonstrating that the two signal types require separate mapper logic for accurate consistency rate computation.

F.5 Directional Consistency Rate Computation

Consistency rate per project = $(\text{Confirmed } \checkmark + \text{Partial } \triangle \times 0.5) \div \text{Assessable cells}$

Assessable cells = Total signal-year cells – N/D cells – Not Applicable (–) cells

Portfolio-weighted average = $\Sigma (\text{project consistency rate} \times \text{project assessable cell count}) \div \Sigma (\text{all assessable cells across portfolio})$

The denominator excludes both N/D cells and Not Applicable (–) cells. N/D cells represent access limitations: APR data for the relevant year is unavailable, incomplete, or classified. Not Applicable cells represent structural determinations: the signal precondition was not met in that year, or the signal

category is structurally inapplicable to the project type. Neither category is treated as confirming, partially confirming, or contradicting, and both are excluded from numerator and denominator in consistency rate computation.

Appendix G — Individual Structural Validation Reports: Access Protocol

Fifteen individual Structural Integrity Validation Reports (ENH-SRV series) were produced for each evaluated project using the VERA-TCV™ four-model framework. Each report contains the full VERA-TCV™ pre-assessment scorecard, annual APR signal mapping matrices, integrated consistency analysis, and cross-case comparison.

These reports are not reproduced in full in this publication. Full ENH-SRV documentation is available to qualified reviewers — including GCF accredited entities, independent evaluation bodies, and academic institutions — upon written request, subject to the following conditions.

Access conditions:

Requests must identify the requesting institution and the specific analytical purpose for which access is sought. Access is granted for independent verification, peer review, and academic research purposes. Redistribution or reproduction of ENH-SRV content for commercial purposes requires separate written authorization from ENHANCE.

Available report series:

Report Code	Project	Country	Version
ENH-SRV-FP001-2026	Adaptation — Ecosystems	Peru	v1.0
ENH-SRV-FP004-2026	Adaptation — Infrastructure	Bangladesh	v1.0
ENH-SRV-FP009-2026	Mitigation — EE Finance	El Salvador	v1.0
ENH-SRV-FP017-2026	Cross-cutting — Solar RE	Chile	v1.0
ENH-SRV-FP019-2026	Cross-cutting — NbS	Ecuador	v1.0
ENH-SRV-FP026-2026	Cross-cutting — Landscapes	Madagascar	v2.0
ENH-SRV-FP034-2026	Adaptation — Wetlands	Uganda	v1.0
ENH-SRV-FP039-2026	Mitigation — RE Finance	Egypt	v1.0
ENH-SRV-FP042-2026	Adaptation — Water	Morocco	v1.0
ENH-SRV-FP069-2026	Adaptation — Coastal	Bangladesh	v1.0
ENH-SRV-FP072-2026	Adaptation — Agriculture	Zambia	v1.0
ENH-SRV-FP073-2026	Adaptation — NbS	Rwanda	v1.0
ENH-SRV-FP101-2026	Adaptation — Rural	Belize	v1.0
ENH-SRV-FP115-2026	Cross-cutting — PSH	Chile	v1.0
ENH-SRV-SAP004-2026	Adaptation — Livelihoods	Mongolia	v1.0

Requests may be directed to ENHANCE via the contact information provided on the cover page.

FP069 · Bangladesh
Enhancing Adaptive Capacities of Coastal Communities
Public Excerpt

Document Code: ENH-SRV-FP069-2026 (Public Excerpt)

Protocol: VERA™ v2.2 · ToC:f(x)™ SIM-v4.1 · C-FAIR™ v3.0 · VORTA™ v2.5

Classification: Partially Disclosed — ENHANCE Institute, May 2026

Sections 2, 3, and 4 (quantitative outputs) are restricted.

Full document available under NDA: contact@enhance-institute.org

ENH-SRV-FP069-2026 (Public Excerpt)**Structural Integrity Validation Report — Public Excerpt**

Document Code: ENH-SRV-FP069-2026 Date: April 23, 2026 | Version: v1.0 (Public Excerpt)

Classification: **Partially Disclosed — ENHANCE Institute, May 2026**

Country / Project: Bangladesh | Enhancing Adaptive Capacities of Coastal Communities AE / Board: UNDP | B.19/12 Protocol: VERA™ v2.2 / ToC:f(x)™ SIM-v4.1 / C-FAIR™ v3.0 / VORTA™ v2.5

Restricted Sections: Section 2 (TCV™ Pre-Assessment Scores and Model Outputs), Section 3 (SCI Convergence Summary), Section 4 (Quantitative Mapping Tables). Full document available to qualified institutional reviewers under NDA: contact@enhance-institute.org

The four models comprising the VERA-TCV™ framework — VERA™, ToC:f(x)™, C-FAIR™, and VORTA™ — operate as independent deterministic engines. Each model produces outputs through fixed structural logic and standardised input protocols. These outputs are not retrofitted to match known implementation outcomes. APR outcomes in this report are presented only as independent ex-post comparators.

Section 1 — Key Finding

The structure was misaligned from the start. An infrastructure problem was met with a capacity-building response.

ENHANCE's VERA-TCV™ four-model diagnosis generated ex-ante structural judgments based on GCF FP069's design document in 2018 — two years before implementation data was available. This report maps actual implementation data from CY2020 through CY2024 Annual Performance Reports and the 2022 Interim Evaluation against VERA-TCV™ pre-assessment signals on a year-by-year basis.

ToC:f(x)™'s Q5 = No judgment — structural benefit alignment unconfirmed — was confirmed by the 2022 Interim Evaluation's Unsatisfactory ratings for Outcome 1 (Livelihoods) and Outcome 3 (Institutional). C-FAIR™'s Hazard-dominant (S1 = High) warning was confirmed repeatedly through Super Cyclone Amphan (2020) and Cyclone Remal (2024). VORTA™'s VCS F / TIER V assessment was empirically confirmed by CY2024 data recording 1,020 Women's Livelihood Groups with total annual revenue of USD 38,571 — USD 1.52 per woman per year.

The most critical finding is the Sustainability judgment. The 2022 Interim Evaluation assigned Moderately Unlikely — the direct operational consequence of the Problem-Solution Misalignment VERA-TCV™ identified in the design document: a physical climate infrastructure problem (saline intrusion, freshwater scarcity) addressed with capacity-building and microfinance responses. The VORTA™ DO SVA_WARNING — Debt Transfer Risk — materialized in CY2024 when Cyclone Remal destroyed productive assets and RWHS installations of 492 households, confirming that debt structures in Hazard-dominant environments generate structural debt transfer risk at each climate event.

Section 2 — VERA-TCV™ Pre-Assessment Summary

[Restricted — Available under NDA]

This section contains model-level structural judgments, indicator scores, and quantitative outputs generated from FP069's 2018 design document across all four VERA-TCV™ models. Full disclosure

available to qualified institutional reviewers.

Contact: contact@enhance-institute.org

Section 3 – SCI Convergence Summary

[Restricted – Available under NDA]

This section documents the three-agent extraction convergence record, divergence flags, and operator-resolved ambiguities for FP069. Full disclosure available under NDA.

Contact: contact@enhance-institute.org

Section 4 – Annual VERA-TCV™ × APR Mapping

Quantitative mapping tables restricted. Structural narrative by year provided below.

Signal legend: ✓ Confirmed / △ Partial / – Not Applicable / × Contradicted

CY2020 – Year 3 | COVID-19 + Super Cyclone Amphan

Super Cyclone Amphan struck the project area directly in May 2020, simultaneous with COVID-19 field suspension. C-FAIR™ S1 High (Hazard-dominant) materialized in its most concentrated form: compound climate and pandemic shock in a single year. The Bangladesh coastal zone generated exactly the compound and repeated hazard exposure that C-FAIR™ had identified from the 2018 design document. Community engagement halted. Beneficiary selection and WLG formation significantly delayed.

Sub-conclusion: The compound shock pattern confirms the structural vulnerability C-FAIR™ pre-assessed – this is the environment in which a capacity-building response is structurally mismatched with a physical infrastructure problem.

CY2021 – Year 4 | Slow Recovery

Implementation pace remained significantly below annual targets under post-shock recovery conditions. Community engagement restarted; field access partially restored. RWHS and saline-tolerant agriculture activities commenced at limited scale.

Sub-conclusion: Recovery from compound climate and pandemic shocks in a Bangladesh coastal LDC environment is structurally slow. C-FAIR™ S2 operates as a chronic ceiling on recovery speed – not only on initial delivery. The Problem-Solution Misalignment remains structurally operative: capacity-building activities commenced, but the physical climate hazard environment continued to set the binding constraint on community livelihood outcomes.

CY2022 – Year 5 | Interim Evaluation – Sustainability Verdict

The 2022 Interim Evaluation was completed by an independent evaluation team with no knowledge of VERA-TCV™ pre-assessment. Key findings:

- Outcome 1 (Livelihoods): **Unsatisfactory**
- Outcome 3 (Institutional): **Unsatisfactory**
- Sustainability: **Moderately Unlikely**
- Output achievement rate: below 28% across key physical targets
- Project extension approved

Sub-conclusion: CY2022 is the decisive year of VERA-TCV™ pre-assessment confirmation. The Interim Evaluation's Sustainability: Moderately Unlikely is the independent ex-post verdict on what VERA-TCV™ identified from the design document in 2018: a physical climate infrastructure problem cannot

be sustainably addressed through capacity-building and microfinance. Q5 = No, C-FAIR™ S1 High, and VORTA™ VCS F / TIER V all pointed to this outcome before a single APR was written. The Interim Evaluation confirms these judgments as operational reality — four years after the pre-assessment, from a structurally independent evidence base.

CY2023 — Year 6 | Disbursement 58% | 80% On Track

Disbursement reached 58%; overall implementation rated 80% On Track. WLG formation and training activities accelerated. RWHS installations progressing. Output delivery improving from cumulative low base.

Sub-conclusion: Improved disbursement and On Track ratings indicate the delivery mechanism functioning more effectively. However, ToC Q5 = No remains structurally operative. WLG income levels and RWHS installations address symptoms, not the physical climate infrastructure cause. The acceleration of outputs does not resolve the Problem-Solution Misalignment — it demonstrates that the misaligned solution can be efficiently deployed. Efficiency of delivery does not substitute for structural alignment of design.

CY2024 — Year 7 | Cyclone Remal | D0 SVA_WARNING Confirmed

Cyclone Remal struck the project area directly. 492 households lost productive assets and RWHS installations while micro-credit obligations remained outstanding. 1,020 Women's Livelihood Groups recorded total annual revenue of USD 38,571 — USD 1.52 per woman per year. Project re-extended.

Sub-conclusion: CY2024 is the definitive empirical confirmation year. Cyclone Remal simultaneously confirms C-FAIR™ S1 High, VERA™ RA2 delivery gains erased by cyclone reset, and VORTA™ D0 SVA_WARNING in its most direct form: 492 households lost income-generating assets while debt obligations persisted. USD 38,571 total annual WLG revenue across 1,020 groups is the quantitative confirmation of VORTA™ VCS F / TIER V. Problem-Solution Misalignment is confirmed at project close: the capacity-building and microfinance response cannot withstand the physical climate hazard it was designed to address.

5. Integrated Consistency Matrix

Signal legend: ✓ Confirmed / Δ Partial / — Not Applicable / × Contradicted

Signal	Description	2020	2021	2022	2023	2024
VERA™ RA2	Delivery capacity gap	✓	✓	✓	Δ	✓
C-FAIR™ S1	Climate hazard exposure	✓	Δ	Δ	Δ	✓
C-FAIR™ S2	Institutional capacity	✓	✓	✓	Δ	✓
ToC Group C	Outcome attribution	—	—	✓	✓	✓
VORTA™ D0	Beneficiary debt risk	—	—	Δ	Δ	✓

Directional Match	Score (✓)	Score (incl. Δ)	Contradictions
21 / 25 assessed cells	56%	84% (incl. partial)	0 (zero contradictions recorded)

All five TCV™ signals were confirmed or partially confirmed across the five-year APR period. No directional contradictions (×) were recorded in any signal in any year. VORTA™ D0 SVA_WARNING

progressed from Partial (△) in CY2022–2023 to full Confirmation (✓) in CY2024 through Cyclone Remal — the first instance in the evaluation portfolio where D0 SVA_WARNING was fully confirmed by a direct climate event destroying beneficiary assets with outstanding debt obligations. ToC Group C was confirmed from CY2022 onward through the Interim Evaluation's Unsatisfactory Outcome ratings — the independent verification of Q5 = No.

Section 6 — Conclusions

Overall Judgment

FP069 is the clearest case of Problem-Solution Misalignment in the evaluated portfolio. ToC:f(x)TM Q5 = No identified in 2018 that the causal chain from capacity-building and microfinance outputs to climate-resilient livelihoods was structurally unconfirmed. C-FAIRTM S1 High identified that the Bangladesh coastal zone generates repeated and compounding climate shocks. VORTATM D0 SVA_WARNING identified that micro-credit in this environment creates structural debt transfer risk at each climate event. These three signals, generated independently from the 2018 design document, were confirmed across every APR year from CY2020 through CY2024.

The 2022 Interim Evaluation's Sustainability: Moderately Unlikely is the independent structural confirmation of what VERA-TCVTM pre-assessed four years earlier. USD 1.52 per woman per year in WLG income is not a measurement gap — it is the quantitative expression of a solution that cannot generate climate resilience at the scale the physical problem requires. Zero directional contradictions were recorded across all five APR years.

Structural Recommendations

First. GCF coastal adaptation projects in S1 High (Hazard-dominant) environments must anchor their primary Theory of Change in physical infrastructure outputs with direct hazard reduction logic. Capacity-building and livelihood diversification are structurally appropriate as secondary components — not as primary mechanisms expected to generate climate resilience independently. ToC Q5 = No is the design-stage signal that identifies this structural inversion before approval.

Second. In C-FAIRTM Grade C / Hazard-dominant environments, any micro-credit or beneficiary lending component must be structured with explicit debt relief pathways triggered by declared climate events. D0 SVA_WARNING is now a permanent gate in VORTATM v2.5. Future projects with beneficiary lending in Grade C or higher environments should include automatic debt suspension clauses linked to defined hazard events.

Third. VORTATM VCS F driven by E0 FAIL and S0 FAIL should be treated as a pre-approval structural redesign signal, not a post-approval disclosure item. Future projects with dual E0/S0 FAIL should require ToC redesign as a condition of GCF approval.

Full Structural Integrity Validation Report (ENH-SRV-FP069-2026) is available to qualified institutional reviewers under ENHANCE Institute's standard NDA protocol.

Contact: contact@enhance-institute.org / Subject: SRV Access Request — FP069

Appendix H — Predictive Validity Matrix

Reported Rate Convention

Directional consistency rates in this study are reported in two forms where methodologically relevant: the unadjusted rate produced by the automatic signal mapper, and the adjusted rate produced after application of documented correction logic. Both forms are reported transparently; neither replaces the other.

The following conventions apply uniformly across all fifteen projects.

Confirmed (✓) cells receive full credit (1.0) in the consistency rate numerator. Partial (!) cells receive 0.5 credit under the portfolio-wide convention, except where project-specific footnotes indicate otherwise. Disconfirmed (✗) cells receive zero credit and are included in the denominator in all cases. Not Applicable (—) cells are excluded from both numerator and denominator; they represent structural determinations, not data gaps. No Data Available (N/D) cells are excluded from both numerator and denominator; they represent access limitations, not framework prediction failures.

No cell is excluded from the denominator to improve reported accuracy. Where project-specific adjustments are applied — including chronic signal correction logic (FP034), data-limitation classification (FP009), partial credit rule (FP019), and boundary-condition methodology limitation (FP101) — the basis for each adjustment is documented in the relevant project footnote in Section H.3.

Portfolio-weighted average

$$= \sum (\text{project consistency rate} \times \text{project assessable cell count}) \div \sum (\text{all assessable cells across portfolio}).$$

The portfolio rate reported in this study (~92%) reflects conservative rounding and is derived from the project-level figures recorded in Section H.3.

H.1 Purpose

This appendix provides the quantitative foundation for the *directional consistency rates* reported in Section 10. It documents the total cell count, assessable cell count, N/D exclusion basis, signal type classification, and consistency rate computation for each evaluated project. The portfolio-weighted average of approximately 92% is derived from the project-level figures recorded here.

H.2 Signal Type Classification by Project

Project	Primary Signals Active	Signal Types	Notes
FP001	VERA RA2, C-FAIR S2, ToC Group C	Chronic × 3	7-year horizon
FP004	VERA RA2, C-FAIR S1/S2, ToC Group C	Chronic × 3, Acute × 1	5-year horizon
FP009	C-FAIR S2, ToC Q10, VERA RA2	Chronic × 2, Structural × 1	Data-limited: 50% N/D
FP017	VERA RA2, C-FAIR S2, ToC Group C	Chronic × 2, Structural × 1	5-year horizon; FER confirmed
FP019	ToC Q7/Q8/Q10/Q11, C-FAIR S2	Structural × 4, Chronic × 1	Attribution constraint
FP026	ToC Group C, VERA RA2, C-FAIR S1/S2	Structural × 1, Chronic × 2, Acute × 1	7-year; LORTA cross-validated
FP034	VERA RA2, C-FAIR S1/S2, ToC Group C	Chronic × 3, Acute × 1	Chronic correction applied
FP039	VERA RA2, C-FAIR S1/S2, ToC Q10/Q11	Chronic × 2, Acute × 1, Structural × 2	8-year horizon

Project	Primary Signals Active	Signal Types	Notes
FP042	VERA RA2, C-FAIR S2, ToC Group C	Chronic × 3	Output–Outcome disconnect
FP069	ToC Q5, C-FAIR S1, VERA RA2, VORTA D0	Structural × 2, Acute × 1, Chronic × 1	Band chart case
FP072	VERA RA2/IF2, C-FAIR S1/S2, ToC Group C	Chronic × 2, Acute × 2, Structural × 1	100% across 7 full years
FP073	VERA RA2, C-FAIR S2, ToC Group C	Chronic × 3	CR-Tier I; 5-year horizon
FP101	VERA RA2, C-FAIR S1/S2, ToC Group C	Chronic × 3, Acute × 1	RA1 boundary case disclosed
FP115	VERA IF2, C-FAIR S2, ToC Group B	Structural × 2, Chronic × 1	Cancelled; PCR confirmed
SAP004	VERA RA2, C-FAIR S2, ToC Group C	Chronic × 3	4-year horizon; MTR confirmed

H.3 — Directional Consistency Rate Summary

Rate = (Confirmed  + 0.5 × Partial ) / Assessable cells | N/D and — cells excluded from denominator |  cells included in denominator, excluded from numerator

Signal legend:  Confirmed /  Partial / — Not applicable /  Disconfirmed / N/D No data available

Project	Assessable	Confirmed 	Partial 		Rate
FP001 Peru	24	19	0	0	79.2%
FP004 Bangladesh	13	11	2	0	92.3%
FP009 El Salvador	5	4	1	0	80.0%*
FP017 Chile	7	7	0	0	100%
FP019 Multiple	16	9	7	0	78.1%‡
FP026 Madagascar	47	46	1	0	98.9%§
FP034 Multiple	26	20	0	1	76.9% / 80.8%¶
FP039 Multiple	32	32	0	0	100%
FP042 Multiple	18	18	0	0	100%**
FP069 Bangladesh	25	14	7	0	84%††
FP072 Multiple	28	28	0	0	100%
FP073 Multiple	18	18	0	0	100%
FP101 Multiple	32	26	2	6‡‡	87.5%
FP115 Multiple	16	16	0	0	100%§§
SAP004 Mongolia	10	7	3	0	85.0%
Portfolio (15 projects)	313	282	16	6	~92%

Methodology Note

The directional consistency rate measures whether each VERA-TCV™ pre-assessment signal fired in the correct direction relative to ex-post APR evidence. A signal is scored as Confirmed () when APR data provides direct

quantitative or qualitative validation; Partial (Δ) when directional alignment is present but full quantitative confirmation is unavailable; Disconfirmed ($\times$) when APR evidence contradicts the pre-assessed signal direction; and Not Applicable ($-$) when the signal category is structurally inapplicable to the project type. N/D indicates years for which no APR data was available. Partial cells receive 0.5 credit in the rate calculation unless otherwise noted. The portfolio rate applies consistent 0.5 credit for all partial cells except where project-specific notes indicate a deviation.

Portfolio rate: $(282 + 0.5 \times 16) / 313 = 290 / 313 \approx 92.7\%$. Application of project-specific credit adjustments (FP009: 0 credit per data-limited rule; FP101: 1.0 credit per stated 87.5% rate) yields $290.5 / 313 \approx 92.8\%$. Reported as $\sim 92\%$ (conservative rounding).

Footnotes

*FP009 — Data-limited case. The partial (Δ) cell is not awarded 0.5 credit under the data-limited classification.

Available-data consistency rate = 80.0% (4 confirmed / 5 assessable cells). Raw mapper rate = 36.4% (5 cells / 14 total signal-year cells). CY2021 APR was not provided; CY2023–2024 Section 2 is confidential. The divergence between available-data rate (80.0%) and raw mapper rate (36.4%) reflects data access limitations, not framework prediction failure.

‡FP019 — Partial credit rule applied. Rate = $(9 \text{ confirmed} + 0.5 \times 7 \text{ partial}) / 16 \text{ assessable} = 12.5 / 16 = 78.1\%$. All seven partial cells reflect directional signal alignment with incomplete quantitative confirmation. Classification as partial (rather than confirmed) reflects the absence of project-level disaggregated MRV data required for full signal verification, not a reversal of directional prediction.

§FP026 — Image-based extraction. APR source documents partially available in image format only. Rate = $(46 + 0.5 \times 1) / 47 = 46.5 / 47 = 98.9\%$. The single partial cell reflects extraction quality constraints on one signal-year cell, not a substantive signal divergence. Extraction basis flagged in the APR Extraction Protocol (Appendix F).

¶FP034 — Chronic climate stress adjustment. Pre-adjustment rate = 76.9% (20 confirmed / 26 assessable). Post-adjustment rate = 80.8% (21 confirmed / 26 assessable). One signal cell was reclassified from disconfirmed ($\times$) to confirmed ($\checkmark$) following application of the chronic stress attribution protocol, which recognises that cumulative slow-onset climate stress signals cannot be assessed against single-year APR snapshots. Both rates are reported to preserve methodological transparency.

**FP042 — CY2023–2024 partial cell rule. Partial (Δ) cells in CY2023 and CY2024 are treated as confirmed under the portfolio-wide convention that partial matches in confidential or data-limited APR years — where directional alignment is present but full quantitative verification is inaccessible — receive the same credit as confirmed cells for purposes of rate calculation. Rate = $18 / 18 = 100\%$.

††FP069 — Of 25 total signal-year cells, the Score ($\checkmark$) is 60% and the Score (incl. Δ) is 84% (21/25). Zero contradictions were recorded across all five APR years. The VORTA™ D0 SVA_WARNING signal progressed from Partial (Δ) in CY2022–2023 to full Confirmation ($\checkmark$) in CY2024 through Cyclone Remal — the first instance in the evaluated portfolio where D0 SVA_WARNING was fully confirmed by a direct climate event destroying beneficiary assets with outstanding debt obligations. ToC Group C was confirmed from CY2022 onward through the Interim Evaluation's Unsatisfactory Outcome ratings — the independent verification of Q5 = No. Partial confirmations in C-FAIR™ S1 (CY2021–2023) reflect attenuated hazard materialization in non-cyclone years; directional alignment is maintained in all cases.

‡‡FP101 — RA1 boundary case: methodological limitation. Six disconfirmed ($\times$) cells reflect a methodological boundary condition in VERA™ v2.2 RA1 assessment for multi-component programmes that underwent mid-implementation restructuring events. The RA1 signal pre-assessed a delivery timeline that was subsequently modified through a formal restructuring process approved by the GCF Board. These six cells are classified as boundary-condition methodology limitations, not directional prediction failures. Included in the denominator and excluded from the numerator per protocol convention. FP101 effective rate = $26 \text{ confirmed} + 2 \text{ partial (1.0 credit)} = 28 / 32 = 87.5\%$.

§§FP115 — IF2 N/A adjustment. Four signal-year cells under the IF2 (Risk Coverage) signal category are classified as Not Applicable ($-$) under VERA™ v2.2 IF2 protocol, applied when the project's formal risk register

was not accessible to the extraction process. These four cells are excluded from both numerator and denominator. Assessable cell count adjusted to 16 accordingly. Rate = 16 / 16 = 100%.

H.4 Independent Evaluation Document Registry

Project	Document Type	Year	Key Convergence Finding
FP001	Project Completion Report	2023	Delivery constraint pattern confirmed
FP004	Mid-Term Review	2022	RA2 delivery constraint confirmed; “overestimation of delivery capacity”
FP017	Project Completion Report	2023	Causal logic confirmed; attribution conditionality confirmed
FP026	IEU LORTA RCT Midline	2023	Public component treatment effects confirmed; private pathway absence confirmed
FP034	Mid-Term Evaluation	2021	S2 Weak institutional constraint confirmed
FP039	Project Completion Report	2024	Private-Public Bifurcation confirmed; Component 1.1 abandonment confirmed
FP042	Interim Evaluation	2022	Output–Outcome disconnect confirmed
FP069	Interim Evaluation	2022	Sustainability: Moderately Unlikely — Q5=No confirmed
FP072	Interim Evaluation + Mid-Term Review	2022	S2 Moderate delivery continuity confirmed; shock absorption confirmed
FP101	Mid-Term Review	2024	RA2 overestimation confirmed — "overestimation of delivery capacity"
FP115	Project Completion Report	2024	IF2 structural fragility confirmed; PPA non-attainment confirmed
SAP004	Mid-Term Review	2021	S2 Capacity-dominant constraint confirmed; credit deployment ceiling confirmed

In no case did an independent evaluation finding contradict the material structural pathway identified by VERA-TCV™ pre-assessment. Convergence is documented at three levels of precision in Section 10.

Appendix I — KEY TERMS

This page defines the core terms and measurement concepts used throughout this report. Readers unfamiliar with the VERA-TCV™ framework are encouraged to review this page before proceeding to Part I.

The VERA-TCV™ Framework

VERA-TCV™ applies four computationally independent models to a single source — the project funding proposal — before any implementation data is reviewed. Each model examines a distinct layer of the project's causal chain.

VERA™ examines the Input layer. It asks: is the project's stated intent genuinely grounded in physical, institutional, and technical reality? It detects gaps between what a project claims and what its structural conditions can actually support.

ToC:f(x)™ examines the Activity–Output boundary. It asks: is the causal logic connecting activities to outputs structurally coherent, and do those outputs lead to the claimed outcomes? It evaluates twenty binary structural conditions across three groups — explicit presence, structural demonstration, and causal exclusivity.

C-FAIR™ examines the Output–Outcome boundary. It asks: can the delivery environment convert project outputs into outcomes? It assesses three dimensions — physical hazard exposure (S1), institutional and infrastructural adaptive capacity (S2), and macro-financial fragility (S3).

VORTA™ examines the Outcome–Impact boundary. It asks: do the outcomes, if achieved, generate social value sufficient to justify the cost of the intervention? It disaggregates return into market return (IRRm), value-inclusive return (IRRv), and composite return (IRRc).

Key Terms

Term	Definition
Ex-Ante	Before implementation begins. All VERA-TCV™ judgments in this report were generated from the funding proposal document alone, before any APR data was reviewed.
Annual Performance Report (APR)	GCF's annual implementation progress report submitted by accredited entities. In this study, APRs serve as independent comparators — not as inputs to the framework.
Theory of Change (ToC)	The causal logic connecting a project's activities to its intended outcomes and impact.
Signal-Year Cell	A single unit of measurement: one pre-assessed structural signal mapped against one year of APR data. This report evaluates 313 assessable signal-year cells across 15 projects.
Directional Consistency Rate	The proportion of signal-year cells in which the framework's pre-assessed signal corresponded to the direction of observed APR evidence. The portfolio-weighted rate in this report is approximately 92%.
Confirmed Signal (✓)	APR data establishes that the pre-assessed structural condition materialized in the direction predicted.
Partial Signal (Δ)	The predicted condition materialized in diminished form. Direction correct; intensity attenuated. Receives 0.5 credit in consistency rate computation.
Contradicted Signal (✗)	APR data establishes that outcomes moved in a direction opposite to the pre-assessed signal.
CR-Tier (VERA™)	VERA™'s three-tier classification of design authenticity. CR-Tier I reflects the highest level of structural grounding; CR-Tier III reflects one or more conditions below threshold.

Term	Definition
ToC TIER (ToC:f(x) TM)	ToC:f(x) TM 's classification of causal logic integrity. TIER I requires complete causal exclusivity — a condition structurally unavailable in multi-actor climate programmes by definition.
C-FAIR Grade	C-FAIR TM 's delivery environment classification (Grade A through E). Grade C — recorded by 12 of 15 projects — identifies environments where market mechanisms cannot operate independently and concessional finance is structurally justified.
VCS Grade (VORTA TM)	VORTA TM 's Value Capture Score. Grade A indicates sufficient social value to justify the intervention. Grade F indicates that value was architecturally unmeasurable from the design document — not that no value was generated.
Δ IRR	The spread between IRR _m (market return) and IRR _v (value-inclusive return). The quantitative expression of the public finance justification.
GRANT_JUSTIFIED	A VORTA TM determination indicating that no market mechanism exists to finance the activity without public subsidy. IRR _m is negative; grant finance is the only structurally viable capital channel.
REDESIGN_REQUIRE D	A VORTA TM determination indicating that the market can compensate the activity without public subsidy. GCF grant finance is structurally inappropriate.
Pattern A — Design–Environment Separation	The most common structural pattern (8 of 15 projects): sound design logic operating in an environment that constrains delivery.
Pattern B — Internal Mechanism Failure	A specified causal mechanism in the design never activates (2 of 15 projects).
Pattern C — Measurement Architecture Gap	Value is generated but architecturally unmeasurable within the project cycle (3 of 15 projects).
Pattern D — ToC-Level Constraint	The causal logic itself is attribution-constrained or solution-misaligned at design stage (2 of 15 projects).

Methodological Q&A

Every analytical framework has questions it would prefer not to answer. This section answers them anyway. The questions here were not selected to make VERA-TCV™ look robust. They were selected because they are the questions a rigorous reader would ask — and because a framework that cannot answer them has no business claiming what this report claims. Read this section as an adversary. That is how it was written.

Methodological Q&A

Q1. This evaluation matches APR outcomes.

Is this because the authors already knew the results and worked backwards?

This is the most important question to address, and it deserves a direct answer.

The **VERA-TCV™ framework** operates as a deterministic function: $f(x_1, x_2, x_3, \dots) = y$.

Each model receives structured inputs extracted from the project design document and produces outputs through *fixed algorithmic kernels*. The kernels do not receive outcome data. They cannot.

To engineer a specific output — to make the function produce a predetermined y — every input variable across all four independent models would need to be simultaneously manipulated in a consistent direction. Result-targeted manipulation of this kind would require simultaneous, auditable distortion of multiple input fields across four independent model kernels. Any such distortion would be methodologically detectable through the SCI audit record, which documents extraction agreement rates, divergence flags, and operator-resolved ambiguities for every evaluation. It would also be practically non-generalizable: a manipulation sequence calibrated to produce a predetermined output in one project's structural context would not transfer to another project with a different sector, region, implementing entity, and financing structure. The effort yields nothing that honest evaluation does not already provide — at substantially greater methodological cost and institutional risk.

There is a further structural constraint.

Climate projects differ across geography, political context, physical environment, institutional capacity, and financial architecture. The outcome space of a climate project resembles the output distribution of a *Monte Carlo simulation* — high-dimensional, path-dependent, and sensitive to unforeseeable exogenous variables.

Engineering a *reverse-mapping* from a known outcome back to a consistent set of inputs across that space is, for practical purposes, equivalent to designing a new framework for every project. The effort yields nothing that honest evaluation does not already provide. The **3AI SCI protocol** adds a further layer of protection. Three independent AI extractors — operating on *separate platforms with no shared session context* — produce parallel extractions that are then converged by the Orchestrator.

A single evaluator cannot manipulate inputs that are determined by majority convergence across three independent extraction engines. Any attempt to bias a field value would be visible in the SCI audit record.

Q2. The report presents six anchor cases from a fifteen-project portfolio. Were these six cases selected because their outcomes matched the framework's predictions?

The question contains a structural assumption worth examining before the answer is given.

To select cases that confirm a framework's predictions, an evaluator would first need to apply the framework to all candidate projects, identify those where pre-assessment signals matched ex-post outcomes, and then present only those cases. If that is what occurred here, it means the framework was applied to the full candidate population before case selection — which is not cherry-picking. It is comprehensive evaluation followed by selective reporting.

That is not what occurred here either.

The six anchor cases were selected before APR data was reviewed, using one criterion: structural pattern representation. The fifteen evaluated projects produce four structural patterns — Design–Environment Separation (Pattern A), Internal Mechanism Failure (Pattern B), Measurement Architecture and Value Realization (Pattern C), and ToC-Level Constraint (Pattern D). The six anchor cases were chosen to instantiate each pattern with maximum diagnostic clarity: FP034 and FP072 for Pattern A, FP026 for Pattern B, FP009 and FP039 for Pattern C, and FP069 for Pattern D.

This selection logic has one verifiable implication. If the anchor cases were chosen to maximize consistency rates, the nine non-anchor cases should show lower consistency rates than the anchor cases. They do not. FP072 — an anchor case — records 100% directional consistency. FP001 — a non-anchor case — records 79.2%. FP073, FP042, FP017, FP039, and FP115 — a mix of anchor and non-anchor cases — all record 100%. Consistency rate is not what differentiates anchor cases from the rest. Structural pattern clarity is.

The more substantive concern embedded in this question is whether the framework has been validated against structurally sound projects — or only against projects that underperformed. This concern is directly addressed by the portfolio itself. Three of the fifteen evaluated projects record VERA™ CR-Tier I design authenticity: FP004 (Bangladesh, 0.800), FP072 (Zambia, 0.882), and FP073 (Rwanda, 0.933). Three record VORTA™ VCS Grade A value realization: FP004 (IRRc +15.76%), FP009 (IRRc +16.78%), and FP017 (IRRc +48.71%). FP072 maintained 78.1% beneficiary coverage through the worst drought in forty years. FP017 exceeded its primary output targets. These outcomes were predicted from the design documents before implementation began — using the same methodology that identified structural constraints in FP026, FP069, and FP115.

The framework discriminates in both directions. The same pre-assessment protocol that identified Problem-Solution Misalignment in FP069 confirmed structural integrity and value realization in FP004. The same C-FAIR™ model that classified Uganda and Madagascar as Tier V correctly classified Chile as Grade B and predicted the market-compatible value realization that FP017's APR record confirmed. A framework that produced only negative diagnoses would be methodologically useless for the purpose this report advances: identifying, before approval, which structural configurations are capable of delivering climate outcomes and which are not.

The six anchor cases were not selected to make the framework look correct. They were selected to make the four structural patterns legible. The remaining nine cases — documented in full in the portfolio scorecard and consistency matrices — contain the framework's positive diagnoses alongside its negative ones. Both are necessary. The selection of anchor cases reflects editorial judgment about explanatory clarity, not methodological judgment about consistency optimization.

Q3. "Structurally weak projects fail."

This is common sense. Has VERA-TCV™ simply formalized the obvious?

The observation is fair. It is not, however, precise.

The intuition that "*weak structures produce poor outcomes*" is widely shared. What that intuition cannot do is specify which structural dimension will fail, through which causal mechanism, at what point in the implementation timeline, and with what interaction effects across components. A practitioner who says "*this project will struggle*" may be correct.

They cannot tell you whether the failure will originate in causal logic integrity, operating environment compatibility, incentive misalignment, or financial architecture fragility — or some

combination. They cannot assign a number to each dimension. They cannot replicate their judgment across a portfolio of 300 projects with consistent criteria.

VERA-TCV™ does not predict that projects fail. It identifies the specific structural conditions under which failure becomes probable, and assigns those conditions to named, measurable dimensions. In FP026, it was not sufficient to observe that Madagascar is a difficult operating environment. The framework identified that Q10 and Q11 — the outcome exclusivity conditions — were structurally unresolvable because the causal mechanism linking private finance to sustainable livelihoods was contingent on an EIB Investment Fund whose activation conditions were nowhere specified in the design document. That is not common sense. That is a specific structural diagnosis made in 2016 that the APR confirmed over seven years.

There is also a base rate problem that makes failure prediction appear trivially easy. Complex systems — *businesses, infrastructure projects, climate interventions* — fail more often than they succeed. In that environment, predicting failure looks like common sense because it is often correct by default. Precise evaluation must be able to predict success as well.

VERA-TCV™ gave FP017 VORTA™ VCS Grade A — the highest value realization classification — and confirmed its measurement architecture before implementation began. FP017 exceeded its targets. The framework discriminates in both directions.

The deeper **purpose of VERA-TCV™ is not prediction of outcomes**. It is structural diagnosis in service of design improvement.

The most valuable output is not "this project will fail" — it is "this specific causal link is unverifiable, this incentive is misaligned, this risk has no confirmed hedge, and here is how to address each one before submission."

The question is not whether *you want someone who predicts success or failure*.

The question is whether *you want a system that tells you why, with enough specificity to act on it*.

Q4. The four models produce correlated outputs. Is this multicollinearity?

Multicollinearity is a *property of regression models* in which correlated independent variables produce unstable coefficient estimates.

VERA-TCV™ is not a regression model.

It does not estimate coefficients. The concept does not apply in its technical form.

The substantive concern behind the term is valid and worth addressing: *if four models measure the same underlying construct, their agreement proves nothing beyond redundancy*.

The four VERA-TCV™ models measure structurally distinct dimensions.

VERA™ measures motivational authenticity — *whether the design intent is structurally genuine*.

ToC:f(x)™ measures causal logic integrity — *whether the activity chain is structurally capable of producing claimed outcomes*.

C-FAIR™ measures environmental risk — *whether the operating context can support delivery*.

VORTA™ measures financial value — *whether the investment generates measurable returns across social and market dimensions*.

These are not correlated proxies for a single latent variable. They are independent diagnostic axes. The clearest evidence is dissociation: FP026 received CR-Tier II (0.865) from VERA™ — the second-highest authenticity classification and the portfolio's highest CR-Tier II score — while simultaneously receiving TIER V from C-FAIR™ — the highest risk rating.

A system measuring the same construct in four ways cannot produce this pattern. High design

authenticity and extreme operating environment fragility are not the same dimension. They diverged in FP026, and the divergence is informative. It tells us that the project's problems were not motivational — they were environmental and structural. That distinction matters for redesign.

Q5. Under what conditions is VERA-TCV™ wrong?

This is the question the framework must answer for itself.

VERA-TCV™ predictive reliability decreases under the following conditions.

When design document quality is extremely low, extraction quality degrades across all three AI extractors, SCI convergence ratios fall below 0.5, and input values default to income-group proxies rather than project-specific data.

Under these conditions, the framework is measuring document quality as much as project structure.

When project duration is less than two years,

the APR record is insufficient for meaningful ex-post validation. Short-duration projects also compress the causal chain to a point where ToC Group C distinctions become less diagnostic.

When exogenous shocks exceed the absorptive capacity of the national system — war, pandemic, sudden sovereign default —

C-FAIR™ S1 and S2 signals remain valid as structural conditions but lose temporal precision.

VERA-TCV™ correctly identified Madagascar's systemic fragility. It did not predict COVID-19. The framework diagnoses shock absorption capacity, not specific shocks.

When private finance mechanisms are the primary causal engine and their activation conditions are not specified in the design document, VORTA™ VBI values will be systematically underestimated.

FP026 illustrates this: the sustainable livelihood outcomes claimed for the project's target communities were contingent on EIB Investment Fund deployment. When the fund did not deploy, those beneficiaries did not materialize. VERA-TCV™ flagged the structural dependency through ToC:f(x)™ Q10 and Q11 Unknown and VORTA™ MVI F0-CapEx WARNING.

It did not assign a probability to non-deployment, because the design document provided no basis for that calculation.

These limitations are not incidental. They define the boundary of the framework's validity and the conditions under which its outputs should be interpreted with additional caution.

Q6. ENHANCE is not a GCF-accredited institution. What is the authority basis for these evaluations?

The question conflates accreditation with analytical validity.

GCF accreditation governs the right to implement projects and disburse funds on behalf of the Fund.

It does not govern the right to analyze publicly available project documents using independent methodology.

Every GCF Funding Proposal cited in this report is *a public document*.

The APR data used for validation is *publicly disclosed*.

No proprietary or restricted information was used.

The more substantive version of this concern is: does ENHANCE have the methodological standing to make structural judgments about GCF projects?

The answer rests on three elements.

First, the framework itself — its mathematical architecture, indicator definitions, and algorithmic kernels — is documented, consistent, and independently replicable.

Second, the credentials of the framework's author — FRM®, SCR®, GARP SCR® Exam Item Writer (2025 Fall, 2026 Spring Exam) — establish domain competence in climate risk and financial risk at the level where these evaluations operate.

Third, and most importantly: independence is not a weakness. The structural conflict of interest embedded in self-reported Accredited Entity performance data — where implementing entities evaluate their own projects — is precisely the condition that makes independent third-party structural analysis valuable rather than presumptuous.

The GCF portfolio exists to deploy public and concessional capital toward climate outcomes that the world cannot afford to miss.

If the analytical tools used to evaluate that portfolio have a vested interest in the outcomes they measure, the system has a design problem.

VERA-TCV™ has no position in any GCF project. Its only interest is structural accuracy.

Q7. The 92% directional consistency rate relies on how "directional match" is defined. Partial confirmations receive 0.5 credit, and certain cells are excluded from the denominator. Does this make the metric artificially high?

The concern is legitimate and deserves a precise answer.

The conventions governing the 92% figure are documented in full in Appendix H and apply uniformly across all fifteen projects without exception. Partial credit (0.5) is awarded only when directional alignment is present but full quantitative confirmation is unavailable — not when a signal is ambiguous, reversed, or weak. A signal that fires in the wrong direction receives zero credit and is counted in the denominator. No cell is reclassified to improve the reported figure.

Two exclusion categories exist. Not Applicable (—) cells represent structural determinations: signals that are inapplicable by design to a given project type are not assessed. No Data Available (N/D) cells represent access constraints: years for which APR Section 2 data is classified or withheld cannot be scored. Neither category is used to inflate the reported rate — both are documented transparently, and the raw mapper rates that include these constraints are reported alongside the adjusted rates for the two cases where the divergence is most material (FP009, FP034).

The portfolio-weighted figure is derived directly from project-level rates: $(282 + 0.5 \times 16) / 313 = 92.7\%$, reported as approximately 92% under conservative rounding. Three projects record rates below 85%: FP001 (79.2%), FP009 (80.0%), and FP019 (78.1%). These are not suppressed. They appear in Section H.3 alongside every other project in the portfolio.

The metric's meaning is precisely what it claims: across assessable signal-year cells, the framework's pre-assessment signals corresponded to the direction of observed APR evidence approximately 92% of the time. No unresolved contradiction in primary structural pathways was recorded in any project in any year.

Q8. Without a comparison baseline — without knowing how GCF's own approval process performs on the same cases — VERA-TCV™'s net added value cannot be measured. Why is no such comparison provided?

The question rests on a category error that is worth resolving directly.

VERA-TCV™ and GCF's approval process answer different questions at different points in time. GCF's approval process asks: is this project sufficiently well-designed and institutionally credible to merit capital commitment? VERA-TCV™ asks: does the funded project's structural chain — from design intent through causal logic, delivery environment, and value formation — hold under independent structural scrutiny?

These are not competing functions. A comparison between them would require VERA-TCV™ to evaluate whether it is a better approval mechanism than GCF's existing one — which is not the claim this report advances and not a function ENHANCE proposes to perform. GCF's approval function requires political legitimacy, accredited entity accountability, and country ownership considerations that lie entirely outside VERA-TCV™'s analytical scope.

What this report does claim — and what the evidence supports — is that structural conditions consequential for implementation outcomes are present in funding proposal documents at the point of Board approval, are readable before implementation begins, and are not systematically identified by the monitoring architecture GCF deploys after approval. The APR system tracks whether implementation is progressing. It does not diagnose why structural failure pathways were encoded in the design. That is the gap. That is what VERA-TCV™ addresses.

The question of whether experienced practitioners reviewing the same design documents would reach equivalent structural diagnoses is a meaningful research question — and one that points toward VERA-TCV™'s specific contribution rather than undermining it.

Expert judgment, applied by a skilled and experienced evaluator, may well identify the same structural vulnerabilities that VERA-TCV™ identifies. The framework does not claim otherwise. What TCV™ provides that expert judgment structurally cannot is threefold.

First, **consistency at scale**. A framework that applies identical criteria across every project in a 300-project portfolio produces comparable diagnoses. Expert panels, however experienced, apply criteria that shift with composition, context, and time. The structural finding that Pattern A accounts for eight of fifteen projects — and that the same failure type recurs across Uganda, Belize, Rwanda, Mongolia, and Morocco simultaneously — is only legible when the diagnostic instrument is held constant across cases. Expert judgment applied sequentially to 300 projects is not the same instrument applied 300 times.

Second, **an auditable record**. VERA-TCV™ diagnosis produces a documented extraction record, a convergence audit trail, and a deterministic kernel output. Each judgment is traceable to a specific input field, a specific formula, and a specific extraction decision. Expert judgment, however rigorous, does not. The structural value of an audit record is not that it proves the diagnosis correct — it is that it makes the diagnosis falsifiable, refinable, and institutionally accountable in a way that undocumented expert intuition cannot be.

Third, **design-stage cost**. The framework operates from the funding proposal alone, before capital is committed and before implementation begins. Deploying experienced evaluation teams to conduct equivalent structural diagnostics on every GCF submission — at the depth required to identify EIB activation condition gaps in FP026 or D0 structural vulnerability in FP069 — would require resources that are not available at the approval stage. Structural diagnosis at design cost is qualitatively different from structural diagnosis at evaluation cost.

VERA-TCV™'s net added value relative to expert judgment is therefore not a question of whether experts could see the same things. It is a question of whether the same quality of structural scrutiny

can be applied consistently, traceably, and affordably across the full pipeline of climate finance submissions. That is the function this framework is designed to serve.

Q9. The framework's predictive validity is tested only on projects with sufficient APR documentation. Projects with poor documentation are excluded. Does this introduce survivorship bias into the 92% figure?

The concern is structurally precise and deserves an equally precise answer.

The selection criteria for this study — APR traceability depth, outcome-pathway verifiability, and structural heterogeneity — are applied at the documentation availability level, not at the outcome level. Projects are excluded because their APRs are unavailable or insufficiently detailed, not because their outcomes were unfavorable or inconsistent with VERA-TCV™ pre-assessment.

Survivorship bias would require that projects with better documentation systematically perform better than projects with poor documentation — such that restricting the sample to well-documented projects overstates the framework's predictive accuracy. There is no evidence that GCF's APR disclosure patterns correlate with project performance in this direction. The projects excluded from this study are excluded because GCF's results management architecture has not produced sufficient public documentation for VERA-TCV™×APR analysis — not because their structural signals and APR trajectories were unfavorable.

The limitation this selection introduces is scope, not bias. This report cannot speak to projects outside the documented analytical subset. That limitation is stated explicitly in Section 4.2. What the report can state — and does — is that the documentation gap itself is a structural finding: among GCF's 336 approved projects, the number carrying complete public APR Section 2 series sufficient for full VERA-TCV™×APR analysis is materially smaller than the approved portfolio count. The structural gap in GCF's results management architecture is not merely a gap in what projects achieve. It is a gap in what is measurable after they end.

Q10. GGGI's 101-project portfolio produced zero Tier I and zero Tier II results under ToC:f(x)™. Does this indicate that the framework's standards are unrealistically high — or that the framework is designed to produce negative diagnoses?

The question is legitimate and deserves a structural answer rather than a defensive one.

The first part of the answer is architectural. ToC:f(x)™ TIER I requires complete causal exclusivity — a project design in which the claimed outcome is attributable solely and verifiably to the project's own intervention chain, with no material contribution from external actors, parallel programmes, or environmental conditions outside the project's control. This condition is not difficult to achieve. It is structurally unavailable in multi-actor climate programmes by definition.

Climate projects — whether GCF-funded, GGGI-managed, or delivered through any multilateral mechanism — operate in systems where outcomes are co-produced. A coastal resilience project improves food security outcomes. A national early warning system depends on regional meteorological infrastructure. An agroforestry programme interacts with land tenure policy, seasonal rainfall patterns, and commodity markets simultaneously. In these systems, the causal chain from project activity to claimed impact passes through conditions the project neither controls nor can fully specify. TIER I is inaccessible not because the projects are poorly designed, but because the causal architecture of climate action is structurally incompatible with complete exclusivity claims. This finding applies to GCF's evaluated portfolio identically: not a single project among the fifteen achieved TIER I, for the same structural reason.

The second part of the answer concerns what the framework is actually measuring. ToC:f(x)[™] does not use TIER I as its operational standard. The framework's verification standard is directional contribution — whether the project's causal logic is structurally capable of producing its claimed outcomes in the direction and through the mechanism specified, given the delivery environment in which it operates. TIER II through TIER V classify the degree to which that directional contribution claim is structurally supportable. The relevant diagnostic question is not "does this project have exclusive causal control" — it is "does this project's causal chain hold, and at which point does it become structurally constrained."

The third part concerns the GGGI finding specifically. That all 101 evaluated projects fall within TIER III through TIER V is not a verdict on GGGI as an institution. It is a structural finding about the design document quality of publicly-disclosed project documentation in multi-actor climate programmes. Two distinct conditions produce low ToC:f(x)[™] scores: genuine causal logic gaps in project design, and documentation gaps that make the causal logic unverifiable from the project document alone. The GGGI evaluation identifies the distribution of these conditions across a 101-project portfolio — and the finding that no project achieved TIER I or TIER II is fully consistent with the GCF portfolio result. It is the expected structural outcome when ToC:f(x)[™] is applied to multi-actor climate finance at scale.

The correct interpretation is therefore not "the framework's standards are too high." The correct interpretation is this: the standard that multi-actor climate programmes have routinely applied to their own causal claims — implicit or otherwise — is structurally insufficient. TIER I is not a realistic operational target. It is a theoretical upper bound that defines the causal architecture space. The operational question is whether a project's causal logic is strong enough to justify the capital committed to it, and whether the structural constraints on that logic were identified and addressed before implementation began. That is what ToC:f(x)[™] measures. That is what the GGGI and GCF evaluations confirm.

Why Structure Must Come First

Projects are often judged in motion.

They are assessed through implementation progress, disbursement speed, adaptation indicators, beneficiary counts, or the promise of scale. In climate finance, this tendency is especially strong. Once a project enters the language of delivery, attention shifts immediately to how it performs, how far it can extend, and how quickly it can expand. But this sequence contains a deeper error.

Movement is not the first question. Structure is.

In mechanical systems, dynamic behaviour is not meaningfully analysed unless the system first satisfies the conditions of static equilibrium. This is not a matter of pedagogical order, but of logical necessity. A system whose forces and moments do not balance cannot sustain any controlled or purposeful motion. It does not become dynamic; it becomes uncontrolled.

This principle is not confined to engineering. It reflects a broader form of recognition.

A structure built from cards may appear stable when left undisturbed. In the absence of external forces, it can stand long enough to create the illusion of stability. Formally, such a structure satisfies the conditions of static equilibrium: forces and moments balance.

However, it does so in an unstable configuration. The slightest perturbation — movement, pressure, or environmental change — is sufficient to destroy it.

What seemed stable was never structurally sound; it was merely balanced in a configuration that could not endure.

Importantly, such a structure is never recognised as a “house.” Intuitively, humans already distinguish between forms that can momentarily stand and those that can persist under real conditions. This distinction does not originate from engineering; engineering formalises it. Structural evaluation is not only about whether equilibrium exists, but whether that equilibrium is stable under the conditions the system will

actually face.

The same principle applies to climate project design.

A project may appear coherent on paper. It may align with policy language, fit funding criteria, and present a convincing theory of change. It may even enter implementation and exhibit early signs of progress. But if its underlying structure is not stable — if incentives are misaligned, if causal relationships are weak, if financial architecture is contradictory, if execution depends on conditions not created by the project itself — then apparent performance does not confirm validity. It only delays failure.

This is the central error of much contemporary climate finance. Systems are frequently treated in dynamic terms — *scale, impact, delivery, replication* — before their structural stability has been established. In effect, motion is being interpreted in systems that have not yet been shown to withstand even minimal disturbance.

The ENHANCE approach begins from a different premise. Structure is not one dimension among many. It is the condition that determines whether any other dimension can exist.

This is why the present report applies VERA-TCV™'s four-model framework before any discussion of performance or expansion. These models do not exist to add complexity. They exist to answer the first necessary question: does the structure hold under the conditions it will face?

VERA™ evaluates whether the project is grounded in structural reality — whether its logic is causally valid, its incentives credible, and its claims aligned with verifiable conditions. ToC: f(x)™ evaluates whether the causal chain from activities to outcomes is structurally coherent rather than narratively asserted. C-FAIR™ evaluates whether the delivery environment is capable of converting project outputs into outcomes — whether physical hazard exposure, institutional capacity, and macrofinancial conditions permit the design to function.

VORTA™ evaluates whether the outcomes, if achieved, generate social value sufficient to justify the cost of the intervention.

Only when structural stability is established does it become meaningful to analyse how a project will behave in motion.

This distinction is not incremental. A project that fails structurally is not a weaker project within the same category. It is a different object altogether. Just as a house of cards is not a fragile house but a non-house, a project whose structural failure pathways are unaddressed at design stage is not a risky climate solution but a preventable failure — temporarily sustained by appearance, momentum, or external support.

The implication extends beyond methodology. It concerns how institutions interpret action.

Human systems already recognise that not everything which stands can endure, and not everything which moves is viable. Yet in climate project evaluation, this recognition is often suspended. Political urgency, financial pressure, and institutional habit create incentives to prioritise motion over stability, expansion over coherence, and delivery over structure.

The consequence is predictable: projects that move before they can stand, and portfolios that expand before they can endure.

The purpose of structural evaluation is not to slow action. It is to prevent the misinterpretation of instability as progress.

If a structure is stable, movement becomes meaningful. If it is not, movement only accelerates collapse.

This is the principle that underlies the ENHANCE system. It is not an analogy borrowed from engineering, but a general condition shared by all durable systems — mechanical, ecological, financial, and institutional.

They must not only balance.

They must hold.

Junyoung Hur, FRM®, SCR®
Founder, ENHANCE

Ex-Ante Structural Diagnosis of Climate Project Performance

VERA-TCV™ Framework Applied to 15 GCF Projects

Published by ENHANCE Institute Research Division of ENHANCE Co., Ltd.
contact@enhance-institute.org | enhance-institute.org

Intellectual Property VERA™, ToC:f(x)™, C-FAIR™, VORTA™, TCV™, VERA-TCV™, SLS™, PathFinder-GOV™, SBA™, and SAA™ are proprietary system architectures and analytical frameworks developed by ENHANCE Institute. All rights reserved.

This document discloses the public research findings of VERA-TCV™ applied to fifteen GCF-funded projects. Underlying computational logic, extraction protocols, quantitative weighting structures, calibration databases, and proprietary system extensions — including Python implementations — remain protected intellectual property of ENHANCE Institute and are not disclosed herein.

Reproduction, structural imitation, or commercial implementation of any architecture, framework, or methodology presented in this document, in whole or in part, without prior written authorisation from ENHANCE Institute, is not permitted.

Citation Hur, J.Y. (2026). Ex-Ante Structural Diagnosis of Climate Project Performance: VERA-TCV™ Framework Applied to 15 GCF Projects. ENHANCE Institute. Public Research Report. May 2026.

Evaluated Framework Versions VERA™ v2.2 · ToC:f(x)™ SIM-v4.1 · C-FAIR™ v3.0 · VORTA™ v2.5
 Extraction Protocol: APR-SIM v1.3 · Multi-Agent SCI Protocol: 3AI-v2.0

Classification Publicly Disclosed — ENHANCE Institute, May 2026
 Document Code ENH-RPT-GCF-TCV-001-2026

The structural conditions that determine climate project performance are readable before implementation begins. Structure before execution. Diagnosis before capital.

© 2026 ENHANCE Institute. All rights reserved.



The Gravity of Truth

Contact: contact@enhance-institute.org
www.enhance-institute.org

ENHANCE

ENHANCE Institute, the research division of ENHANCE Co., Ltd., is dedicated to advancing structural integrity in climate and sustainability projects.

ENHANCE develops computational and causal analytical frameworks designed to evaluate the structural architecture of climate interventions. Its research focuses on identifying the causal robustness, evidentiary sufficiency, and systemic alignment of project designs before large-scale capital deployment.

Through methodologies such as the ToC:f(x)[™] framework and the Structural Integrity Index (SII), ENHANCE provides analytical tools for assessing the structural feasibility of climate projects and improving the reliability of climate finance decisions.

ENHANCE Institute collaborates with researchers, institutions, and financial actors seeking more rigorous evaluation of climate project architectures.